Probability for Finance

Studiabile — Chapters 1–6, through HW 61

Theory, Expanded Proofs, Worked Homework, Past and AI-Generated Exams, and More

These unrevised notes were generated using artificial intelligence, following the student's preferred style and, where possible, the style of Professor Tiziano De Angelis and former tutor Andrea Bovo. Additional proofs and explanations not presented or proved by the professor have been included at the student's discretion. The material has not undergone a complete independent review and may contain errors.

These notes follow the lecturer's order and explain the hypotheses and steps of each proof. Every homework retains its own statement and solution beside the relevant theory. A later homework is anticipated only when the lecturer's own proof takes that exact fact as given; its proof remains at the homework itself. All mathematical displays are unnumbered.

Contents

Theory, Expanded Proofs, Worked Homework, Past and AI-Generated Exams, and More . . . 1

1. Measure Spaces . . . 5
- Measure theory and continuous functions . . . 5
- Def 1.1. Algebra . . . 5
- HW 1 — Closure under finite intersections . . . 6
- HW 2 — Finite unions and intersections . . . 6
- Def 1.2. σ-algebra . . . 6
- HW 3 — Countable intersections . . . 6
- Measurable spaces, generated sigma-algebras and Borel sets . . . 6
  - Notation: . . . 6
  - Measurable sets: . . . 7
  - Measurable space: . . . 7
- Prop 1.1 . . . 7
- Set functions . . . 8
- Additive and countably additive set functions . . . 8
- HW 4 — Finite additivity by induction . . . 8
- HW 5 — Countable additivity implies additivity . . . 8
- Measure . . . 9
  - Measure space: . . . 9
  - Null set: . . . 9
  - Complete measure space: . . . 9
- Definition 1.3. π-system . . . 9
- HW 6 — Lower rays form a pi-system . . . 9
- Lemma 1.2 . . . 9
- Theorem 1.3 (Carathéodory's Theorem) . . . 9
- Lebesgue measure . . . 9
- HW 7 — The interval algebra . . . 10
- HW 8 — Finite additivity of interval length . . . 11
- HW 9 — Borel singletons . . . 11
- HW 10 — Lebesgue measure of a singleton . . . 12
- Lemma 1.4 . . . 12
- HW 11 — Subadditivity and the union formula . . . 12

HW 12 — Monotonicity of measures . . . 13
HW 13 — Measure of a difference . . . 13
Lemma 1.5. Continuity of measure from below . . . 13
HW 14 — Disjoint pieces . . . 13
HW 15 — Unions of the disjoint pieces . . . 14
Lemma 1.6. Continuity of measure from above . . . 14
HW 16 — Continuity from above . . . 14
Hint: . . . 14
Lemma 1.7 . . . 14
HW 17 — Rational points form a null set . . . 15
2. Events . . . 16
Probability spaces . . . 16
Example 1 (Simple coin tossing models) . . . 16
HW 18 — Sure events and null complements . . . 16
Proposition 2.1 . . . 16
Def lim inf and lim sup . . . 17
HW 19 — Tail infima . . . 17
Definition of convergence through liminf and limsup . . . 17
HW 20 — An oscillating sequence . . . 17
HW 21 — Threshold properties of limits . . . 17
HW 22 — Negation, suprema and infima . . . 18
Def limsup of collection of sets . . . 19
HW 23 — Measurability of the limsup event . . . 19
Lemma 2.2 (Reverse Fatou's lemma for sets) . . . 19
Lemma 2.3 (First Borel–Cantelli's lemma) . . . 19
Def liminf of collection of sets . . . 20
HW 24 — Measurability of the liminf event . . . 20
HW 25 — Complement relation of event limits . . . 20
Lemma 2.4 (Fatou's lemma for sets) . . . 20
HW 26 — Fatou inequality by complements . . . 21
Hint: . . . 21
Definition of indicator function . . . 21
HW 27 — Indicators and limsup . . . 21
Hint: . . . 21
3. Random Variables . . . 23
Random variables and pre-images . . . 23
Def — pre-image of a set . . . 23
Def 3.1 (Measurable functions) . . . 23
Notation: . . . 23
Borel-measurable functions: . . . 23
Properties of pre-image mapping . . . 23
HW 28 — Pre-images of complements and intersections . . . 23
Lemma 3.1 — The pre-image of Borel sets form a σ-algebra . . . 24
Rmk: . . . 24
Problem: . . . 24
Lemma 3.2. Measurability through a generating class . . . 24
Remarks . . . 25
HW 29 — A Borel indicator . . . 25
HW 30 — Equivalent threshold tests . . . 26
Lemma 3.3. Composition of functions . . . 26

Lemma 3.4 . . . 27
Lemma 3.5. Measurability of infimum, supremum, liminf and limsup . . . 27
HW 31 — Superlevel sets of an infimum . . . 28
Corollary 3.6 — Measurability of the Convergence Set . . . 28
Def: Random variable . . . 29
Example 1(b) — Frequency event . . . 29
Def Law of a random variable . . . 29
HW 32 — Pre-image of a countable union . . . 30
Why is it sufficient to specify the law on a π-system . . . 31
Definition CDF . . . 31
Properties of cdf . . . 31
HW 33 — CDF tails with mass at infinity . . . 32
HW 34 — Left limits and atoms . . . 33
Definition — Probability Density Function (pdf) . . . 34
Properties of pdf . . . 34
Theorem – Existence of a r.v. with a prescribed CDF . . . 34
Construction — Inverse Transform Method . . . 35
Lemma 3.7 — The σ-Algebra Generated by a Random Variable . . . 35
HW 35 — Prove That $\pi(Y)$ Is a π-System on Ω . . . 37
Properties — Measurable Functions of Random Variables . . . 37
Financial Interpretation . . . 38

4. Independence . . . 39
Definition 4.1 — Independent Sigma-Algebras . . . 39
Definition 4.2 — Independent Random Variables . . . 39
Definition 4.3 — Independent Events . . . 39
HW 36 — σ-Algebra Generated by an Indicator Function . . . 39
Connection with Independence of Events . . . 40
Lemma 4.1 — Independence of σ-Algebras Generated by π-Systems . . . 40
HW 37 — Prove That Two Set Functions Define Measures . . . 41
Corollary — Characterisation of Independence via Threshold Events . . . 42
Generalisation — Independence of Finitely Many Random Variables . . . 43
HW 38 — Independence of Three σ-Algebras Generated by π-Systems . . . 43
HW 38 — Part II: Why Is the Assumption $\Omega \in \mathcal{I}_k$ Necessary? . . . 45
Counterexample 1 — Pairwise Independence Does Not Imply Joint Independence . . . 45
Counterexample 2 — Necessity of $\Omega \in \mathcal{I}_k$ . . . 46
Lemma 4.2 — Second Borel–Cantelli's Lemma . . . 48
Example 2 (Williams, Example 4.4) — Application of the Borel–Cantelli Lemmas . . . 50
HW 39 — Formula (4.3) . . . 53
Fundamental Question for Modelling . . . 54
Stochastic Processes . . . 54
Existence of Stochastic Processes . . . 54
Example 3 — Markov Chains . . . 55
A Monkey Types Shakespeare . . . 57
HW 40 — Set-Theoretic Verification of the Events in the Monkey Example . . . 59
HW 41 — Independence of Products of Random Variables . . . 62

5. Integration . . . 64
Motivation — Expectation as an Integral . . . 64
Recall — Riemann Integral . . . 64

Notation — Lebesgue Integral with Respect to a Measure . . . 64
Example — Counting Measure and Infinite Series . . . 65
Integrals of Positive Simple Functions . . . 65
Definition — Integral of an Indicator Function . . . 65
Definition — Positive Simple Function . . . 65
HW 42 — Measurability of a Simple Function . . . 65
Different Representations and Canonical Form of Simple Functions . . . 66
Definition — Integral of a Positive Simple Function . . . 66
Observation — The Convention $0 \cdot \infty = 0$ . . . 66
HW 43 — Well-Definedness of the Integral of Simple Functions . . . 67
HW 44 — Properties of Integrals of Simple Functions . . . 69
Recall — μ-almost everywhere. . . . 69
Example — Equality Lebesgue-Almost Everywhere . . . 70
Definition — Integral of a Positive Measurable Function . . . 71
Observation — Agreement with the Simple Integral . . . 71
HW 45 — Monotonicity of the Positive Lebesgue Integral . . . 71
Lemma 5.1 — Zero Integral Implies Zero Almost Everywhere . . . 71
HW 46 — Level-Set Representation of $\{f > 0\}$ . . . 72
Example — Application of Lemma 5.1 . . . 73
Observation — Reduction from Almost-Everywhere to Pointwise Convergence 73
Theorem 5.2 — Monotone Convergence Theorem (MON) . . . 74
HW 47 — Increasing Sets in MON . . . 75
Approximation by Positive Simple Functions . . . 76
Lemma 5.3 — Equality Almost Everywhere . . . 77
Lemma 5.4 — Fatou's Lemma for Functions . . . 77
Lemma 5.5 — Reverse Fatou's Lemma for Functions . . . 78
Observation. . . . 78
HW 48 — Proof of Reverse Fatou's Lemma for Functions . . . 78
Lemma 5.6 — Linearity of the Positive Lebesgue Integral . . . 79
Positive and Negative Parts of a Measurable Function . . . 80
HW 49 — Positive and Negative Parts . . . 80
Definition — Lebesgue Integral of a Signed Measurable Function . . . 80
Definition 5.1 — L^1-Space . . . 81
HW 50 — Absolute Value of the Integral . . . 81
HW 51 — Closure of L^1 under Linear Combinations . . . 81
Linearity of the Integral on L^1 . . . 82
Theorem 5.7 — Dominated Convergence (DOM) . . . 82
HW 52 — Why the Domination Hypothesis Is Necessary . . . 83
The Standard Machinery . . . 84
A Measure Defined by a Nonnegative Density . . . 84
HW 53 — Countable Additivity of λ . . . 84
Theorem 5.8 — Radon–Nikodym Theorem . . . 85
Recall. . . . 85
Summary . . . 86

6. Expectation . . . 89
Recap — Properties of Expectation . . . 89
Monotone Convergence (MON, Theorem 5.2). . . . 89
Fatou's Lemma (Lemma 5.4). . . . 89
Dominated Convergence (DOM, Theorem 5.7). . . . 89
Lemma 6.1 — Markov's Inequality . . . 89

Example 4 — Applications of Markov's Inequality . . . 90
Properties — Nonnegative Random Variables . . . 90
Finite expectation implies almost-sure finiteness. . . . 90
Expectation of a nonnegative series. . . . 90
Summable expectations imply almost-sure convergence. . . . 91
Application to indicators — Borel–Cantelli I. . . . 91
HW 54 — Proofs of Properties (a)–(c) . . . 91
Theorem 6.2 — Jensen's Inequality . . . 92
Definition — L^p Spaces of Random Variables . . . 92
HW 55 — The L^p Norm . . . 92
Lemma 6.3 — Monotonicity of L^p-Norms . . . 93
HW 56 — Closure of L^p under Linear Combinations . . . 94
Remark — Vector Space Structure of L^p . . . 94
Theorem 6.4 — Cauchy–Schwarz Inequality . . . 94
HW 57 — The Triangle Inequality in L^2 via Cauchy–Schwarz . . . 95
Variance and Covariance . . . 95
Scalar Product in L^2 . . . 96
Pythagoras' theorem. . . . 96
HW 58 — Proof of Pythagoras' Theorem . . . 96
Completeness — Cauchy Sequences in L^p . . . 96
Lemma 6.5 — Completeness of L^p . . . 97
HW 59 — Geometric Series . . . 97
Lemma 6.6 — Expectation with Respect to the Law of a Random Variable . . . 98
Special Case — Expectation Using a Probability Density . . . 99
Lemma 6.7 — Hölder's and Minkowski's Inequalities . . . 99
Hölder's inequality: . . . 99
Minkowski's inequality: . . . 99
Lemma 6.8 — Chebyshev's Inequality . . . 100
HW 60 — Proof of Chebyshev's Inequality . . . 100
Theorem 6.9 — Independence, Expectation, and Variance . . . 101
Remark — Dependence Does Not Ensure Integrability of a Product . . . 102
Introduction — A Strong Law of Large Numbers . . . 102
Theorem 6.10 — A Strong Law of Large Numbers . . . 102
HW 61 — Strong Law with a Nonzero Common Mean . . . 103

1. Measure Spaces

Measure theory and continuous functions

A function f from a topological set X into another topological set Y (denoted as $f : X \to Y$) is said to be continuous if the pre-images of open sets in Y are open sets in X. In formulae $f^{-1}(A) = \{x \in X : f(x) \in A\}$ is open in X whenever A is open in Y.

In Measure Theory the focus is on measurable functions. A function $f : X \to Y$ is measurable if $f^{-1}(A) = \{x \in X : f(x) \in A\}$ is a measurable set in X whenever A is a measurable set in Y.

Def 1.1. Algebra

Let S be a set. A collection Σ_0 of subsets of S is called an algebra if

(i) $S \in \Sigma_0$,

(ii) $A \in \Sigma_0 \Rightarrow A^c \in \Sigma_0$,

(iii) $A, B \in \Sigma_0 \Rightarrow A \cup B \in \Sigma_0$.

HW 1 — Closure under finite intersections

Show that $A, B \in \Sigma_0 \Rightarrow A \cap B \in \Sigma_0$.

Proof.

Let $A, B \in \Sigma_0$. It is enough to show that $(A \cap B)^c \in \Sigma_0$ since Σ_0 is an algebra.

$$(A \cap B)^c = \{x : x \notin (A \cap B)\} = \{x : x \notin A \text{ or } x \notin B\} = A^c \cup B^c.$$

By the properties **(ii)** and **(iii)** of an algebra we have that $(A \cap B)^c \in \Sigma_0$ and so $(A \cap B) \in \Sigma_0$ by **(ii)**.

HW 2 — Finite unions and intersections

Show that $(A_n)_{n=1}^N \subseteq \Sigma_0$ implies $\bigcup_{n=1}^N A_n \in \Sigma_0$ and $\bigcap_{n=1}^N A_n \in \Sigma_0$.

Proof.

By definition of algebra we know that $A, B \in \Sigma_0 \Rightarrow A \cup B \in \Sigma_0$. We prove the result by induction on N. **Base case:** $N = 1$ is trivial. **Inductive hypothesis:** Assume that for some $N \geq 1$, $\bigcup_{n=1}^N A_n \in \Sigma_0$. **Inductive step:** We want to prove that $\bigcup_{n=1}^{N+1} A_n \in \Sigma_0$. Notice that

$$\bigcup_{n=1}^{N+1} A_n = \left(\bigcup_{n=1}^{N} A_n \right) \cup A_{N+1} \in \Sigma_0.$$

Therefore by inductive hypothesis and by assumption $A_{N+1} \in \Sigma_0$ the closure of an algebra under finite unions holds. For finite intersection by De Morgan's law

$$\bigcap_{n=1}^{N} A_n = \left(\bigcup_{n=1}^{N} A_n^c \right)^c.$$

Hence by the closure under complements $A_n^c \in \Sigma_0$ for $n = 1, \ldots, N$ and by the closure under finite unions also $\bigcup_{n=1}^N A_n^c \in \Sigma_0$ and therefore

$$\left(\bigcup_{n=1}^{N} A_n^c \right)^c = \bigcap_{n=1}^{N} A_n \in \Sigma_0$$

by closure under complements.

Def 1.2. σ-algebra

Let S be a set. A collection Σ of subsets of S is called a σ-algebra if

(i) $S \in \Sigma$,

(ii) $A \in \Sigma \Rightarrow A^c \in \Sigma$,

(iii) $(A_n)_{n=1}^\infty \subseteq \Sigma \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \Sigma$.

HW 3 — Countable intersections

Show that $(A_n)_{n=1}^\infty \subseteq \Sigma \Rightarrow \bigcap_{n=1}^\infty A_n \in \Sigma$.

Proof.

Let $(A_n)_{n \in \mathbb{N}} \subseteq \Sigma$. Since Σ is a σ-algebra it is closed under complements. Therefore $A_n^c \in \Sigma \; \forall n \in \mathbb{N}$. By closure under countable unions $\bigcup_{n=1}^\infty A_n^c \in \Sigma$ and therefore its complement $(\bigcup_{n=1}^\infty A_n^c)^c \in \Sigma$, which is $\bigcap_{n=1}^\infty A_n$ by De Morgan's law.

Measurable spaces, generated sigma-algebras and Borel sets

In the probabilistic formulation of a random experiment we will use σ-algebras to describe all possible events. However, working with σ-algebras is difficult because it is often difficult to describe ALL sets in a σ-algebra. It is often easier to handle algebras and we will soon learn about π-systems.

Notation: From now on S will always denote a set, Σ_0 an algebra of subsets of S and Σ a σ-algebra of subsets of S.

Measurable sets: any $A \in \Sigma$ is said to be a Σ-measurable set.

Measurable space: a pair (S, Σ) is called a measurable space.

Letting $\mathcal{C}$ be a class of subsets of S we denote by $\sigma(\mathcal{C})$ the smallest σ-algebra that contains all sets in $\mathcal{C}$. We call it the σ-algebra generated by $\mathcal{C}$. Notice that, in particular, if $\mathcal{C} \subseteq \Sigma$ then $\sigma(\mathcal{C}) \subseteq \Sigma$.

Given a topological space X the Borel σ-algebra $\mathcal{B}(X)$ is the smallest σ-algebra that contains all the open sets of X. That is

$$\mathcal{B}(X) = \sigma(\text{open sets in } X).$$

In particular we will often work with $X = \mathbb{R}$ and $\mathcal{B}(\mathbb{R})$ generated by open sets on the real line. In that context a very useful class of subsets is the following (example of π-system)

$$\pi(\mathbb{R}) = \{(-\infty, x],\ x \in \mathbb{R}\}.$$

Indeed it is easy to describe all the set contained in $\pi(\mathbb{R})$ and the next proposition allow us to connect with $\mathcal{B}(\mathbb{R})$.

Prop 1.1

It holds $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$.

Proof.

To prove the proposition we will show that $\sigma(\pi(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$ and $\sigma(\pi(\mathbb{R})) \supseteq \mathcal{B}(\mathbb{R})$. A generic set in $\pi(\mathbb{R})$ is of the form $(-\infty, x]$ for some $x \in \mathbb{R}$ and it can be written as

$$(-\infty, x] = \bigcap_{n=1}^{\infty} (-\infty, x + \tfrac{1}{n}) \in \mathcal{B}(\mathbb{R}),$$

A point $y \leq x$ belongs to every open ray on the right. Conversely, if $y > x$, choose n with $1/n < y - x$; then $y \notin (-\infty, x + 1/n)$. This proves the equality. Each ray is open, and the Borel sigma-algebra is closed under countable intersections, proving the final membership. *Every element of $\pi(\mathbb{R})$ is also an element of $\mathcal{B}(\mathbb{R})$.* Therefore $\pi(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R})$ which implies $\sigma(\pi(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$. *For the reverse inclusion, we show that every open set belongs to $\sigma(\pi(\mathbb{R}))$.* We notice that any open set in $\mathbb{R}$ can be written as countable union of disjoint open intervals. Therefore it suffices to show that any open interval (a, b), allowing infinite endpoints, belongs to $\sigma(\pi(\mathbb{R}))$. If both endpoints are infinite, the interval is $\mathbb{R}$, which belongs to every σ-algebra on $\mathbb{R}$. For the remaining cases the following identities apply, omitting a redundant ray when an endpoint is infinite. Notice that

$$(a, b) = (-\infty, b) \cap (a, \infty),$$
$$(a, +\infty) = (-\infty, a]^c \in \sigma(\pi(\mathbb{R})),$$

because it is the complement of a set in $\sigma(\pi(\mathbb{R}))$.

$$(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b - \tfrac{1}{n}] \in \sigma(\pi(\mathbb{R}))$$

For this equality, every point in a ray on the right is strictly below b. If $y < b$, choose n such that $1/n \leq b - y$; then $y \leq b - 1/n$ and belongs to the union. Each closed ray belongs to the generating class, so countable-union closure proves the membership. Since $(a, +\infty)$ and $(-\infty, b) \in \sigma(\pi(\mathbb{R}))$ then $(a, b) \in \sigma(\pi(\mathbb{R}))$. For an arbitrary open set $A \subseteq \mathbb{R}$ we have a sequence $(a_n, b_n)_{n \in \mathbb{N}}$ such that

$$A = \bigcup_{n=1}^{\infty} (a_n, b_n) \in \sigma(\pi(\mathbb{R})),$$

where the final membership is by countable union of elements of a σ-alg. Therefore any open set in $\mathbb{R}$ belongs to $\sigma(\pi(\mathbb{R}))$ and thus we conclude that $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\pi(\mathbb{R}))$.

Set functions

A set function is a mapping from an algebra to the positive extended real line $\mu_0 : \Sigma_0 \to [0, +\infty]$.

Additive and countably additive set functions

A set function $\mu_0 : \Sigma_0 \to [0, +\infty]$ is additive if $\mu_0(\emptyset) = 0$ and

$$\mu_0(A \cup B) = \mu_0(A) + \mu_0(B)$$

whenever $A, B \in \Sigma_0$ are disjoint. A set function is countably additive if $\mu_0(\emptyset) = 0$ and for any $(A_n)_{n=1}^{\infty} \subseteq \Sigma_0$ such that $A_i \cap A_j = \emptyset \ \forall i \neq j$ and $\bigcup_{n=1}^{\infty} A_n \in \Sigma_0$ it holds

$$\mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

HW 4 — Finite additivity by induction

Let μ_0 be an additive set function. Show that for any $(A_n)_{n=1}^{N} \subseteq \Sigma_0$ such that $A_i \cap A_j = \emptyset$ for all $i \neq j$, it holds

$$\mu_0\left(\bigcup_{n=1}^{N} A_n\right) = \sum_{n=1}^{N} \mu_0(A_n).$$

Proof.

We prove the statement by induction on N. **Base case:** $N = 1$ is trivial $\mu_0(A_1) = \mu_0(A_1)$. **Inductive hypothesis:** Assume the statement holds for some $N \geq 1$. **Inductive step:** We want to prove that the statement holds for $N + 1$.

Notice that

$$\bigcup_{n=1}^{N+1} A_n = \bigcup_{n=1}^{N} A_n \cup A_{N+1} \in \Sigma_0$$

since Σ_0 is an algebra (finite unions).

$$\left(\bigcup_{n=1}^{N} A_n\right) \cap A_{N+1} = \emptyset$$

since sets are pairwise disjoint. By additivity of μ_0

$$\begin{aligned}\mu_0\left(\bigcup_{n=1}^{N+1} A_n\right) &= \mu_0\left(\bigcup_{n=1}^{N} A_n \cup A_{N+1}\right) = \mu_0\left(\bigcup_{n=1}^{N} A_n\right) + \mu_0(A_{N+1}) \\ &= \sum_{n=1}^{N} \mu_0(A_n) + \mu_0(A_{N+1}) = \sum_{n=1}^{N+1} \mu_0(A_n).\end{aligned}$$

The first equality uses the decomposition of the union, the second finite additivity for the two disjoint measurable sets, and the third the inductive hypothesis. The final equality is the definition of the finite sum. This completes the induction.

HW 5 — Countable additivity implies additivity

Show that a countably additive set function is also additive.

Proof.

Let $\mu_0 : \Sigma_0 \to [0, +\infty]$ a countably additive set function. By definition of countable additivity, $\mu_0(\emptyset) = 0$. Let $A, B \in \Sigma_0$ be two disjoint sets, we define the sequence $(C_n)_{n\in\mathbb{N}}$ as follows: $C_1 = A$, $C_2 = B$, $C_n = \emptyset \ \forall n \geq 3$. Notice that $(C_n)_{n\in\mathbb{N}}$ are pairwise disjoint and $\bigcup_{n=1}^{\infty} C_n = A \cup B \in \Sigma_0$. Therefore by countable additivity

$$\mu_0(A \cup B) = \mu_0\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} \mu_0(C_n) = \mu_0(A) + \mu_0(B) + \sum_{n=3}^{\infty} \mu_0(\emptyset) = \mu_0(A) + \mu_0(B).$$

The first equality uses the union of the C_n, the second countable additivity, and the remaining equalities use their definitions and $\mu_0(\emptyset) = 0$.

Measure

Given a measurable space (S, Σ) a function $\mu : \Sigma \to [0, +\infty]$ is called measure if it is countably additive.

Measure space: The triple (S, Σ, μ) define a measure space.

A measure is finite if $\mu(S) < \infty$.

A measure is σ-finite if there exists $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$ such that $\bigcup_{n=1}^{\infty} A_n = S$ and $\mu(A_n) < \infty$ for each $n \in \mathbb{N}$.

Null set: a set $A \in \Sigma$ is a null set if $\mu(A) = 0$. Actually, to be formal the collection of null sets is $\mathcal{N} = \{A \subseteq S : A \subseteq B \in \Sigma,\ \mu(B) = 0\}$. Notice that in the definition of $\mathcal{N}$ the sets A are not necessarily measurable but they are subsets of measurable sets.

Complete measure space: a measure space is complete if $\mathcal{N} \subseteq \Sigma$.

A statement $H(s)$ about points $s \in S$ is said to hold almost everywhere (almost surely in case of probability measures) if its failure set is contained in a measurable set of measure zero. In particular, when the failure set is measurable, this means $\mu(F) = 0$, where $F = \{s \in S : H(s) \text{ is false}\}$.

Definition 1.3. π-system

A collection $\mathcal{I}$ of subsets of S such that $I_1, I_2 \in \mathcal{I} \Rightarrow I_1 \cap I_2 \in \mathcal{I}$ is called π-system.

HW 6 — Lower rays form a pi-system

Prove that $\pi(\mathbb{R})$ is a π-system. Recall the definition of $\pi(\mathbb{R}) = \{(-\infty, x],\ x \in \mathbb{R}\}$.

Proof.

By definition a collection $\mathcal{I}$ is a π-system if $A, B \in \mathcal{I} \Rightarrow A \cap B \in \mathcal{I}$. Let $A, B \in \pi(\mathbb{R})$ then there exists $x, y \in \mathbb{R}$ such that $A = (-\infty, x]$, $B = (-\infty, y]$.

$$A \cap B = (-\infty, \min\{x, y\}] \in \pi(\mathbb{R}).$$

Indeed, a real number belongs to both lower rays exactly when it is at most both endpoints, equivalently at most their minimum. The resulting ray is again an element of $\pi(\mathbb{R})$.

Lemma 1.2

Let $\Sigma = \sigma(\mathcal{I})$ where $\mathcal{I}$ is a π-system on S. Suppose that μ_1 and μ_2 are measures on (S, Σ) such that $\mu_1(S) = \mu_2(S) < \infty$ and $\mu_1(I) = \mu_2(I)$ for every $I \in \mathcal{I}$. Then $\mu_1(A) = \mu_2(A)\ \forall A \in \Sigma$.

Theorem 1.3 (Carathéodory's Theorem)

Let S be a set, Σ_0 an algebra on S and $\Sigma = \sigma(\Sigma_0)$. If a set function $\mu_0 : \Sigma_0 \to [0, +\infty]$ is countably additive, then there exists a measure $\mu : \Sigma \to [0, +\infty]$ such that $\mu(A) = \mu_0(A)$ for any $A \in \Sigma_0$. Moreover if $\mu_0(S) < \infty$ then μ is uniquely defined. We say that μ in the theorem above is an extension of μ_0 to the σ-algebra Σ.

Lebesgue measure

Let $S = (0, 1]$, $\Sigma = \mathcal{B}((0, 1])$ and

$$\Sigma_0 = \left\{ F : F = \bigcup_{i=1}^{n} (a_i, b_i],\ \text{for pairwise disjoint sets } (a_i, b_i] \subseteq (0, 1] \text{ and some } n \in \mathbb{N} \right\}.$$

Then $\sigma(\Sigma_0) = \Sigma$. Define μ_0 for $F \in \Sigma_0$ as $\mu_0(F) = \sum_{i=1}^{n}(b_i - a_i)$. μ_0 is countably additive on Σ_0 and $\mu_0(S) = 1 \Rightarrow \exists$ a measure μ on Σ that extends μ_0. We call it Lebesgue measure, denoted Leb. An analogous construction defines Lebesgue measure on $\mathbb{R}$ and on $\mathbb{R}^d$. The countable-additivity step is taken as given here; HW 8 asks only for finite additivity.

HW 7 — The interval algebra

Let $S = (0, 1]$ and let Σ_0 be the collection of finite unions of pairwise disjoint intervals $(a_i, b_i] \subseteq S$, with $0 \leq a_i \leq b_i \leq 1$. Prove that Σ_0 is an algebra on S.

Proof.

First,

$$\emptyset \in \Sigma_0, \qquad S = (0, 1] \in \Sigma_0,$$

since $(x, x] = \emptyset$ for every $x \in (0, 1]$, while S is itself an interval of the required form.

Let $A \in \Sigma_0$ be nonempty. Without loss of generality, we can assume that

$$A = \bigcup_{i=1}^{n} (a_i, b_i],$$

with

$$0 \leq a_1 < b_1 \leq a_2 < b_2 \leq \cdots \leq a_n < b_n \leq 1,$$

since the nonempty intervals are pairwise disjoint and can be ordered by their left endpoints. We obtain

$$A^c = (0, a_1] \cup \left(\bigcup_{i=1}^{n-1} (b_i, a_{i+1}] \right) \cup (b_n, 1].$$

These intervals are precisely the gaps between the intervals forming A, including possible gaps at the endpoints of S. Since they are finitely many and pairwise disjoint, $A^c \in \Sigma_0$. If $A = \emptyset$, then $A^c = S \in \Sigma_0$.

Finally, let $A, B \in \Sigma_0$, with

$$A = \bigcup_{i=1}^{n} (a_i, b_i], \qquad B = \bigcup_{j=1}^{m} (c_j, d_j].$$

The intervals from the two collections may overlap. To obtain a finite partition, collect all their endpoints a_i, b_i, c_j, d_j, together with 0 and 1, remove repetitions and arrange them in increasing order:

$$0 = t_0 < t_1 < \cdots < t_r = 1.$$

Define

$$D_\ell = (t_{\ell-1}, t_\ell], \qquad \ell = 1, \ldots, r.$$

These intervals are pairwise disjoint and satisfy

$$S = \bigcup_{\ell=1}^{r} D_\ell,$$

so they form a finite partition of S.

The same construction applies to any finite collection of sets in Σ_0. Thus, in subsequent arguments, we may work without loss of generality with a common finite partition of S into intervals of this form, with each original set represented as a union of its pieces.

Moreover, every original nonempty interval $(a_i, b_i]$ is a union of consecutive D_ℓ. Indeed, since all its endpoints were selected, there exist $p < q$ such that $a_i = t_p$ and $b_i = t_q$. Hence,

$$(a_i, b_i] = \bigcup_{\ell=p+1}^{q} D_\ell.$$

The equality holds because consecutive half-open intervals include their right endpoints and exclude their left endpoints, so no points are lost or counted twice.

The same argument applies to every interval forming B. Consequently, $A \cup B$ is a finite union of some of the D_ℓ, which are pairwise disjoint intervals of the required form. Therefore,

$$A \cup B \in \Sigma_0.$$

Thus, Σ_0 contains S and is closed under complements and finite unions, so it is an algebra on $(0, 1]$.

HW 8 — Finite additivity of interval length

On the interval algebra Σ_0 from HW 7, define $\mu_0(A) = \sum_{i=1}^{n}(b_i - a_i)$ for a disjoint representation $A = \bigcup_{i=1}^{n}(a_i, b_i]$. Prove that μ_0 is well defined and finitely additive.

Proof.

First, for every $A \in \Sigma_0$ represented by pairwise disjoint intervals,

$$A = \bigcup_{i=1}^{n}(a_i, b_i], \qquad \mu_0(A) = \sum_{i=1}^{n}(b_i - a_i) \geq 0.$$

The inequality follows because $b_i \geq a_i$ for every i. Moreover,

$$\mu_0(\emptyset) = \mu_0((x, x]) = x - x = 0.$$

We also verify that μ_0 does not depend on the chosen disjoint interval representation. Given two representations of A, their common endpoint partition, constructed as in HW 7, yields

$$\sum_{i=1}^{n}(b_i - a_i) = \sum_{\ell:\, D_\ell \subseteq A}(t_\ell - t_{\ell-1}) = \sum_{j=1}^{m}(d_j - c_j).$$

Both equalities follow by splitting the intervals of each representation into consecutive partition intervals and adding their lengths.

Finally, let $A, B \in \Sigma_0$ be disjoint, with

$$A = \bigcup_{i=1}^{n}(a_i, b_i], \qquad B = \bigcup_{j=1}^{m}(c_j, d_j].$$

Since $A \cap B = \emptyset$, every interval of A is disjoint from every interval of B. Define

$$e_k = \begin{cases} a_k, & k = 1, \ldots, n, \\ c_{k-n}, & k = n+1, \ldots, n+m, \end{cases} \qquad f_k = \begin{cases} b_k, & k = 1, \ldots, n, \\ d_{k-n}, & k = n+1, \ldots, n+m. \end{cases}$$

Then

$$A \cup B = \bigcup_{k=1}^{n+m}(e_k, f_k],$$

where all intervals are pairwise disjoint. Hence,

$$\mu_0(A \cup B) = \sum_{k=1}^{n+m}(f_k - e_k) = \sum_{i=1}^{n}(b_i - a_i) + \sum_{j=1}^{m}(d_j - c_j) = \mu_0(A) + \mu_0(B).$$

The first equality uses the disjoint representation just constructed; the second separates the first n and last m terms; the last follows from the definition of μ_0. Repeating the argument proves additivity for any finite family of pairwise disjoint sets.

HW 9 — Borel singletons

Prove that $\{x\} \in \mathcal{B}(\mathbb{R})$ $\forall x \in \mathbb{R}$.

Proof.

The Borel sigma algebra contains all open intervals and is closed under countable intersections. Let $x \in \mathbb{R}$, we have that $\{x\} = \bigcap_{n=1}^{\infty}(x - \frac{1}{n}, x + \frac{1}{n})$. We know that for all $n \in \mathbb{N}$, $(x - \frac{1}{n}, x + \frac{1}{n}) \in \mathcal{B}(\mathbb{R})$

and countable intersection of measurable sets is a measurable set. For the set equality, x belongs to every interval. Conversely, if $y \neq x$, choose n such that $1/n < |y - x|$; then y is outside the corresponding interval. Thus the intersection is exactly $\{x\}$.

HW 10 — Lebesgue measure of a singleton

Prove that $\text{Leb}(\{x\}) = 0 \ \forall x \in \mathbb{R}$.

Proof.

Let $x \in \mathbb{R}$. From HW 9 we know that $\{x\} \in \mathcal{B}(\mathbb{R})$. Notice that for every $n \in \mathbb{N}$, $\{x\} \subseteq (x - \frac{1}{n}, x]$ and $(x - \frac{1}{n}, x] = (x - \frac{1}{n}, x) \cup \{x\}$ where the two sets are disjoint and belong to $\mathcal{B}(\mathbb{R})$. By additivity of measure we have

$$\frac{1}{n} = \text{Leb}((x - \tfrac{1}{n}, x]) = \text{Leb}((x - \tfrac{1}{n}, x)) + \text{Leb}(\{x\}) \geq \text{Leb}(\{x\}),$$

where the last inequality follows from the non-negativity of measures. Therefore $0 \leq \text{Leb}(\{x\}) \leq \frac{1}{n}$. Since this inequality holds for every $n \in \mathbb{N}$ by letting $n \to +\infty$ and applying the squeeze thm we obtain $\text{Leb}(\{x\}) = 0$. Moreover if a singleton had measure $a > 0$, choosing $n > \frac{1}{a}$ would contradict $a \leq \frac{1}{n}$ hence its measure is zero.

Lemma 1.4

Let (S, Σ, μ) be a measure space. Then

(a) $\mu(A \cup B) \leq \mu(A) + \mu(B) \qquad \forall A, B \in \Sigma,$

(b) $\mu\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \mu(A_i) \qquad \forall (A_i)_{i=1}^{n} \subseteq \Sigma,$

(c) If $\mu(S) < \infty$ then $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) \quad \forall A, B \in \Sigma.$

HW 11 — Subadditivity and the union formula

Prove all three assertions of Lemma 1.4: binary subadditivity, finite subadditivity, and the union formula under $\mu(S) < \infty$.

Proof.

(a) Let $A, B \in \Sigma$, we have that $A \setminus B = A \cap B^c \in \Sigma$, $A \cup B = (A \setminus B) \cup B$ and $(A \setminus B) \cap B = \emptyset$. It follows that

$$\begin{aligned}\mu(A \cup B) &= \mu((A \setminus B) \cup B) = \mu(A \setminus B) + \mu(B) \leq \mu(A \setminus B) + \mu(A \cap B) + \mu(B) \\ &= \mu((A \setminus B) \cup (A \cap B)) + \mu(B) = \mu(A) + \mu(B).\end{aligned}$$

The first two equalities use the disjoint decomposition of $A \cup B$, the inequality uses $\mu(A \cap B) \geq 0$, and the last two equalities use the disjoint decomposition of A.

(b) Finite sub-additivity. We prove the statement by induction on n. **Base case:** $n = 1$. Statement is trivial $\mu(A_1) = \mu(A_1)$. **Inductive hypothesis:** $n \geq 1$, $\mu(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} \mu(A_i)$. **Inductive step:** $n + 1$. Notice that $\bigcup_{i=1}^{n+1} A_i = (\bigcup_{i=1}^{n} A_i) \cup A_{n+1}$. By prop (a) we have

$$\mu\left(\bigcup_{i=1}^{n+1} A_i\right) \leq \mu\left(\bigcup_{i=1}^{n} A_i\right) + \mu(A_{n+1}) \leq \sum_{i=1}^{n} \mu(A_i) + \mu(A_{n+1}) = \sum_{i=1}^{n+1} \mu(A_i),$$

where the first inequality uses part (a), the second the inductive hypothesis, and the equality the definition of the finite sum.

(c) $\mu(A \cup B) = \mu(A \setminus B) + \mu(B)$, since $A \setminus B$ and B are disjoint. $\mu(A) = \mu(A \setminus B) + \mu(A \cap B)$ since $A \setminus B$ and $A \cap B$ are disjoint. Adding $\mu(B)$ we obtain

$$\mu(A) + \mu(B) = \mu(A \setminus B) + \mu(A \cap B) + \mu(B) = \mu(A \cup B) + \mu(A \cap B),$$

where the first equality uses the disjoint decomposition of A, and the second uses the disjoint decomposition of $A \cup B$ from part (a). Since $\mu(S) < \infty$ for every $E \in \Sigma$, $\mu(E) < \infty$, in fact $\mu(S) = \mu(E) + \mu(E^c)$. Therefore subtracting $\mu(A \cap B)$ from from both sidess we obtain

$$\mu(A) + \mu(B) - \mu(A \cap B) = \mu(A \cup B).$$

HW 12 — Monotonicity of measures

Prove that if $A, B \in \Sigma$, $A \subseteq B$ then $\mu(A) \leq \mu(B)$.

Proof.

Notice that $B = (B \cap A) \cup (B \cap A^c)$ and that $(B \cap A) \cap (B \cap A^c) = \emptyset$. It follows

$$\mu(B) = \mu((B \cap A) \cup (B \cap A^c)) = \mu(B \cap A) + \mu(B \cap A^c) \geq \mu(B \cap A) = \mu(A),$$

because $A \subseteq B$. The second equality uses additivity on the two disjoint measurable pieces, the inequality their nonnegative measures, and the last equality $A \cap B = A$.

HW 13 — Measure of a difference

The operation of "subtracting" two sets is denoted by $A \setminus B$ and it means $A \cap B^c$. Show that for a finite measure μ, if $B \subseteq A$ then $\mu(A \setminus B) = \mu(A) - \mu(B)$.

Proof.

We obtain

$$\mu(A) = \mu((A \cap B) \cup (A \cap B^c)) = \mu(A \cap B) + \mu(A \cap B^c) = \mu(A \cap B) + \mu(A \setminus B).$$

Knowing that $A \cap B = B$ it follows $\mu(A) = \mu(B) + \mu(A \setminus B)$ and so $\mu(A \setminus B) = \mu(A) - \mu(B)$ if $\mu(B) < \infty$. The decomposition is disjoint and measurable. Finiteness of the measure ensures that the subtraction is between finite quantities; no $\infty - \infty$ expression is used.

Lemma 1.5. Continuity of measure from below

Let (S, Σ, μ) be a measure space. If $(A_n)_{n \in \mathbb{N}} \subseteq \Sigma$ such that $A_n \subseteq A_{n+1}$ and $A_n \uparrow A$ as $n \to +\infty$ (in the sense that $A = \bigcup_{n=1}^{\infty} A_n$). Then $\mu(A_n) \uparrow \mu(A)$ as $n \to +\infty$.

Proof.

We relabel sets to make them disjoint with: $B_1 = A_1$, $B_2 = A_2 \setminus B_1$, ..., $B_n = A_n \setminus \bigcup_{i=1}^{n-1} B_i$. Each B_n is measurable, being a difference of measurable sets. As we will see in HW 14, the sets $(B_n)_{n \in \mathbb{N}}$ are pairwise disjoint. As we will see in HW 15, $\bigcup_{i=1}^{n} B_i = A_n$ and $\bigcup_{i=1}^{\infty} B_i = A$. We obtain

$$\mu(A_n) = \mu\left(\bigcup_{i=1}^{n} B_i\right) = \sum_{i=1}^{n} \mu(B_i) \leq \sum_{i=1}^{n+1} \mu(B_i) = \mu\left(\bigcup_{i=1}^{n+1} B_i\right) = \mu(A_{n+1}).$$

Here the union identities are supplied by HW 15, the finite-sum equalities use additivity on the disjoint sets from HW 14, and the inequality uses $\mu(B_{n+1}) \geq 0$. Therefore the sequence $\mu(A_n)$, $n \in \mathbb{N}$ is non decreasing hence it admits a limit which we can now compute

$$\lim_{n \to +\infty} \mu(A_n) = \lim_{n \to +\infty} \sum_{i=1}^{n} \mu(B_i) = \sum_{i=1}^{\infty} \mu(B_i) = \mu\left(\bigcup_{i=1}^{\infty} B_i\right) = \mu(A),$$

The first equality substitutes the finite-sum identity, the second is the definition of a nonnegative series, the third countable additivity for the disjoint measurable B_i, and the last $\bigcup_i B_i = A$ from HW 15.

HW 14 — Disjoint pieces

Prove that $B_i \cap B_j = \emptyset \ \forall i \neq j$ where $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{k=1}^{n-1} B_k$, $n \geq 2$.

Proof.

We prove by induction on n that $B_1, \ldots, B_n$ are pairwise disjoint. **Base case:** $n = 1$ is trivial. **Inductive hypothesis:** $n \geq 1$. $B_1, \ldots, B_n$ are pairwise disjoint i.e. $B_i \cap B_j = \emptyset \ \forall 1 \leq i < j \leq n$.

Inductive step: By definition $B_{n+1} = A_{n+1} \setminus \bigcup_{k=1}^{n} B_k$. Therefore $B_{n+1} \cap (\bigcup_{k=1}^{n} B_k) = \emptyset$. Since $B_i \subseteq \bigcup_{k=1}^{n} B_k \ \forall i = 1, \dots, n$ we have that $B_{n+1} \cap B_i = \emptyset \ \forall i = 1, \dots, n$. Moreover by inductive hyp $B_1, \dots, B_n$ are already pairwise disjoint hence $B_1, \dots, B_{n+1}$ are pairwise disjoint.

HW 15 — Unions of the disjoint pieces

Prove that $\bigcup_{i=1}^{n} B_i = A_n$ **(i)** and $\bigcup_{i=1}^{\infty} B_i = A$.

Proof.

Recall that $(A_n)_{n\in\mathbb{N}}$ is non decreasing sequence of sets such that $A_n \subseteq A_{n+1}$; $A = \bigcup_{i=1}^{\infty} A_n$. We define $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{j=1}^{n-1} B_j$, $n \geq 2$. We prove **(i)** by induction: $N = 1$ is trivial. **Inductive hypothesis:** assume that statement hold for $n \geq 1$, i.e. $\bigcup_{i=1}^{n} B_i = A_n$. **Inductive step:** $B_{n+1} = A_{n+1} \setminus \bigcup_{i=1}^{n} B_i$, by inductive hyp we obtain $B_{n+1} = A_{n+1} \setminus A_n$. Therefore

$$\bigcup_{i=1}^{n+1} B_i = \left(\bigcup_{i=1}^{n} B_i \right) \cup B_{n+1} = A_n \cup (A_{n+1} \setminus A_n) = A_{n+1},$$

where the last equality follows from $A_n \subseteq A_{n+1}$. Therefore by induction $\bigcup_{i=1}^{n} B_i = A_n \ \forall n \in \mathbb{N}$ **(i)**. For **(ii)** we know that $A_n \uparrow A$ and $A = \bigcup_{i=1}^{\infty} A_n$ by ass. Therefore

$$\bigcup_{i=1}^{\infty} B_i = \bigcup_{n=1}^{\infty} \left(\bigcup_{i=1}^{n} B_i \right) = \bigcup_{n=1}^{\infty} A_n = A,$$

where the first equality follows the fact that B_i belongs to all finite unions with $n \geq i$ and the second follow the first part of HW 15 **(i)**. The last follow the assumption $A_n \uparrow A$.

Lemma 1.6. Continuity of measure from above

Let (S, Σ, μ) be a measure space. If $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$ such that $A_n \supseteq A_{n+1}$ and $A_n \downarrow A$ as $n \to +\infty$ (in the sense that $A = \bigcap_{n=1}^{\infty} A_n$). Then $\mu(A_n) \downarrow \mu(A)$ as $n \to +\infty$ provided that $\mu(A_m) < \infty$ for some $m \in \mathbb{N}$.

HW 16 — Continuity from above

Statement. Prove Lemma 1.6. **Hint:** take $B_n = A_k \setminus A_{k+n}$ so that $B_n \uparrow A_k \setminus A$ and use HW 13.

Proof.

Let A_m such that $\mu(A_m) < \infty$ and define a sequence $B_k = A_m \setminus A_{m+k}$. Since $A_{n+1} \subseteq A_n \ \forall n \in \mathbb{N}$ we have $B_{k+1} \supseteq B_k \ \forall k \in \mathbb{N}$. Moreover we have that $B_k \uparrow B$ with $B = A_m \setminus A = A_m \setminus \bigcap_{n=1}^{\infty} A_n$. By Lemma 1.5 we have

$$\lim_{k\to+\infty} \mu(B_k) = \mu(B) = \mu(A_m \setminus A) = \mu(A_m) - \mu(A).$$

By definition we have

$$\lim_{k\to+\infty} \mu(B_k) = \lim_{k\to+\infty} \mu(A_m \setminus A_{m+k}) = \mu(A_m) - \lim_{k\to+\infty} \mu(A_{m+k}).$$

Combining these two equations we obtain $\mu(A_m) - \mu(A) = \mu(A_m) - \lim_{k\to+\infty} \mu(A_{m+k})$.

We need the condition $\mu(A_n) < \infty$, if we don't impose that we would have this counter-example: Take $S = [0, +\infty)$, $\Sigma = \mathcal{B}([0, +\infty))$ and $\mu = \text{Leb}$. Take $A_n = [n, +\infty)$ so that $A_n \downarrow \emptyset$, notice that $\mu(A_n) = \infty \ \forall n$ and so the lemma fails. All subtractions are legitimate because $A \subseteq A_{m+k} \subseteq A_m$ and $\mu(A_m) < \infty$, so the difference formula from HW 13 applies. Cancelling the common finite quantity gives $\lim_{k\to\infty} \mu(A_{m+k}) = \mu(A)$. Omitting the finitely many initial indices does not change the limit; monotonicity of A_n and HW 12 show that $\mu(A_n)$ decreases to this value.

Lemma 1.7

Countable union of null set is a null set.

Proof.

Let $(N_n)_{n\in\mathbb{N}} \subseteq \Sigma$ be such that $\mu(N_n) = 0$ for all n's. Set $\bar{N} = \bigcup_{n=1}^{\infty} N_n$ and $\bar{N}^k = \bigcup_{n=1}^{k} N_n$. Then by Lemma 1.5 and sub additivity of the measure

$$\mu(\bar{N}) = \lim_{k\to+\infty} \mu(\bar{N}^k) \leq \lim_{k\to+\infty} \sum_{n=1}^{k} \mu(N_n) = 0.$$

Here $\bar{N}^k \uparrow \bar{N}$, so the first equality uses continuity from below. The inequality uses finite subadditivity at each k, and the last equality uses that every summand is zero. Nonnegativity then gives $\mu(\bar{N}) = 0$.

For null sets N_n that are not assumed measurable, choose $B_n \in \Sigma$ such that $N_n \subseteq B_n$ and $\mu(B_n) = 0$, as allowed by the definition of a null set. We obtain

$$\bigcup_{n=1}^{\infty} N_n \subseteq \bigcup_{n=1}^{\infty} B_n, \qquad \mu\left(\bigcup_{n=1}^{\infty} B_n\right) = 0.$$

The inclusion follows from $N_n \subseteq B_n$ for every n, and the equality follows from the measurable case just proved. Thus $\bigcup_n N_n$ is a null set in the general sense, whether or not it is measurable; no measure is assigned to a non-measurable set.

HW 17 — Rational points form a null set

Take the measure space $([0,1], \mathcal{B}([0,1]), \text{Leb})$. Prove that $[0,1]\cap\mathbb{Q} \in \mathcal{B}([0,1])$ and $\text{Leb}([0,1]\cap\mathbb{Q}) = 0$.

Proof.

Recall that $\mathbb{Q}$ is countable and any singleton $\in \mathcal{B}(\mathbb{R})$. We can write $[0,1]\cap\mathbb{Q} = \{q_1, q_2, \ldots\} = \bigcup_{n=1}^{\infty}\{q_n\}$ where $q_n \in [0,1]\cap\mathbb{Q}$. Since $\{q_n\} \in \mathcal{B}([0,1])$ and $\mathcal{B}([0,1])$ is closed under countable unions then $[0,1]\cap\mathbb{Q} = \bigcup_{n=1}^{\infty}\{q_n\} \in \mathcal{B}([0,1])$. By HW 10 we know that $\text{Leb}(\{q_n\}) = 0\ \forall n \in \mathbb{N}$, moreover since q_n's are distinct are pairwise disjoint. Therefore by countable additivity

$$\text{Leb}([0,1]\cap\mathbb{Q}) = \text{Leb}\left(\bigcup_{n=1}^{\infty}\{q_n\}\right) = \sum_{n=1}^{\infty} \text{Leb}(\{q_n\}) = 0.$$

2. Events

Probability spaces

A mathematical model for an experiment with uncertain (random) outcome is given by probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where

Ω is the sample space (the set S from the previous section). A sample point is a point in Ω, $\omega \in \Omega$ and it represents (loosely speaking) a possible scenario in the experiment.

$\mathcal{F}$ is the collection of all events (the σ-algebra Σ from the previous section). An event $E \in \mathcal{F}$ is a collection of sample points, i.e. is a collection of all possible scenarios that may lead to the observed outcome in the experiment.

$\mathbb{P}$ is the probability measure that assigns to each event a probability of occurring. The probability $\mathbb{P}$ is a measure on $\mathcal{F}$ with $\mathbb{P}(\Omega) = 1$. (Notice that $\mathbb{P} : \mathcal{F} \to [0, 1]$.)

Example 1 (Simple coin tossing models)

(a) Toss a coin twice: $\Omega = \{HH, HT, TH, TT\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$, the collection of all unions of singleton outcomes, closed under intersections and complements.

(b) Toss a coin infinitely many times: $\Omega = \{H, T\}^{\mathbb{N}} \Rightarrow \omega \in \Omega$ is a sequence $\omega = (\omega_1, \omega_2, \omega_3, \ldots)$ with $\omega_i = H$ or T. In principle one could consider $\mathcal{F} = \mathcal{P}(\Omega)$ where $\mathcal{P}(\Omega)$ is the collection of ALL subsets of Ω and it is called the power set. It turns out that this is not necessary and we can take the easier σ-algebra

$$\mathcal{F} = \sigma(\{\omega \in \Omega : \omega_n = W\},\ n \in \mathbb{N},\ W \in \{H, T\}).$$

To convince ourselves that this is a good choice we will work towards showing that the set

$$F := \left\{\omega \in \Omega : \lim_{n \to +\infty} \frac{\#(k \leq n : \omega_k = H)}{n} = \frac{1}{2}\right\}$$

belongs to $\mathcal{F}$ and, under the independent fair-coin probability model, $\mathbb{P}(F) = 1$. This is telling us that indeed the Law of Large Numbers (LLN) holds as it should in any sensible mathematical model for random experiments!

Now we prove a result that is in some sense a converse of Lemma 1.7. It tells us that countable intersection of sure events give us again a sure event. We need a small result first.

HW 18 — Sure events and null complements

For any $A \in \mathcal{F}$ show that $\mathbb{P}(A) = 1 \iff \mathbb{P}(A^c) = 0$.

Proof.

Let $A \in \mathcal{F}$, $A^c \in \mathcal{F}$. Notice that $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$. We obtain $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$, where the first equality is normalization and the second additivity on the two disjoint complementary events. **($\Rightarrow$)** If we assume that $\mathbb{P}(A) = 1$, then $\mathbb{P}(A^c) = 0$.

($\Leftarrow$) If we assume that $\mathbb{P}(A^c) = 0$ then $\mathbb{P}(A)$ has to be 1.

Proposition 2.1

If $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$ are such that $\mathbb{P}(A_n) = 1$ for all $n \in \mathbb{N}$, then $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = 1$.

Proof.

Since $A_n \in \mathcal{F} \Rightarrow A_n^c \in \mathcal{F}$ then $(\bigcap_{n=1}^{\infty} A_n)^c = \bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F}$ and

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = 1 \iff \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n^c\right) = 0.$$

Since $\mathbb{P}(A_n) = 1 \iff \mathbb{P}(A_n^c) = 0$ then $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n^c) = 0$ by Lemma 1.7 which concludes the proof.

Def lim inf and lim sup

Given a sequence $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$ we define

$$\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m$$

(i.e. the largest minorant in the limit),

$$\limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m$$

(i.e. the smallest majorant in the limit). Notice that the sequence $y_n := \sup_{m\geq n} x_m$ is decreasing as $y_n = \sup_{m\geq n} x_m \geq \sup_{m\geq n+1} x_m = y_{n+1}$. Then $\inf_{n\geq 1} y_n = \lim_{n\to+\infty} y_n$ and

$$\limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m = \lim_{n\to+\infty} \sup_{m\geq n} x_m.$$

HW 19 — Tail infima

Show that $\liminf_{n\to+\infty} x_n = \lim_{n\to+\infty} \inf_{m\geq n} x_m$.

Proof.

Let $y_n = \inf_{m\geq n} x_m$. Notice that $\{x_m : m \geq n+1\} \subseteq \{x_m : m \geq n\}$. Therefore $\inf_{m\geq n} x_m \leq \inf_{m\geq n+1} x_m$ which implies $y_n \leq y_{n+1}$. Hence $(y_n)_{n\in\mathbb{N}}$ is a non-decreasing sequence, by monotonicity the limit existss in the extended real line and $\lim_{n\to+\infty} y_n = \sup_{n\geq 1} y_n$. Substituting the definition of y_n we conclude the proof.

Definition of convergence through liminf and limsup

Given $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}$, the values of $\limsup x_n$ and $\liminf x_n$ are always well-defined (not necessarily finite) whereas the value of $\lim x_n$ may not exist. We say that a limit exist if

$$\liminf_{n\to+\infty} x_n = \limsup_{n\to+\infty} x_n =: \lim_{n\to+\infty} x_n.$$

The common value may belong to $\overline{\mathbb{R}} = [-\infty, +\infty]$. Convergence to a finite real number additionally requires that this common value is finite.

HW 20 — An oscillating sequence

Consider the sequence $x_n = (-1)^n$, $n \in \mathbb{N}$, calculate its liminf and limsup. Does the limit exists?

Proof.

Notice that $x_n = \begin{cases} -1 & n \text{ odd}, \\ 1 & n \text{ even}. \end{cases}$ Therefore $\inf_{m\geq n} x_m = -1$ and $\sup_{m\geq n} x_m = 1$. Therefore $\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m = \sup_{n\geq 1}(-1) = -1$ and in the same way $\limsup_{n\to+\infty} x_n = 1$. Therefore they don't coincide and the limit does not exist.

HW 21 — Threshold properties of limits

Prove the following properties of liminf and limsup.

(i) If $z > \limsup_{n\to+\infty} x_n$, then there exists $N \in \mathbb{N}$ such that $x_n < z \ \forall n \geq N$.

(ii) If $z < \liminf_{n\to+\infty} x_n$ then there exists $N \in \mathbb{N}$ such that $x_n > z \ \forall n \geq N$.

(iii) If $z < \limsup_{n\to+\infty} x_n$ then there are infinitely many $n \in \mathbb{N}$ such that $x_n > z$.

(iv) If $z > \liminf_{n\to+\infty} x_n$ then there are infinitely many $n \in \mathbb{N}$ such that $x_n < z$.

Proof.

(i) Assume that $z > \limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m := \inf_{n\geq 1} y_n$, where $y_n := \sup_{m\geq n} x_m$. By definition of infimum, if $\inf_{n\geq 1} y_n \in \mathbb{R}$ we have $\forall \varepsilon > 0 \ \exists N = N(\varepsilon) \in \mathbb{N}$ such that $y_N < \inf_{n\geq 1} y_n + \varepsilon$. Combining these two inequalities it exist $\varepsilon > 0$ sufficiently small so that $y_N \leq \inf_{n\geq 1} y_n + \varepsilon < z$. Restoring the definition of y_N we obtain

$$x_m \leq \sup_{j\geq N} x_j = y_N < z \quad \forall m \geq N \quad \Rightarrow \quad x_m < z \quad \forall m \geq N,$$

where the first inequality follows from the definition of supremum.

(ii) Assume that $z < \liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m := \sup_{n\geq 1} y_n$, where $y_n := \inf_{m\geq n} x_m$. By definition of supremum if $\sup_{n\geq 1} y_n \in \mathbb{R}$ we have $\forall \varepsilon > 0\ \exists N = N(\varepsilon) \in \mathbb{N}$ such that $y_N > \sup_{n\geq 1} y_n - \varepsilon$. Since $z < \sup_{n\geq 1} y_n$ we can choose $\varepsilon > 0$ sufficiently small such that $z < \sup_{n\geq 1} y_n - \varepsilon$. Combining these two inequalities we obtain $z \leq \sup_{n\geq 1} y_n - \varepsilon < y_N$. Restoring the definition of y_N we obtain

$$x_m \geq \inf_{j\geq N} x_j = y_N > z \quad \forall m \geq N.$$

Hence $x_m > z\ \forall m \geq N$.

For the tail infima in **(ii)**, if $\sup_{n\geq 1} y_n = +\infty$ its definition directly guarantees an N such that $y_N > z$ and the conclusion follows in the same way.

For the tail suprema $y_n = \sup_{m\geq n} x_m$ in **(i)**, if $\inf_{n\geq 1} y_n = -\infty$ then by definition of infimum $\forall z \in \mathbb{R}$ there exists $N \in \mathbb{N}$ such that $y_N < z$ and restoring def of y_N we obtain $x_m < z\ \forall m \geq N$.

(iii) Assume that $z < \limsup_{n\to+\infty} x_n = \inf_{n\geq 1} \sup_{m\geq n} x_m = \inf_{n\geq 1} y_n$. By the definition of infimum we have $z < \inf_{n\geq 1} y_n \leq y_n\ \forall n \in \mathbb{N}$. Restoring the definition of y_n we obtain $z < \sup_{m\geq n} x_m\ \forall n \in \mathbb{N}$. Taking $n = 1$ by the definition of supremum there exists $n_1 \in \mathbb{N}$ such that $x_{n_1} > z$. Now taking $n = n_1 + 1$ we have $z < \sup_{m\geq n_1+1} x_m$. Hence by definition of supremum there exists $n_2 > n_1$ such that $x_{n_2} > z$. Repeating this argument we obtain a strictly increasing sequence of indices $n_1 < n_2 < n_3 < \cdots$ such that $x_{n_j} > z\ \forall j \in \mathbb{N}$. Therefore $x_n > z$ for infinitely many $n \in \mathbb{N}$.

(iv) Assume that $z > \liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m := \sup_{n\geq 1} y_n$. By the definition of supremum we have $z > \sup_{n\geq 1} y_n \geq y_n\ \forall n \in \mathbb{N}$. Restoring the definition of y_n we obtain $z > \inf_{m\geq n} x_m\ \forall n \in \mathbb{N}$. Taking $n = 1$ by the definition of infimum there exists $n_1 \in \mathbb{N}$ such that $x_{n_1} < z$. Now taking $n = n_1 + 1$ we have $z > \inf_{m\geq n_1+1} x_m$. Hence by the definition of infimum there exists $n_2 > n_1$ such that $x_{n_2} < z$. Repeating this argument we obtain a strictly increasing sequence of indices $n_1 < n_2 < n_3 < \cdots$ such that $x_{n_j} < z\ \forall j \in \mathbb{N}$. Therefore $x_n < z$ for infinitely many $n \in \mathbb{N}$.

HW 22 — Negation, suprema and infima

Let $(x_\alpha)_{\alpha\in\mathcal{I}}$ be a family of real numbers ($\mathcal{I} \neq \emptyset$ is not necessarily countable, e.g., $\mathcal{I} = [0, 1]$). Prove that $\sup_{\alpha\in\mathcal{I}} x_\alpha = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha)$ **(i)**. Deduce that $\liminf_{n\to+\infty} x_n = -\limsup_{n\to+\infty}(-x_n)$ **(ii)**.

Proof.

(i) Let $M = \inf_{\alpha\in\mathcal{I}}(-x_\alpha)$. We distinguish two cases.

$M \in \mathbb{R}$. By definition of infimum we have $M \leq -x_\alpha\ \forall \alpha \in \mathcal{I}$ and for every $\varepsilon > 0$ there exists $\beta \in \mathcal{I}$ such that $-x_\beta \leq M + \varepsilon$. Multiplying both inequalities by (-1) we obtain $-M \geq x_\alpha\ \forall \alpha \in \mathcal{I}$ and $x_\beta \geq -M - \varepsilon$. Therefore $-M$ is an upper bound of $(x_\alpha)_{\alpha\in\mathcal{I}}$ and for every $\varepsilon > 0$ there exists $\beta \in \mathcal{I}$ such that $-M - \varepsilon \leq x_\beta$. Hence by definition of supremum $\sup_{\alpha\in\mathcal{I}} x_\alpha = -M = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha)$.

$M = -\infty$. By definition of infimum, for every $K \in \mathbb{R}$ there exists $\beta \in \mathcal{I}$ such that $-x_\beta < -K$. Multiplying by (-1) we obtain $x_\beta > K$. Therefore $(x_\alpha)_{\alpha\in\mathcal{I}}$ is not bounded above and hence $\sup_{\alpha\in\mathcal{I}} x_\alpha = +\infty$. Since $M = -\infty$ we conclude that $\sup_{\alpha\in\mathcal{I}} x_\alpha = +\infty = -M = -\inf_{\alpha\in\mathcal{I}}(-x_\alpha)$.

$M = +\infty$ is impossible since $\mathcal{I} \neq \emptyset$ and every $x_\alpha \in \mathbb{R}$.

(ii) By definition $\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m$. Notice that

$$\liminf_{n\to+\infty} x_n = \sup_{n\geq 1} \inf_{m\geq n} x_m = \sup_{n\geq 1} \left(-\sup_{m\geq n} (-x_m) \right) = -\inf_{n\geq 1} \sup_{m\geq n} (-x_m) = -\limsup_{n\to+\infty} (-x_n),$$

where the second equality follows by **(i)** taking $\inf_{m\geq n} x_m = -\sup_{m\geq n}(-x_m)$ and the third equality follows by **(i)** taking $\sup_{n\geq 1}(-a_n) = -\inf_{n\geq 1} a_n$ with $a_n := \sup_{m\geq n}(-x_m)$. The same reversal of order exchanges arbitrary upper bounds with lower bounds even when the bounds are infinite. Thus both applications of the sup/inf identity remain valid for the extended-real tail values.

Def limsup of collection of sets

Let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ and define

$$\begin{aligned}\limsup_{n\to+\infty} E_n &:= \{\omega \in \Omega : \omega \in E_n \text{ for infinitely many } n\text{'s}\} \\ &= \{\omega \in \Omega : \text{for any } n \in \mathbb{N} \text{ there is } m \geq n \text{ such that } \omega \in E_m\} = \bigcap_{n\geq 1} \bigcup_{m\geq n} E_m.\end{aligned}$$

For fixed n, $\bigcup_{m\geq n} E_m$ contains exactly those ω that belong to at least one E_m with $m \geq n$. Intersecting over all n requires such occurrences arbitrarily far out in the sequence. Notice also that $\limsup_{n\to+\infty} E_n \in \mathcal{F}$ because it is countable intersection of countable unions of sets in $\mathcal{F}$.

HW 23 — Measurability of the limsup event

Prove that $\limsup_{n\to+\infty} E_n \in \mathcal{F}$.

Proof.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. By definition $\limsup_{n\to+\infty} E_n = \bigcap_{n\geq 1} \bigcup_{m\geq n} E_m$. Since $\mathcal{F}$ is a σ-algebra is closed under countable unions therefore $\bigcup_{m\geq n} E_m \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Moreover by HW 3, a σ-algebra is also closed under countable intersections hence $\bigcap_{n\geq 1}(\bigcup_{m\geq n} E_m) \in \mathcal{F}$.

Lemma 2.2 (Reverse Fatou's lemma for sets)

Let $(\Omega, \mathcal{F}, \mathbb{P})$. Given $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ we have

$$\mathbb{P}\left(\limsup_{n\to+\infty} E_n\right) \geq \limsup_{n\to+\infty} \mathbb{P}(E_n).$$

Proof.

Define $G_m := \bigcup_{n\geq m} E_n$. Notice that $G_{m+1} \subseteq G_m$ $\forall m \in \mathbb{N}$. Indeed by definition $G_m = \bigcup_{n\geq m} E_n = E_m \cup (\bigcup_{n\geq m+1} E_n) = E_m \cup G_{m+1} \supseteq G_{m+1}$. Therefore $(G_m)_{m\in\mathbb{N}}$ is a non-increasing sequence of sets and $G_m \downarrow G$ where $G = \bigcap_{m=1}^{\infty} G_m = \bigcap_{m=1}^{\infty} \bigcup_{n\geq m} E_n = \limsup_{n\to+\infty} E_n$. Since $\mathbb{P}$ is a probability measure we have $\mathbb{P}(G_m) \leq 1 < \infty$ $\forall m \in \mathbb{N}$. Hence by Lemma 1.6 (continuity from above) $\mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m)$. Moreover by definition of G_m we have $E_n \subseteq G_m$ $\forall n \geq m$. By the monotonicity of probability measures (HW 12), $\mathbb{P}(E_n) \leq \mathbb{P}(G_m)$ $\forall n \geq m$. Taking the limit as $m \to +\infty$ we have $\lim_{m\to+\infty} \sup_{n\geq m} \mathbb{P}(E_n) \leq \lim_{m\to+\infty} \mathbb{P}(G_m)$.

Recalling the definition of limsup for numerical sequences and using Lemma 1.6 we conclude that

$$\mathbb{P}\left(\limsup_{n\to+\infty} E_n\right) = \mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m) \geq \lim_{m\to+\infty} \sup_{n\geq m} \mathbb{P}(E_n) = \limsup_{n\to+\infty} \mathbb{P}(E_n).$$

Hence $\mathbb{P}(\limsup_{n\to+\infty} E_n) \geq \limsup_{n\to+\infty} \mathbb{P}(E_n)$.

Lemma 2.3 (First Borel–Cantelli's lemma)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be such that $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$. Then $\mathbb{P}(\limsup_{n\to+\infty} E_n) = 0$ (i.e. almost every ω belongs to only finitely many E_n's).

Proof.

Define $G_m := \bigcup_{n\geq m} E_n$. Notice that $G_{m+1} \subseteq G_m$ $\forall m \in \mathbb{N}$. Indeed by definition $G_m = E_m \cup G_{m+1}$. Therefore (G_m) is a non-increasing sequence of sets and $G_m \downarrow G$ where $G = \bigcap_{m=1}^{\infty} G_m = \limsup_{n\to+\infty} E_n$. Since $\mathbb{P}(G_m) \leq 1 < \infty$ by Lemma 1.6 (continuity from above) we obtain $\mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m)$. Now for each fixed $m \in \mathbb{N}$ we define $G_m^k := \bigcup_{n=m}^{m+k} E_n$. Notice that $G_m^k \subseteq G_m^{k+1}$ since $G_m^{k+1} = G_m^k \cup E_{m+k+1}$. Therefore $(G_m^k)_{k\in\mathbb{N}}$ is a non-decreasing and $G_m^k \uparrow G_m = \bigcup_{n\geq m} E_n$. Hence by Lemma 1.5 (continuity from below) $\mathbb{P}(G_m) = \lim_{k\to+\infty} \mathbb{P}(G_m^k)$. Moreover by finite sub-

additivity of probability measures (Lemma 1.4) $\mathbb{P}(G_m^k) = \mathbb{P}(\bigcup_{n=m}^{m+k} E_n) \leq \sum_{n=m}^{m+k} \mathbb{P}(E_n)$. Taking the limit as $k \to +\infty$ we obtain

$$\mathbb{P}(G_m) = \lim_{k\to+\infty} \mathbb{P}(G_m^k) \leq \lim_{k\to+\infty} \sum_{n=m}^{m+k} \mathbb{P}(E_n) = \sum_{n=m}^{\infty} \mathbb{P}(E_n).$$

Combining this inequality with the previous result we have

$$\mathbb{P}\left(\limsup_{n\to+\infty} E_n\right) = \mathbb{P}(G) = \lim_{m\to+\infty} \mathbb{P}(G_m) \leq \lim_{m\to+\infty} \sum_{n=m}^{\infty} \mathbb{P}(E_n) = 0,$$

where the last equality follows from the ass. $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$. Finally, since probabilities are non-negative $0 \leq \mathbb{P}(\limsup_{n\to+\infty} E_n) \leq 0$. Therefore $\mathbb{P}(\limsup_{n\to+\infty} E_n) = 0$.

Def liminf of collection of sets

Let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ and define

$$\begin{aligned}\liminf_{n\to+\infty} E_n &:= \{\omega \in \Omega : \omega \in E_n \text{ for all but finitely many } n\text{'s}\} \\ &= \{\omega \in \Omega : \exists N_\omega \in \mathbb{N} \text{ such that } \omega \in E_n \ \forall n \geq N_\omega\} = \bigcup_{n\geq 1} \bigcap_{m\geq n} E_m.\end{aligned}$$

Notice that $\bigcap_{m\geq n} E_m$ contains those ω's that belong to all sets E_m for $m \geq n$. Taking the union allows to include all those ω's that belong to all but finitely many sets E_n. Notice that $\liminf_{n\to+\infty} E_n \in \mathcal{F}$ because it is countable union of countable intersections of sets in $\mathcal{F}$.

HW 24 — Measurability of the liminf event

Prove that $\liminf_{n\to+\infty} E_n \in \mathcal{F}$.

Proof.

Define $Y_n := \bigcap_{m\geq n} E_m = E_n \cap (\bigcap_{m\geq n+1} E_m) = E_n \cap Y_{n+1}$. Therefore $Y_n \subseteq Y_{n+1}$ $\forall n \in \mathbb{N}$, hence $(Y_n)_{n\in\mathbb{N}}$ is a non-decreasing sequence of sets. Moreover since $E_m \in \mathcal{F}$ $\forall m \in \mathbb{N}$ and $\mathcal{F}$ is a σ-algebra, by HW 3 it follows that $Y_n = \bigcap_{m\geq n} E_m \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Finally since $\mathcal{F}$ is closed under countable unions, we obtain

$$\liminf_{n\to+\infty} E_n = \bigcup_{n=1}^{\infty} Y_n = \bigcup_{n=1}^{\infty} \left(\bigcap_{m\geq n} E_m\right) \in \mathcal{F}.$$

HW 25 — Complement relation of event limits

Prove that $\liminf_{n\to+\infty} E_n = (\limsup_{n\to+\infty} E_n^c)^c$.

Proof.

We obtain

$$\begin{aligned}\left(\limsup_{n\to+\infty} E_n^c\right)^c &= \left(\bigcap_{n=1}^{\infty} \bigcup_{m\geq n} E_m^c\right)^c = \bigcup_{n=1}^{\infty} \left(\bigcup_{m\geq n} E_m^c\right)^c \\ &= \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} (E_m^c)^c = \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} E_m = \liminf_{n\to+\infty} E_n,\end{aligned}$$

where the second and third equalities follow De Morgan's laws, the fourth from double complement property and the last from the definition of liminf.

Lemma 2.4 (Fatou's lemma for sets)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Prove that

$$\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) \leq \liminf_{n\to+\infty} \mathbb{P}(E_n).$$

HW 26 — Fatou inequality by complements

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Prove Lemma 2.4, namely that

$$\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) \leq \liminf_{n\to+\infty} \mathbb{P}(E_n).$$

Hint: follow the steps in the proof of Lemma 2.2.

Proof.

We obtain

$$\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) = \mathbb{P}\left(\left(\limsup_{n\to+\infty} E_n^c\right)^c\right) = 1 - \mathbb{P}\left(\limsup_{n\to+\infty} E_n^c\right) \leq 1 - \limsup_{n\to+\infty} \mathbb{P}(E_n^c),$$

The first equality is HW 25, the second is the complement identity from HW 18, and the inequality is Lemma 2.2 applied to (E_n^c), after multiplying by -1 and adding 1. Hence

$$\begin{aligned}\mathbb{P}\left(\liminf_{n\to+\infty} E_n\right) &\leq 1 - \limsup_{n\to+\infty} \mathbb{P}(E_n^c) = 1 - \limsup_{n\to+\infty}(1 - \mathbb{P}(E_n)) \\ &= 1 - \left(1 + \limsup_{n\to+\infty}(-\mathbb{P}(E_n))\right) = -\limsup_{n\to+\infty}(-\mathbb{P}(E_n)) = \liminf_{n\to+\infty} \mathbb{P}(E_n),\end{aligned}$$

The first equality again uses HW 18; the next two use the constant-shift rule for limsup and cancel 1, and the last is HW 22.

Definition of indicator function

The indicator function of a set $E \in \mathcal{F}$ is defined as

$$\mathbf{1}_E(\omega) = \begin{cases} 1 & \text{if } \omega \in E, \\ 0 & \text{if } \omega \notin E. \end{cases}$$

For example, if we take $\Omega = [0,1]$ (i.e. ω's are numbers in $[0,1]$) and $\mathcal{F} = \mathcal{B}([0,1])$, then $\mathbf{1}_{[\frac{1}{2},1]}(\omega)$ is a function which is zero when $\omega \in [0, \frac{1}{2})$ and it is equal to one when $\omega \in [\frac{1}{2}, 1]$.

HW 27 — Indicators and limsup

(Ex. 2.9 from the book.) Given $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Prove that, for every $\omega \in \Omega$

$$\mathbf{1}_{\limsup_{n\to+\infty} E_n}(\omega) = \limsup_{n\to+\infty} \mathbf{1}_{E_n}(\omega).$$

Hint: follow the intermediate steps set out below. **(i)** prove that $\mathbf{1}_{E_i\cap E_j} = \mathbf{1}_{E_i} \cdot \mathbf{1}_{E_j} = \min\{\mathbf{1}_{E_i}, \mathbf{1}_{E_j}\}$; **(ii)** prove that $\mathbf{1}_{E_i\cup E_j} = \mathbf{1}_{E_i} + \mathbf{1}_{E_j} - \mathbf{1}_{E_i\cap E_j} = \max\{\mathbf{1}_{E_i}, \mathbf{1}_{E_j}\}$; **(iii)** use **(i)** and **(ii)** to conclude.

Proof.

(i) *We prove the intersection identity by its two possible values.*

$$\mathbf{1}_{A\cap B}(\omega) = \begin{cases} 0 & \text{if } \omega \notin A\cap B \iff \omega \notin A \text{ or } \omega \notin B \\ & \quad \iff \mathbf{1}_A(\omega)\mathbf{1}_B(\omega) = 0 = \min\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}, \\ 1 & \text{if } \omega \in A\cap B \iff \omega \in A \text{ and } \omega \in B \\ & \quad \iff \mathbf{1}_A(\omega)\mathbf{1}_B(\omega) = 1 = \min\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}. \end{cases}$$

(ii) *We prove the union identity by its two possible values.*

$$\mathbf{1}_{A\cup B}(\omega) = \begin{cases} 0 & \text{if } \omega \notin A \text{ and } \omega \notin B \\ & \quad \iff \mathbf{1}_A(\omega) + \mathbf{1}_B(\omega) - \mathbf{1}_{A\cap B}(\omega) = 0 = \max\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}, \\ 1 & \text{if } \omega \in A \text{ or } \omega \in B \\ & \quad \iff \mathbf{1}_A(\omega) + \mathbf{1}_B(\omega) - \mathbf{1}_{A\cap B}(\omega) = 1 = \max\{\mathbf{1}_A(\omega), \mathbf{1}_B(\omega)\}. \end{cases}$$

The same pointwise reasoning applies to countably many indicators. An intersection indicator equals one exactly when all the indicators equal one; a union indicator equals one exactly when at least one equals one. Since every value is either zero or one, infimum and supremum are attained. Thus

$$\mathbf{1}_{\bigcap_{n=1}^{\infty} A_n}(\omega) = \inf_{n\geq 1} \mathbf{1}_{A_n}(\omega) = \min_{n\geq 1} \mathbf{1}_{A_n}(\omega), \qquad \mathbf{1}_{\bigcup_{n=1}^{\infty} A_n}(\omega) = \sup_{n\geq 1} \mathbf{1}_{A_n}(\omega) = \max_{n\geq 1} \mathbf{1}_{A_n}(\omega).$$

For simplicity define $F_n := \bigcup_{m\geq n} E_m$ and let $\omega \in \Omega$, we have

$$\begin{aligned}\mathbf{1}_{\limsup_{n\to+\infty} E_n}(\omega) &= \mathbf{1}_{\bigcap_{n\geq 1} F_n}(\omega) = \inf_{n\geq 1} \mathbf{1}_{F_n}(\omega) \\ &= \inf_{n\geq 1} \mathbf{1}_{\bigcup_{m\geq n} E_m}(\omega) = \inf_{n\geq 1} \sup_{m\geq n} \mathbf{1}_{E_m}(\omega) = \limsup_{n\to+\infty} \mathbf{1}_{E_n}(\omega).\end{aligned}$$

In the chain, the first equality substitutes the limsup event, the second uses the intersection identity, the third substitutes F_n, the fourth uses the union identity, and the last is the numerical definition of limsup. Each step holds for the same arbitrary ω.

3. Random Variables

Random variables and pre-images

Random variables are measurable functions.

Def — pre-image of a set

Let (S, Σ) be a measurable space and let $h : S \to \mathbb{R}$ be a function. We define the pre-image of a set $A \subseteq \mathbb{R}$ as $h^{-1}(A) = \{s \in S : h(s) \in A\}$.

Def 3.1 (Measurable functions)

A function $h : S \to \mathbb{R}$ is Σ-measurable (also denoted $h \in \mathrm{m}\Sigma$) if $h^{-1}(A) \in \Sigma \ \forall A \in \mathcal{B}(\mathbb{R})$. Equivalently we say that $h \in \mathrm{m}\Sigma$ if $h^{-1} : \mathcal{B}(\mathbb{R}) \to \Sigma$.

Notation: we use $(\mathrm{m}\Sigma)^+$ for the class of non-negative Σ-measurable functions. We use $\mathrm{b}\Sigma$ for the class of bounded Σ-measurable functions. **Borel-measurable functions:** If S is a topological space and $\Sigma = \mathcal{B}(S)$, then $\mathrm{m}\Sigma$ is the class of Borel-measurable functions. For example if $S = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$ and $h : \mathbb{R} \to \mathbb{R}$ is such $h \in \mathrm{m}\Sigma$, then h is Borel-measurable.

Properties of pre-image mapping

Let $h : S \to \mathbb{R}$ and let $(A_\alpha)_{\alpha \in \mathcal{I}}$ be a family of subsets of $\mathbb{R}$ with $\mathcal{I}$ possibly uncountable (i.e. $\mathcal{I} = [0, 1]$). Then

$$\textbf{(i)} \quad h^{-1}\left(\bigcup_{\alpha \in \mathcal{I}} A_\alpha\right) = \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha),$$

$$\textbf{(ii)} \quad h^{-1}(A^c) = (h^{-1}(A))^c,$$

$$\textbf{(iii)} \quad h^{-1}\left(\bigcap_{\alpha \in \mathcal{I}} A_\alpha\right) = \bigcap_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha).$$

Proof (i).

We first prove that $h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha) \subseteq \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$. Let $s_0 \in h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha)$. By definition of pre-image we have $h(s_0) \in \bigcup_{\alpha \in \mathcal{I}} A_\alpha$. By definition of union there exists $\alpha_0 \in \mathcal{I}$ such that $h(s_0) \in A_{\alpha_0}$. Therefore $s_0 \in h^{-1}(A_{\alpha_0}) \subseteq \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$, where the membership $(\in)$ follows from the definition of pre-image and the inclusion follows the fact that $h^{-1}(A_{\alpha_0})$ is one of the sets in the union. Since s_0 was arbitrary it follows that $h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha) \subseteq \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$. For the reverse inclusion, let $s_0 \in \bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha)$. By definition of union, there exists $\alpha_0 \in \mathcal{I}$ such that $s_0 \in h^{-1}(A_{\alpha_0})$. By definition of pre-image we obtain $h(s_0) \in A_{\alpha_0}$. Therefore $h(s_0) \in A_{\alpha_0} \subseteq \bigcup_{\alpha \in \mathcal{I}} A_\alpha$
where the inclusion follows from the definition of union. Hence by definition of pre-image $s_0 \in h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha)$. Since s_0 is arbitrary we obtain $\bigcup_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha) \subseteq h^{-1}(\bigcup_{\alpha \in \mathcal{I}} A_\alpha)$. Combining the two inclusions we conclude the proof.

HW 28 — Pre-images of complements and intersections

For $h : S \to \mathbb{R}$, prove properties **(ii)** and **(iii)**:

$$h^{-1}(A^c) = (h^{-1}(A))^c, \qquad h^{-1}\left(\bigcap_{\alpha \in \mathcal{I}} A_\alpha\right) = \bigcap_{\alpha \in \mathcal{I}} h^{-1}(A_\alpha).$$

Complements on the left are taken in $\mathbb{R}$ and those on the right in S.

Proof.

(ii) Let $A \subseteq \mathbb{R}$, *we first prove that* $h^{-1}(A^c) \subseteq (h^{-1}(A))^c$. Let $s_0 \in h^{-1}(A^c)$. By definition of pre-image we have $h(s_0) \in A^c \Rightarrow h(s_0) \notin A$. Therefore $s_0 \notin h^{-1}(A) \Rightarrow s_0 \in (h^{-1}(A))^c$ where the first statement follows from the definition of pre-image and the second from the definition of complement. Since s_0 is arbitrary it follows that $h^{-1}(A^c) \subseteq (h^{-1}(A))^c$. For the reverse inclusion

let $s_0 \in (h^{-1}(A))^c \Rightarrow s_0 \notin h^{-1}(A)$. By the definition of pre-image this implies that $h(s_0) \notin A$. Therefore $h(s_0) \in A^c \Rightarrow s_0 \in h^{-1}(A^c)$ where the first membership follows from the definition of complement and the second from the definition of pre-image. Since s_0 was arbitrary we obtain $(h^{-1}(A))^c \subseteq h^{-1}(A^c)$. Combining the two inclusionss we conclude the proof.

(iii) Let $(A_\alpha)_{\alpha\in\mathcal{I}}$ be a family of subsets of $\mathbb{R}$ where $\mathcal{I}$ is possibly uncountable. *We first prove that* $h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha) \subseteq \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$. Let $s_0 \in h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha)$. By definition of pre-image we have $h(s_0) \in \bigcap_{\alpha\in\mathcal{I}} A_\alpha$. By definition of intersection this means that $h(s_0) \in A_\alpha \ \forall\alpha \in \mathcal{I}$. Therefore $s_0 \in h^{-1}(A_\alpha) \ \forall\alpha \in \mathcal{I}$ where the membership follows the def of pre-image. Hence $s_0 \in \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$ by def of intersection. Since s_0 is arbitrary it follows that $h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha) \subseteq \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$. For the reverse inclusion, let $s_0 \in \bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha)$. By definition of intersection we have $s_0 \in h^{-1}(A_\alpha) \ \forall\alpha \in \mathcal{I}$. By definition of pre-image it follows that $h(s_0) \in A_\alpha \ \forall\alpha \in \mathcal{I}$. Therefore by def of intersection $h(s_0) \in \bigcap_{\alpha\in\mathcal{I}} A_\alpha$ hence $s_0 \in h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha)$. Since s_0 is arbitrary we obtain $\bigcap_{\alpha\in\mathcal{I}} h^{-1}(A_\alpha) \subseteq h^{-1}(\bigcap_{\alpha\in\mathcal{I}} A_\alpha)$. Combining the two inclusions we conclude the proof.

Lemma 3.1 — The pre-image of Borel sets form a σ-algebra

Let $h : S \to \mathbb{R}$, we want to prove that $h^{-1}(\mathcal{B}(\mathbb{R})) := \{h^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}$ is a σ-algebra on S.

Proof.

By definition of σ-algebra we must verify the following three prop.

(i) $\emptyset \in h^{-1}(\mathcal{B}(\mathbb{R}))$. Notice that $\emptyset, \mathbb{R} \in \mathcal{B}(\mathbb{R})$ since $\mathcal{B}(\mathbb{R})$ is a σ-algebra on $\mathbb{R}$. Moreover by definition of pre-image $h^{-1}(\emptyset) = \emptyset$ and $h^{-1}(\mathbb{R}) = S$. Therefore $\emptyset, S \in h^{-1}(\mathcal{B}(\mathbb{R}))$.

(ii) Closure under complements. Let $H \in h^{-1}(\mathcal{B}(\mathbb{R}))$. By definition of this class, there exists $A \in \mathcal{B}(\mathbb{R})$ such that $H = h^{-1}(A)$. Since $\mathcal{B}(\mathbb{R})$ is a σ-algebra, it is closed under complements. Therefore $A^c \in \mathcal{B}(\mathbb{R})$. Now consider the complement of H:

$$H^c = (h^{-1}(A))^c = h^{-1}(A^c) \in h^{-1}(\mathcal{B}(\mathbb{R})),$$

where the first equality follows from the definition of H, the second from prop **(ii)** HW 28 and the last membership follows from $A^c \in \mathcal{B}(\mathbb{R})$. Therefore $H \in h^{-1}(\mathcal{B}(\mathbb{R})) \Rightarrow H^c \in h^{-1}(\mathcal{B}(\mathbb{R}))$.

(iii) Closure under countable unions. Let $(H_n)_{n\in\mathbb{N}} \subseteq h^{-1}(\mathcal{B}(\mathbb{R}))$. By definition of this class $\forall n \in \mathbb{N}$ $\exists A_n \in \mathcal{B}(\mathbb{R})$ such that $H_n = h^{-1}(A_n)$. Since $\mathcal{B}(\mathbb{R})$ is a σ-alg. it is closed under countable unions therefore $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}(\mathbb{R})$. Now consider the union of the sets H_n:

$$\bigcup_{n=1}^{\infty} H_n = \bigcup_{n=1}^{\infty} h^{-1}(A_n) = h^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) \in h^{-1}(\mathcal{B}(\mathbb{R})),$$

where the first equality follows from the def of H_n, the second from prop **(i)** of the pre-image mapping and the last membership follows from $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}(\mathbb{R})$. Therefore $(H_n)_{n\in\mathbb{N}} \subseteq h^{-1}(\mathcal{B}(\mathbb{R})) \Rightarrow \bigcup_{n=1}^{\infty} H_n \in h^{-1}(\mathcal{B}(\mathbb{R}))$. Since all three prop are satisfied $h^{-1}(\mathcal{B}(\mathbb{R}))$ is a σ-alg on S.

Rmk: It should be clear that h is measurable respect to $h^{-1}(\mathcal{B}(\mathbb{R}))$. Moreover for any $h : S \to \mathbb{R}$ we can construct a σ-alg on S with respect to which h is measurable. The smallest of such σ-algebras is exactly $h^{-1}(\mathcal{B}(\mathbb{R}))$.

Problem: According to Def 3.1 in order to check whether a function is Σ-measurable or not one need to verify the pre-images of ALL Borel sets are in Σ. Checking every Borel set directly is generally impractical; the next lemma provides a generating-class test.

Lemma 3.2. Measurability through a generating class

Let (S, Σ) be a measurable space, $h : S \to \mathbb{R}$ and let $\mathcal{C} \subseteq \mathcal{B}(\mathbb{R})$ be such that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$. If $h^{-1}(A) \in \Sigma \ \forall A \in \mathcal{C}$ then h is Σ-measurable i.e. $h^{-1}(A) \in \Sigma \ \forall A \in \mathcal{B}(\mathbb{R})$.

Proof.

Define $\mathcal{E} := \{A \in \mathcal{B}(\mathbb{R}) : h^{-1}(A) \in \Sigma\}$. By definition of $\mathcal{E}$ and by our assumption we have $\mathcal{C} \subseteq \mathcal{E} \subseteq \mathcal{B}(\mathbb{R})$. *We want to prove that* $\mathcal{E}$ *is a* σ*-algebra on* $\mathbb{R}$. By definition of σ-alg we must verify the three prop.

(i) $\emptyset$ and $\mathbb{R}$ belong to $\mathcal{E}$. Notice that by definition of pre-image $h^{-1}(\emptyset) = \emptyset$ and $h^{-1}(\mathbb{R}) = S$. Since Σ is a σ-algebra on S we have $\emptyset, S \in \Sigma$. Therefore $h^{-1}(\emptyset) \in \Sigma$ and $h^{-1}(\mathbb{R}) \in \Sigma$ hence by definition of $\mathcal{E}$, $\emptyset, \mathbb{R} \in \mathcal{E}$.

(ii) Closure under complements. Let $A \in \mathcal{E}$, by def of $\mathcal{E}$ we have $A \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(A) \in \Sigma$. Since both $\mathcal{B}(\mathbb{R})$ and Σ are σ-algebras they are closed under complements. Therefore $A^c \in \mathcal{B}(\mathbb{R})$ and $(h^{-1}(A))^c \in \Sigma$. Now by prop **(ii)** of HW 28 $h^{-1}(A^c) = (h^{-1}(A))^c \in \Sigma$ where the equality follows from the property of pre-images of complements and the last membership follows from the closure of Σ under complements. Since $A^c \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(A^c) \in \Sigma$ by definition of $\mathcal{E}$, $A^c \in \mathcal{E}$. Hence $\mathcal{E}$ is closed under complements.

(iii) Closure under countable unions. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{E}$, by definition of $\mathcal{E}$, $A_n \in \mathcal{B}(\mathbb{R})$ and $h^{-1}(A_n) \in \Sigma\ \forall n \in \mathbb{N}$. Since $\mathcal{B}(\mathbb{R})$ is a σ-algebra, it is closed under countable unions. Therefore $\bigcup_{n=1}^{\infty} A_n \in \mathcal{B}(\mathbb{R})$. Moreover by property **(i)** of the pre-image mapping

$$h^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} h^{-1}(A_n) \in \Sigma$$

where the equality follows from property **(i)** of pre-images and the last membership follows from the closure of Σ under countable unions, since $h^{-1}(A_n) \in \Sigma$ for every $n \in \mathbb{N}$. Hence by definition of $\mathcal{E}$, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{E}$, thus $\mathcal{E}$ is closed under countable unions. Since all three prop are satisfied $\mathcal{E}$ is a σ-alg on $\mathbb{R}$. Moreover since $\mathcal{C} \subseteq \mathcal{E}$ and $\mathcal{E}$ is a σ-algebra, by the definition of the generated σ-algebra $\sigma(\mathcal{C}) \subseteq \mathcal{E}$. Therefore $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subseteq \mathcal{E} \subseteq \mathcal{B}(\mathbb{R})$ hence $\mathcal{E} = \mathcal{B}(\mathbb{R})$. By def of $\mathcal{E}$ this implies $h^{-1}(A) \in \Sigma\ \forall A \in \mathcal{B}(\mathbb{R})$ therefore $h \in \mathrm{m}\Sigma$.

Remarks

There are some important consequences of the above lemma:

Proof.

(i) If S is a topological space and $h : S \to \mathbb{R}$ is continuous then h is Borel measurable (i.e. $h^{-1}(A) \in \mathcal{B}(S)\ \forall A \in \mathcal{B}(\mathbb{R})$). Indeed, $h^{-1}(A)$ is an open set in S whenever A is an open set in $\mathbb{R}$, due to continuity. Take the generating class of open sets in $\mathbb{R}$. Their pre-images are open in S, hence in $\mathcal{B}(S)$, and they generate $\mathcal{B}(\mathbb{R})$. Lemma 3.2 therefore applies with domain sigma-algebra $\mathcal{B}(S)$.

(ii) Given a measurable space (S, Σ) a function $h : S \to \mathbb{R}$ is Σ-measurable if and only if $\{h \leq c\} := \{s \in S : h(s) \in (-\infty, c]\} = h^{-1}((-\infty, c]) \in \Sigma$ for all $c \in \mathbb{R}$. Indeed, if $h \in \mathrm{m}\Sigma$, then $h^{-1}((-\infty, c]) \in \Sigma$ for all $c \in \mathbb{R}$ because $(-\infty, c] \in \mathcal{B}(\mathbb{R})$. If instead $h^{-1}((-\infty, c]) \in \Sigma$ for all $c \in \mathbb{R}$, then we apply the lemma upon recalling the π-system $\pi(\mathbb{R})$ and the fact that $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$. This is a clever way to test measurability.

HW 29 — A Borel indicator

Let $S = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$ and $h(s) = \mathbf{1}_{[0,1]}(s)$. Prove that h is Borel-measurable. Hint: first you need to show that $[0, 1] \in \mathcal{B}(\mathbb{R})$.

Proof.

First, $[0, 1] = \mathbb{R} \setminus ((-\infty, 0) \cup (1, +\infty)) \in \mathcal{B}(\mathbb{R})$, since the set being complemented is open. Recall that by consequence of **(ii)** of Lemma 3.2 a function $h : \mathbb{R} \to \mathbb{R}$ is Borel measurable if and only if $h^{-1}((-\infty, c]) \in \mathcal{B}(\mathbb{R})\ \forall c \in \mathbb{R}$. By definition of indicator function $h(s) = \begin{cases} 0 & s \notin [0, 1], \\ 1 & s \in [0, 1]. \end{cases}$ Therefore by definition of pre-image

$$h^{-1}((-\infty, c]) = \{s \in \mathbb{R} : h(s) \leq c\} = \begin{cases} \emptyset & c < 0, \\ (-\infty, 0) \cup (1, +\infty) & 0 \leq c < 1, \\ \mathbb{R} & c \geq 1. \end{cases}$$

Indeed if $c < 0$ neither 0 nor 1 satisfies $h(s) \leq c$ so the pre-image is empty. If $0 \leq c < 1$ only $h(s) = 0$ satisfies the inequality which occurs when $s \notin [0, 1]$. Finally if $c \geq 1$, both value satisfies the inequality so the pre-image is $\mathbb{R}$. Now since $\mathcal{B}(\mathbb{R})$ is a σ-algebra we have $\emptyset, \mathbb{R} \in \mathcal{B}(\mathbb{R})$. Moreover since $(-\infty, 0) \cup (1, +\infty)$ are open sets, their union is also open. Therefore by definition of σ-algebra $(-\infty, 0) \cup (1, +\infty) \in \mathcal{B}(\mathbb{R})$. Hence in all three cases $h^{-1}((-\infty, c]) \in \mathcal{B}(\mathbb{R})$ $\forall c \in \mathbb{R}$, therefore by consequence **(ii)** of Lemma 3.2 $h \in \mathrm{m}\mathcal{B}(\mathbb{R})$.

HW 30 — Equivalent threshold tests

Prove that $h : S \to \mathbb{R}$ is Σ-measurable if and only if each of the following holds:

$$\begin{aligned} &\textbf{(i)} \quad \{h < c\} := h^{-1}((-\infty, c)) \in \Sigma \quad \forall c \in \mathbb{R}, \\ &\textbf{(ii)} \quad \{h > c\} := h^{-1}((c, +\infty)) \in \Sigma \quad \forall c \in \mathbb{R}, \\ &\textbf{(iii)} \quad \{h \geq c\} := h^{-1}([c, +\infty)) \in \Sigma \quad \forall c \in \mathbb{R}. \end{aligned}$$

Proof.

(⇒) We assume that h is Σ-measurable then by Definition 3.1 $\forall A \in \mathcal{B}(\mathbb{R})$, $h^{-1}(A) \in \Sigma$. Since $(-\infty, c)$ and $(c, +\infty)$ are open sets they are Borel and $[c, +\infty) = (-\infty, c)^c$ hence is Borel so their pre-image belong to Σ.

(⇐)

(i) Assume that $\{h < c\} \in \Sigma$ $\forall c \in \mathbb{R}$. Notice that $(-\infty, c] = \bigcap_{n=1}^{\infty}(-\infty, c + \frac{1}{n})$. Therefore

$$\begin{aligned} \{h \leq c\} = h^{-1}((-\infty, c]) &= h^{-1}\left(\bigcap_{n=1}^{\infty}(-\infty, c + \tfrac{1}{n})\right) \\ &= \bigcap_{n=1}^{\infty} h^{-1}((-\infty, c + \tfrac{1}{n})) = \bigcap_{n=1}^{\infty}\{h < c + \tfrac{1}{n}\} \in \Sigma, \end{aligned}$$

where the first equality follows from the definition of pre-image, the second from the identity above, the third from property **(iii)** of HW 28 and the fourth from the definition of $\{h < c\}$. The last membership follows from the closure under countable intersections since $\{h < c + \frac{1}{n}\} \in \Sigma$ $\forall n \in \mathbb{N}$ by assumption. Hence $\{h \leq c\} \in \Sigma$ $\forall c \in \mathbb{R}$ and by Lemma 3.2 h is Σ-measurable.

(ii) Assume that $\{h > c\} \in \Sigma$ $\forall c \in \mathbb{R}$. Notice that $(-\infty, c] = (c, +\infty)^c$. Therefore

$$\{h \leq c\} = h^{-1}((-\infty, c]) = h^{-1}((c, +\infty)^c) = (h^{-1}((c, +\infty)))^c = \{h > c\}^c \in \Sigma,$$

where the first equality follows from the definition of pre-image, the second from the complement identity, the third from prop **(ii)** of HW 28, the fourth from the definition of $\{h > c\}$. The last membership follows closure of Σ under complements since $\{h > c\} \in \Sigma$ by assumption. Hence by Lemma 3.2 h is Σ-measurable.

(iii) Assume that $\{h \geq c\} \in \Sigma$ $\forall c \in \mathbb{R}$. Notice that $(-\infty, c) = [c, +\infty)^c$. Therefore

$$\{h < c\} = h^{-1}((-\infty, c)) = h^{-1}([c, +\infty)^c) = (h^{-1}([c, +\infty)))^c = \{h \geq c\}^c \in \Sigma,$$

where the first equality follows from the definition of pre-image, the second from the complement identity, the third from property **(ii)** of HW 28 and the fourth from the definition of $\{h \geq c\}$. The last membership follows from the closure of Σ under complements since $\{h \geq c\} \in \Sigma$ by assumption. Hence condition **(i)** holds and by the first part of the proof h is Σ-measurable. Therefore each of the three condition is equivalent to Σ-measurability.

Lemma 3.3. Composition of functions

Let $h : S \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}$ be such that $h \in \mathrm{m}\Sigma$ and $f \in \mathrm{m}\mathcal{B}$. Then $g := f \circ h : S \to \mathbb{R}$ (i.e. $g(s) = f(h(s))$, $s \in S$) is a Σ-measurable functions.

Proof.

For any $A \in \mathcal{B}(\mathbb{R})$ we have

$$g^{-1}(A) = \{s \in S : f(h(s)) \in A\} = \{s \in S : h(s) \in f^{-1}(A)\} = h^{-1}(f^{-1}(A)) \in \Sigma,$$

The first equality uses $g(s) = f(h(s))$, the second the definition of the pre-image under f, and the third that under h. Since $f^{-1}(A)$ is Borel by measurability of f, measurability of h gives the final membership.

Lemma 3.4

If $h, h_1, h_2 \in \mathrm{m}\Sigma$ then **(i)** $h_1 + h_2 \in \mathrm{m}\Sigma$, **(ii)** $h_1 \cdot h_2 \in \mathrm{m}\Sigma$, **(iii)** $\lambda h \in \mathrm{m}\Sigma$ for any $\lambda \in \mathbb{R}$.

Proof.

(i) Recall by HW 30 it is sufficient to show that $\{h_1 + h_2 > c\} \in \Sigma \ \forall c \in \mathbb{R}$. Notice that if $h_1(s) + h_2(s) > c$ then $c - h_2(s) < h_1(s)$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$ there exists $q \in \mathbb{Q}$ such that $h_1(s) > q > c - h_2(s)$. Therefore

$$\{h_1 + h_2 > c\} = \bigcup_{q\in\mathbb{Q}} \{s \in S : h_1(s) > q > c - h_2(s)\} = \bigcup_{q\in\mathbb{Q}} (\{h_1 > q\} \cap \{h_2 > c - q\}) \in \Sigma.$$

Since h_1, h_2 are Σ-measurable by HW 30 we have $\{h_1 > q\} \in \Sigma$ and $\{h_2 > c-q\} \in \Sigma$. The second equality rewrites $q > c - h_2(s)$ as $h_2(s) > c - q$. Finite-intersection closure and then countable-union closure of Σ, using countability of $\mathbb{Q}$, give the final membership. The rational-union identity in **(i)** also has the reverse implication: if $h_1(s) > q$ and $h_2(s) > c-q$, adding gives $h_1(s)+h_2(s) > c$. Thus no point is added by that union. Since $c \in \mathbb{R}$ was arbitrary, HW 30 gives $h_1 + h_2 \in \mathrm{m}\Sigma$.

(iii) By HW 30 it is sufficient to prove that $\{\lambda h > c\} \in \Sigma \ \forall c \in \mathbb{R}$. We distinguish three cases. Case 1. $\lambda = 0$. Notice that $\{\lambda h > c\} = \{s \in S : 0 > c\} = \begin{cases} \emptyset & c \geq 0, \\ S & c < 0. \end{cases}$ Since $\emptyset, S \in \Sigma$ it follows that $\{\lambda h > c\} \in \Sigma$. Case 2. $\lambda > 0$. Notice that $\{\lambda h > c\} = \{s \in S : \lambda h(s) > c\} = \{s \in S : h(s) > \frac{c}{\lambda}\} \in \Sigma$ since $h \in \mathrm{m}\Sigma$. Case 3. $\lambda < 0$. Notice that $\{\lambda h > c\} = \{s \in S : h(s) < \frac{c}{\lambda}\} \in \Sigma$ by HW 30. Hence in all three cases $\{\lambda h > c\} \in \Sigma$ therefore by HW 30 $\lambda h \in \mathrm{m}\Sigma$.

(ii) Let $h_1, h_2 \in \mathrm{m}\Sigma$. Notice that the function $f(x) = x^2$ is continuous and therefore is Borel-measurable by consequence **(i)** of Lemma 3.2. Hence by Lemma 3.3 $h_1^2, h_2^2 \in \mathrm{m}\Sigma$ and, applying the same composition lemma to the measurable sum from **(i)**, $(h_1 + h_2)^2 \in \mathrm{m}\Sigma$. Now notice that $h_1 h_2 = \frac{1}{2}(h_1 + h_2)^2 - \frac{1}{2}(h_1^2 + h_2^2)$, since sum, difference and scalar multiples are Σ-measurable also $h_1 h_2 \in \mathrm{m}\Sigma$.

Lemma 3.5. Measurability of infimum, supremum, liminf and limsup

Let (S, Σ) be a measurable space and let $(h_n)_{n\in\mathbb{N}} \subseteq \mathrm{m}\Sigma$. The infimum, supremum and limits below may take values in $\overline{\mathbb{R}} = [-\infty, +\infty]$. For a map $u : S \to \overline{\mathbb{R}}$, measurability means that $\{u \leq c\} \in \Sigma$ for every $c \in \mathbb{R}$; the equivalent threshold tests from HW 30 apply in this extended setting as well. Memberships in $\mathrm{m}\Sigma$ within this lemma are understood in this extended sense. Whenever one of these functions is finite at every point, it belongs to the real-valued class of Definition 3.1. Then **(i)** $\inf_{n\in\mathbb{N}} h_n \in \mathrm{m}\Sigma$, $\sup_{n\in\mathbb{N}} h_n \in \mathrm{m}\Sigma$; **(ii)** $\liminf_{n\to+\infty} h_n \in \mathrm{m}\Sigma$, $\limsup_{n\to+\infty} h_n \in \mathrm{m}\Sigma$.

Proof.

(i) Let $h(s) := \inf_{n\in\mathbb{N}} h_n(s)$. Recall that by HW 30 to prove that h is Σ-measurable it is sufficient to show that $\{h \geq c\} \in \Sigma \ \forall c \in \mathbb{R}$. As we will see in HW 31, $\{h \geq c\} = \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. Each set in the intersection is measurable by HW 30, so closure under countable intersections gives $\{h \geq c\} \in \Sigma$. By HW 22 $\sup_{n\in\mathbb{N}} h_n = -\inf_{n\in\mathbb{N}}(-h_n)$, since $h_n \in \mathrm{m}\Sigma$ by Lemma 3.4 **(iii)**, $-h_n \in \mathrm{m}\Sigma$ and by the first part of the proof $\inf_{n\in\mathbb{N}}(-h_n) \in \mathrm{m}\Sigma$. Negation also preserves extended-real measurability, since $\{-u > c\} = \{u < -c\}$ for every real threshold c. Consequently $-\inf_{n\in\mathbb{N}}(-h_n) = \sup_{n\in\mathbb{N}} h_n$ is measurable.

(ii) By definition of liminf $\liminf_{n\to+\infty} h_n(s) = \sup_{n\geq 1} \inf_{m\geq n} h_m(s)$. For every $n \in \mathbb{N}$ define $\ell_n(s) := \inf_{m\geq n} h_m(s)$. Notice that ℓ_n is the infimum of a countable family of Σ-measurable functions therefore by prop **(i)** $\ell_n \in \mathrm{m}\Sigma \ \forall n \in \mathbb{N}$. Therefore also $\sup_{n\geq 1} \ell_n(s) \in \mathrm{m}\Sigma$ thanks to **(i)** hence $\liminf h_n(s) \in \mathrm{m}\Sigma$. By HW 22 $\limsup_{n\to+\infty} h_n(s) = -\liminf_{n\to+\infty}(-h_n(s))$ and the same threshold test for negation shows that $\limsup_{n\to+\infty} h_n$ is measurable.

HW 31 — Superlevel sets of an infimum

For $h = \inf_n h_n$, prove that $\{h \geq c\} = \bigcap_{n=1}^{\infty} \{h_n \geq c\}$.

Proof.

We first prove that $\{h \geq c\} \subseteq \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. Let $s_0 \in \{h \geq c\}$. By definition of this set we have $h(s_0) = \inf_{n\in\mathbb{N}} h_n(s_0) \geq c$. By definition of infimum $\inf_{n\in\mathbb{N}} h_n(s_0) \leq h_n(s_0)\ \forall n \in \mathbb{N}$. Therefore $h_n(s_0) \geq \inf_{n\in\mathbb{N}} h_n(s_0) \geq c\ \forall n \in \mathbb{N}$ hence $s_0 \in \{h_n \geq c\}$ which implies $s_0 \in \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. Since s_0 was arbitrary $\bigcap_{n=1}^{\infty} \{h_n \geq c\} \supseteq \{h \geq c\}$. For the reverse inclusion let $s_0 \in \bigcap_{n=1}^{\infty} \{h_n \geq c\}$. By definition of intersection we have $s_0 \in \{h_n \geq c\}\ \forall n \in \mathbb{N}$. Therefore by definition of these sets $h_n(s_0) \geq c\ \forall n \in \mathbb{N}$. Notice that c is a lower bound for the family $\{h_n(s_0), n \in \mathbb{N}\}$. Since the infimum is the greatest lower bound we obtain $h(s_0) = \inf_{n\in\mathbb{N}} h_n(s_0) \geq c$ hence $s_0 \in \{h \geq c\}$. Since s_0 was arbitrary $\bigcap_{n=1}^{\infty} \{h_n \geq c\} \subseteq \{h \geq c\}$. Combining the two inclusions we conclude the proof.

Corollary 3.6 — Measurability of the Convergence Set

Let $(h_n)_{n\in\mathbb{N}} \subset m\Sigma$. Then $\{s \in \mathcal{S} : \lim_{n\to\infty} h_n(s) \text{exists}\} \in \Sigma$.

Expanded proof.

Let

$$\underline{h} = \liminf_{n\to\infty} h_n,\ \overline{h} = \limsup_{n\to\infty} h_n.$$

By Lemma 3.5 these are extended-real measurable, and $\underline{h} \leq \overline{h}$ pointwise. We follow the lecturer's idea of testing their difference at zero, first restricting to where the subtraction is defined.

Part 1 — Both liminf and limsup are finite real numbers. Let $C = \{s \in \mathcal{S} : \underline{h}(s), \overline{h}(s) \in \mathbb{R}\}$. This set is measurable: since $\underline{h} \leq \overline{h}$, $C = \bigcup_{k\geq 1}(\{\underline{h} \geq -k\} \cap \{\overline{h} \leq k\}) \in \Sigma$. On C both functions are real-valued, so we can use the lecturer's subtraction argument. Define $g(s) := \underline{h}(s) - \overline{h}(s)$ for $s \in C$. By Lemma 3.5 the two functions are measurable, and Lemma 3.4**(i)**,**(iii)** makes their difference g measurable on C.

We show that finite real convergence is equivalent to $g = 0$. A real sequence converges to a finite limit if and only if its liminf and limsup coincide in $\mathbb{R}$. Since both values are finite on C, this is equivalent to their difference being zero. Therefore

$$E_{\text{fin}} = \{s \in C : \underline{h}(s) - \overline{h}(s) = 0\} = \{s \in C : g(s) = 0\} = g^{-1}(\{0\}) \in \Sigma.$$

The first equality is the finite liminf/limsup convergence criterion, the second substitutes the definition of g, and the third is the definition of pre-image. Since $\{0\} \in \mathcal{B}(\mathbb{R})$ and g is measurable on the measurable set C, its pre-image belongs to Σ.

Part 2 — Both liminf and limsup equal $+\infty$. *We show that convergence to* $+\infty$ *is a measurable event.* The inequality $\underline{h} \leq \overline{h}$ implies that $\underline{h}(s) = +\infty$ forces $\overline{h}(s) = +\infty$. Consequently,

$$E_+ := \{s : \underline{h}(s) = \overline{h}(s) = +\infty\} = \{\underline{h} = +\infty\} = \bigcap_{k\geq 1} \{\underline{h} > k\} \in \Sigma.$$

The first equality uses the inequality between liminf and limsup. For the second, an extended real number equals $+\infty$ exactly when it exceeds every positive integer. Each level set is measurable by Lemma 3.5; closure under countable intersections gives the final membership. If $\underline{h}(s) = +\infty$, the identity $\underline{h}(s) = \sup_N \inf_{n\geq N} h_n(s)$ shows that, for every real M, some tail infimum exceeds M. Hence $h_n(s) > M$ throughout that tail, which is convergence to $+\infty$. Conversely, convergence to $+\infty$ puts $h_n(s) > M + 1$ eventually for every M. The corresponding tail infimum is at least $M + 1$, so $\underline{h}(s) > M$ for every M, and therefore $\underline{h}(s) = +\infty$.

Part 3 — Both liminf and limsup equal $-\infty$. *We show that convergence to* $-\infty$ *is a measurable event.* Again $\underline{h} \leq \overline{h}$, so $\overline{h}(s) = -\infty$ forces $\underline{h}(s) = -\infty$. Thus

$$E_- := \{s : \underline{h}(s) = \overline{h}(s) = -\infty\} = \{\overline{h} = -\infty\} = \bigcap_{k\geq 1} \{\overline{h} < -k\} \in \Sigma.$$

The first equality uses the same order inequality. The second characterizes $-\infty$ by all negative integer thresholds. Each level set is measurable by Lemma 3.5, so its countable intersection belongs to Σ. If $\overline{h}(s) = -\infty$, the identity $\overline{h}(s) = \inf_N \sup_{n \geq N} h_n(s)$ gives, for every real M, a tail supremum below M. Every term of that tail is then below M, which is convergence to $-\infty$. Conversely, that convergence gives $h_n(s) < M - 1$ eventually for every M. The tail supremum is at most $M - 1$, so $\overline{h}(s) < M$ for every M, and therefore $\overline{h}(s) = -\infty$.

Combining the three parts, the event of convergence in $\overline{\mathbb{R}}$ is

$$\{s : \lim_{n\to\infty} h_n(s) \text{ exists in } \overline{\mathbb{R}}\} = E_{\text{fin}} \cup E_+ \cup E_- \in \Sigma.$$

For convergence to a finite real number, the event is E_{fin} alone. No difference of two infinite values is used. QED

Def: Random variable

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ a r.v. X is a $\mathcal{F}$-measurable function i.e. $X : \Omega \to \mathbb{R}$ and $X^{-1} : \mathcal{B}(\mathbb{R}) \to \mathcal{F}$.

Ex 1b proof

Consider the infinite coin tossing where $\Omega = \{H, T\}^{\mathbb{N}}$ and $\mathcal{F} = \sigma(\{\omega \in \Omega : \omega_n = W\},\ n \in \mathbb{N},\ W \in \{H, T\})$. For every $n \in \mathbb{N}$ define $X_n(\omega) = \begin{cases} 1 & \text{if } \omega_n = H, \\ 0 & \text{if } \omega_n = T, \end{cases}$ and $S_n = X_1 + \cdots + X_n$. Prove that $F := \{\omega \in \Omega : \lim_{n\to+\infty} \frac{S_n(\omega)}{n} = \frac{1}{2}\} \in \mathcal{F}$.

Proof.

We first prove that X_n is $\mathcal{F}$-measurable $\forall n \in \mathbb{N}$. By HW 30 it is sufficient to show that $\{X_n \leq c\} \in \mathcal{F}\ \forall c \in \mathbb{R}$. By definition of X_n we have

$$\{X_n \leq c\} = \begin{cases} \emptyset & c < 0, \\ \{\omega \in \Omega : \omega_n = T\} & 0 \leq c < 1, \\ \Omega & c \geq 1. \end{cases}$$

Notice that $\emptyset, \Omega \in \mathcal{F}$ and $\{\omega \in \Omega : \omega_n = T\} \in \mathcal{F}$ by def of $\mathcal{F}$ in example 1b. Therefore $\{X_n \leq c\} \in \mathcal{F}\ \forall c \in \mathbb{R}$ hence $X_n \in \mathrm{m}\mathcal{F}\ \forall n \in \mathbb{N}$. Now by Lemma 3.4 **(i)** $S_n := \sum_{k=1}^n X_k \in \mathrm{m}\mathcal{F}$ $\forall n \in \mathbb{N}$. Moreover since $\frac{1}{n} \in \mathbb{R}$, by Lemma 3.4 **(iii)** $\frac{1}{n} S_n \in \mathrm{m}\mathcal{F}\ \forall n \in \mathbb{N}$. *We prove now that $F \in \mathcal{F}$.* Recall that a sequence of real numbers converges to $a \in \mathbb{R}$ if and only if its limsup and liminf are both equal to a. Therefore

$$F = \{\omega \in \Omega : \lim_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} = \{\omega \in \Omega : \limsup_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} \cap \{\omega \in \Omega : \liminf_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\}.$$

Define $U(\omega) := \limsup_{n\to+\infty} \frac{S_n(\omega)}{n}$ and $L(\omega) := \liminf_{n\to+\infty} \frac{S_n(\omega)}{n}$. Since $\frac{S_n}{n} \in \mathrm{m}\mathcal{F}\ \forall n \in \mathbb{N}$ by Lemma 3.5 **(ii)** $U, L \in \mathrm{m}\mathcal{F}$. Notice that both functions are real valued since $0 \leq \frac{S_n(\omega)}{n} \leq 1\ \forall \omega$. Moreover $\{\frac{1}{2}\} \in \mathcal{B}(\mathbb{R})$ since it is a singleton. In particular

$$A := \{\omega \in \Omega : \limsup_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} = U^{-1}(\{\tfrac{1}{2}\}) \in \mathcal{F},$$
$$B := \{\omega \in \Omega : \liminf_{n\to+\infty} \tfrac{S_n(\omega)}{n} = \tfrac{1}{2}\} = L^{-1}(\{\tfrac{1}{2}\}) \in \mathcal{F}.$$

Hence recalling $F = A \cap B$ since $\mathcal{F}$ is closed under intersection we conclude that $F \in \mathcal{F}$. As a consequence, since $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure and $F \in \mathcal{F}$, the probability $\mathbb{P}(F)$ is well-defined.

Def Law of a random variable

On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ let X be a real-valued random variable. The law of X is a probability measure on $\mathcal{B}(\mathbb{R})$ i.e. a mapping $\mathcal{L}_X : \mathcal{B}(\mathbb{R}) \to [0, 1]$ defined as $\mathcal{L}_X := \mathbb{P} \circ X^{-1}$.

That is, for any $A \in \mathcal{B}(\mathbb{R})$ $\mathcal{L}_X(A) := \mathbb{P}(X \in A)$. The diagram below should help your intuition: $\mathcal{B}(\mathbb{R}) \xrightarrow{X^{-1}} \mathcal{F} \xrightarrow{\mathbb{P}} [0,1]$.

Proof.

Notice that $\mathcal{L}_X$ is well defined. Indeed since X is $\mathcal{F}$-measurable by definition 3.1, $X^{-1}(A) \in \mathcal{F}$ $\forall A \in \mathcal{B}(\mathbb{R})$. Therefore $\mathcal{L}_X(A) = \mathbb{P}(X^{-1}(A)) \in [0,1]$. *We now verify the properties of a probability measure.*

(i) *We first prove that* $\mathcal{L}_X(\emptyset) = 0$. We obtain

$$\mathcal{L}_X(\emptyset) = \mathbb{P}(X^{-1}(\emptyset)) = \mathbb{P}(\emptyset) = 0$$

where the first equality follows from the definition of the law X, the second from the fact that pre-image of the empty set is empty, the last from properties of the probability measure $\mathbb{P}$.

(ii) *We now prove that* $\mathcal{L}_X(\mathbb{R}) = 1$. We obtain

$$\mathcal{L}_X(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$$

where the first equality is the definition of $\mathcal{L}_X$, the second follows from the fact that $X : \Omega \to \mathbb{R}$ so $X(\omega) \in \mathbb{R}$ $\forall \omega \in \Omega$ and the last equality follows from the fact that $\mathbb{P}$ is a probability measure.

(iii) *Finally we prove countable additivity.* Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{B}(\mathbb{R})$ be a sequence of pairwise disjoint sets namely $A_i \cap A_j = \emptyset$ $\forall i \neq j$. *We want to prove that*

$$\mathcal{L}_X\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n).$$

As we will see in HW 32, $X^{-1}(\bigcup_{n=1}^{\infty} A_n) = \bigcup_{n=1}^{\infty} X^{-1}(A_n)$. We use that identity without proving it here. First notice that the pre-images $X^{-1}(A_n)$ are also pairwise disjoint. Indeed for $i \neq j$ by property **(iii)** of HW 28

$$X^{-1}(A_i) \cap X^{-1}(A_j) = X^{-1}(A_i \cap A_j) = X^{-1}(\emptyset) = \emptyset$$

Moreover since X is $\mathcal{F}$-measurable $X^{-1}(A_n) \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Therefore we can apply the countable additivity of $\mathbb{P}$ to the sequence $(X^{-1}(A_n))_{n\in\mathbb{N}}$. We obtain

$$\mathcal{L}_X\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(X^{-1}\left(\bigcup_{n=1}^{\infty} A_n\right)\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} X^{-1}(A_n)\right) = \sum_{n=1}^{\infty} \mathbb{P}(X^{-1}(A_n)) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n)$$

where the first equality follows from the definition of $\mathcal{L}_X$, the second from HW 32, the third from the countable additivity of $\mathbb{P}$ since the pre-images are pairwise disjoint, and the last from the definition of $\mathcal{L}_X$. Hence $\mathcal{L}_X(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mathcal{L}_X(A_n)$. Therefore $\mathcal{L}_X$ is countably additive. Since $\mathcal{L}_X(\emptyset) = 0$, $\mathcal{L}_X(\mathbb{R}) = 1$ and $\mathcal{L}_X$ is countably additive on $\mathcal{B}(\mathbb{R})$ we conclude that $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$.

HW 32 — Pre-image of a countable union

Prove that if $(A_n)_{n\in\mathbb{N}}$ is a family of pairwise disjoint subsets of $\mathbb{R}$ then

$$\{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\} = \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}.$$

Proof.

We first prove that $\{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\} \subseteq \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$. Let $\omega_0 \in \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. By definition of this set we have $X(\omega_0) \in \bigcup_{n=1}^{\infty} A_n$. By definition of union there exists $n_0 \in \mathbb{N}$ such that $X(\omega_0) \in A_{n_0}$. n_0 is unique since A_n are pairwise disjoint. Therefore $\omega_0 \in \{\omega \in \Omega : X(\omega) \in A_{n_0}\} \subseteq \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$ where the first membership follows from the definition of the set and the inclusion follows from the fact that $\{\omega \in \Omega : X(\omega) \in A_{n_0}\}$ is one of the sets of the union. Since ω_0 was arbitrary we obtain $\{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\} \subseteq \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$. For the reverse inclusion we let $\omega_0 \in \bigcup_{n=1}^{\infty} \{\omega \in \Omega : X(\omega) \in A_n\}$.

By definition of union there exists $n_0 \in \mathbb{N}$ such that $\omega_0 \in \{\omega \in \Omega : X(\omega) \in A_{n_0}\}$. By definition of this set we have $X(\omega_0) \in A_{n_0}$. Therefore $X(\omega_0) \in A_{n_0} \subseteq \bigcup_{n=1}^{\infty} A_n$. Hence $\omega_0 \in \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. Since ω_0 was arbitrary it follows that $\bigcup_{n=1}^{\infty}\{\omega \in \Omega : X(\omega) \in A_n\} \subseteq \{\omega \in \Omega : X(\omega) \in \bigcup_{n=1}^{\infty} A_n\}$. Combining the two inclusions we conclude the proof.

Why is it sufficient to specify the law on a π-system

Since $\mathcal{L}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$ we may ask whether it is necessary to specify $\mathcal{L}_X(A)$ $\forall$ Borel set A in order to uniquely determine the law of X. The answer is no. By Lemma 1.2 it is sufficient to specify the values of $\mathcal{L}_X$ on a π-system generating $\mathcal{B}(\mathbb{R})$. Recall that by Prop 1.1 $\pi(\mathbb{R}) = \{(-\infty, c] : c \in \mathbb{R}\}$ is a π-system and $\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R})$. Moreover by Lemma 1.2 if two finite measures on a σ-algebra generated by a π-system have the same total mass and coincide on that π-system, then they coincide on the entire σ-algebra. In particular let μ_1 and μ_2 be two probability measures on $\mathcal{B}(\mathbb{R})$ such that $\mu_1((-\infty, c]) = \mu_2((-\infty, c])$ $\forall c \in \mathbb{R}$. Since both measures are probability measures $\mu_1(\mathbb{R}) = \mu_2(\mathbb{R}) = 1 < \infty$. Therefore all the assumption of Lemma 1.2 are satisfied and we conclude that $\mu_1(A) = \mu_2(A)$ $\forall A \in \mathcal{B}(\mathbb{R})$. Hence a probability measure on $\mathcal{B}(\mathbb{R})$ is uniquely determined by its values on the intervals $(-\infty, c]$ for every $c \in \mathbb{R}$. In particular the law of a random variable can be uniquely determined without specifying its value on every Borel set. It is sufficient to know $\mathcal{L}_X((-\infty, c])$ $\forall c \in \mathbb{R}$. This motivates the definition of the cumulative distribution function.

Definition CDF

Let $X : \Omega \to \mathbb{R}$ be a random variable. The CDF of X is the function $F_X : \mathbb{R} \to [0, 1]$ defined by $F_X(c) := \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X \leq c)$ $\forall c \in \mathbb{R}$. Indeed by definition of the law of X

$$F_X(c) = \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X^{-1}((-\infty, c])) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq c\}),$$

where the second equality follows from the definition of $\mathcal{L}_X$ and the third from the definition of pre-image.

Since the intervals $(-\infty, c]$ form a π-system generating $\mathcal{B}(\mathbb{R})$ by Lemma 1.2 the CDF uniquely determines the law $\mathcal{L}_X$. In particular $F_X(c) = F_Y(c)$ $\forall c \in \mathbb{R} \iff \mathcal{L}_X = \mathcal{L}_Y$. Thus two random variables have the same law exactly when they have the same CDF.

Properties of cdf

The cdf enjoys the following properties: **(a)** Range and monotonicity $F_X : \mathbb{R} \to [0, 1]$, $F_X(y) \leq F_X(z)$ $\forall y \leq z$.

(b) Limit at infinity. If $\mathbb{P}(|X| < \infty) = 1$ then $\lim_{z \to +\infty} F_X(z) = 1$, $\lim_{z \to -\infty} F_X(z) = 0$.

(c) Right-continuity $\forall z \in \mathbb{R}$, $\lim_{\varepsilon \downarrow 0} F_X(z + \varepsilon) = F_X(z)$.

Proof.

(a) By definition of the CDF $F_X(z) = \mathbb{P}(X \leq z)$. Since $\mathbb{P}$ is a probability measure we have $0 \leq \mathbb{P}(X \leq z) \leq 1$, $\forall z \in \mathbb{R}$. Therefore $0 \leq F_X(z) \leq 1$ $\forall z \in \mathbb{R}$ hence $F_X : \mathbb{R} \to [0, 1]$. For monotonicity, let $y, z \in \mathbb{R}$ such that $y \leq z$. Notice that $\{X \leq y\} \subseteq \{X \leq z\}$. Indeed if $\omega \in \{X \leq y\}$ then $X(\omega) \leq y \leq z$ therefore $\omega \in \{X \leq z\}$. Since X is $\mathcal{F}$-measurable both events belong to $\mathcal{F}$ and by monotonicity of probability HW 12 we obtain $\mathbb{P}(X \leq y) \leq \mathbb{P}(X \leq z)$. Hence by definition of cdf $F_X(y) \leq F_X(z)$.

(b) For the positive tail, $\forall n \in \mathbb{N}$ define $A_n := \{\omega \in \Omega : X(\omega) \leq n\}$. Since X is $\mathcal{F}$-measurable by definition 3.1 $A_n \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Notice that $A_n \subseteq A_{n+1}$ $\forall n \in \mathbb{N}$, because $X(\omega) \leq n$ implies $X(\omega) \leq n + 1$. Moreover $\bigcup_{n=1}^{\infty} A_n = \{\omega \in \Omega : X(\omega) < +\infty\}$. Since $\mathbb{P}(|X| < \infty) = 1$, we have $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) = 1$. Therefore by Lemma 1.5 (continuity of probability from below)

$$\lim_{n \to +\infty} F_X(n) = \lim_{n \to +\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = 1,$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.5 and third from assumption. Since F_X is non-decreasing by prop (a) the convergence $F_X(n) \to 1$ implies that $\lim_{z\to+\infty} F_X(z) = 1$. For the negative tail, $\forall n \in \mathbb{N}$ define $B_n := \{\omega \in \Omega : X(\omega) \leq -n\}$.
Again since X is measurable, $B_n \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Notice that $B_n \supseteq B_{n+1}$ $\forall n \in \mathbb{N}$ because $X(\omega) \leq -(n+1)$ implies $X(\omega) \leq -n$. Moreover $\bigcap_{n=1}^{\infty} B_n = \{\omega \in \Omega : X(\omega) = -\infty\}$. By assumption $\mathbb{P}(|X| < \infty) = 1$, $\mathbb{P}(\bigcap_{n=1}^{\infty} B_n) = 0$. Since the events B_n form a non-increasing sequence and $\mathbb{P}(B_1) \leq 1 < \infty$, we may apply Lemma 1.6 (continuity from above). Therefore

$$\lim_{n\to+\infty} F_X(-n) = \lim_{n\to+\infty} \mathbb{P}(B_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = 0,$$

where the first equality follows from the definition of CDF, the second from Lemma 1.6 and third from ass. Since F_X is non-decreasing by prop (a) the convergence $F_X(-n) \to 0$ implies that $\lim_{z\to-\infty} F_X(z) = 0$.
(c) Fix an arbitrary z. $\forall n \in \mathbb{N}$ define $A_n := \{\omega \in \Omega : X(\omega) \leq z + \frac{1}{n}\}$. Since X is $\mathcal{F}$-measurable, $A_n \in \mathcal{F}$ $\forall n \in \mathbb{N}$. Notice that $A_n \supseteq A_{n+1}$ $\forall n \in \mathbb{N}$ because $z + \frac{1}{n+1} \leq z + \frac{1}{n}$. *We now show that* $\bigcap_{n=1}^{\infty} A_n = \{X \leq z\}$ *by showing both inclusions. First we show that* $\bigcap_{n=1}^{\infty} A_n \subseteq \{X \leq z\}$. Let $\omega_0 \in \bigcap_{n=1}^{\infty} A_n$, by definition of intersection $\omega_0 \in A_n$ $\forall n \in \mathbb{N}$. Therefore by definition of A_n, $X(\omega_0) \leq z + \frac{1}{n}$ $\forall n \in \mathbb{N}$. Since $\frac{1}{n} \to 0$ we obtain $X(\omega_0) \leq z$. Hence $\omega_0 \in \{X \leq z\}$. Since ω_0 was arbitrary we conclude that $\bigcap_{n=1}^{\infty} A_n \subseteq \{X \leq z\}$. *For the reverse inclusion, we show that* $\{X \leq z\} \subseteq \bigcap_{n=1}^{\infty} A_n$. Let $\omega_0 \in \{X \leq z\}$. By definition of this set $X(\omega_0) \leq z$. Therefore $X(\omega_0) \leq z < z + \frac{1}{n}$ $\forall n \in \mathbb{N}$. Hence $\omega_0 \in A_n$ $\forall n \in \mathbb{N}$. By definition of intersection $\omega_0 \in \bigcap_{n=1}^{\infty} A_n$. Since ω_0 was arbitrary we obtain $\{X \leq z\} \subseteq \bigcap_{n=1}^{\infty} A_n$. Combining the two inclusionss we conclude that $\bigcap_{n=1}^{\infty} A_n = \{X \leq z\}$. Now since $(A_n)_{n\in\mathbb{N}}$ is a decreasing sequence of events and $\mathbb{P}(A_1) \leq 1 < \infty$, by Lemma 1.6 we obtain

$$\lim_{n\to+\infty} F_X(z + \tfrac{1}{n}) = \lim_{n\to+\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \mathbb{P}(X \leq z) = F_X(z),$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.6, the third from the equality of sets proved above and the last the definition of F_X. Finally, to obtain the limit for arbitrary $\varepsilon \downarrow 0$, let $\delta > 0$. From the preceding convergence, there exists $N \in \mathbb{N}$ such that $F_X(z + \frac{1}{N}) < F_X(z) + \delta$. For every $0 < \varepsilon \leq \frac{1}{N}$, by monotonicity of F_X, $F_X(z) \leq F_X(z+\varepsilon) \leq F_X(z + \frac{1}{N}) < F_X(z) + \delta$. Since $\delta > 0$ was arbitrary we conclude that

$$\lim_{\varepsilon\downarrow 0} F_X(z + \varepsilon) = F_X(z).$$

Since $z \in \mathbb{R}$ was arbitrary, F_X is right-continuous.

HW 33 — CDF tails with mass at infinity

Let X be an extended-real random variable. Prove if $\mathbb{P}(X = +\infty) = p$ and $\mathbb{P}(X = -\infty) = q$ then $\lim_{z\to+\infty} F_X(z) = 1 - p$ and $\lim_{z\to-\infty} F_X(z) = q$.

Proof.

Recall that by definition of CDF $F_X(z) = \mathbb{P}(X \leq z)$, $z \in \mathbb{R}$. For the positive tail, $\forall n \in \mathbb{N}$ define $A_n := \{X \leq n\}$. Notice that $A_n \subseteq A_{n+1}$ $\forall n \in \mathbb{N}$ since $X(\omega) \leq n \Rightarrow X(\omega) \leq n + 1$. Therefore $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events. Moreover $\bigcup_{n=1}^{\infty} A_n = \{X < \infty\}$. Indeed, $X(\omega) \leq n$ for at least one $n \in \mathbb{N}$ if and only if $X(\omega) < +\infty$. Hence by Lemma 1.5 continuity of probability from below we obtain

$$\lim_{n\to+\infty} F_X(n) = \lim_{n\to+\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}(X < +\infty) = 1 - \mathbb{P}(X = +\infty) = 1 - p,$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.5, the third from the equality of sets above and the fourth from the complement rule for probability.

Since F_X is non-decreasing by prop (a) of the CDF, and $F_X(n) \leq F_X(z) \leq 1-p \ \forall z \geq n$ we conclude that $\lim_{z\to+\infty} F_X(z) = 1-p$. For the negative tail, $\forall n \in \mathbb{N}$ define $B_n := \{X \leq -n\}$. Notice that $B_n \supseteq B_{n+1} \ \forall n \in \mathbb{N}$ since $X(\omega) \leq -(n+1) \Rightarrow X(\omega) \leq -n$. Therefore $(B_n)_{n\in\mathbb{N}}$ is a decreasing sequence of events. Moreover $\bigcap_{n=1}^{\infty} B_n = \{X = -\infty\}$. Indeed $X(\omega) \leq -n \ \forall n \in \mathbb{N} \iff X(\omega) = -\infty$. Hence, since $\mathbb{P}(B_1) \leq 1 < \infty$ by Lemma 1.6 (continuity of probability from above) we obtain

$$\lim_{n\to+\infty} F_X(-n) = \lim_{n\to+\infty} \mathbb{P}(B_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} B_n\right) = \mathbb{P}(X = -\infty) = q,$$

where the first equality follows from the definition of the CDF, the second from Lemma 1.6, the third from the equality of sets above and the last from the assumption. Since F_X is non-decreasing by prop (a) and $q \leq F_X(z) \leq F_X(-n) \ \forall z \leq -n$ we conclude that $\lim_{z\to-\infty} F_X(z) = q$.

HW 34 — Left limits and atoms

Prove that $\forall z \in \mathbb{R}$, $\{X \leq z-\varepsilon\} \uparrow \{X < z\}$ as $\varepsilon \downarrow 0$. Then conclude that $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = F_X(z) - \mathcal{L}_X(\{z\})$.

Proof.

Fix an arbitrary $z \in \mathbb{R}$. For every $n \in \mathbb{N}$ define $A_n := \{\omega \in \Omega : X(\omega) \leq z - \frac{1}{n}\}$. Since X is $\mathcal{F}$-measurable we have $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$. Notice that $A_n \subseteq A_{n+1} \ \forall n \in \mathbb{N}$. Indeed let $\omega_0 \in A_n$. By definition of A_n $X(\omega_0) \leq z - \frac{1}{n}$, since $z - \frac{1}{n} < z - \frac{1}{n+1}$ we obtain $X(\omega_0) \leq z - \frac{1}{n} < z - \frac{1}{n+1}$. Therefore $\omega_0 \in A_{n+1}$. Since ω_0 was arbitrary the inclusion follows. Hence $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events, *we now prove that* $\bigcup_{n=1}^{\infty} A_n = \{X < z\}$. We establish the equality by proving both inclusionss. First inclusion. Let $\omega_0 \in \bigcup_{n=1}^{\infty} A_n$. By definition of union, there exists $n_0 \in \mathbb{N}$ such that $\omega_0 \in A_{n_0}$. By definition of A_{n_0}, $X(\omega_0) \leq z - \frac{1}{n_0} < z$. Therefore $X(\omega_0) < z$ which implies that $\omega_0 \in \{X < z\}$. Since ω_0 was arbitrary we conclude that $\bigcup_{n=1}^{\infty} A_n \subseteq \{X < z\}$. Reverse inclusion. Let $\omega_0 \in \{X < z\}$. By definition of this set $X(\omega_0) < z$. Therefore $z - X(\omega_0) > 0$. Since $\frac{1}{n} \to 0$ as $n \to +\infty$ there exists $n_0 \in \mathbb{N}$ such that $\frac{1}{n_0} < z - X(\omega_0)$. Rearranging this inequality we obtain $X(\omega_0) < z - \frac{1}{n_0}$ which implies $X(\omega_0) \leq z - \frac{1}{n_0}$ hence $\omega_0 \in A_{n_0}$ and consequently $\omega_0 \in \bigcup_{n=1}^{\infty} A_n$. Since ω_0 was arbitrary we conclude that $\{X < z\} \subseteq \bigcup_{n=1}^{\infty} A_n$. Combining the two inclusionss, we obtain $\bigcup_{n=1}^{\infty} A_n = \{X < z\}$. Since $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events we have proved that $\{X \leq z-\varepsilon\} \uparrow \{X < z\}$ as $\varepsilon \downarrow 0$. Since $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence of events, by Lemma 1.5 we obtain

$$\lim_{n\to+\infty} F_X(z - \tfrac{1}{n}) = \lim_{n\to+\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}(X < z)$$

where the first equality follows from the definition of the CDF and of A_n, the second from Lemma 1.5 and the third from the equality of sets proved above. Moreover since F_X is non-decreasing for every $0 < \varepsilon \leq \frac{1}{n}$, $F_X(z - \frac{1}{n}) \leq F_X(z-\varepsilon) \leq \mathbb{P}(X < z)$. Since the left-hand side converges to $\mathbb{P}(X < z)$ as $n \to +\infty$ we conclude that $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = \mathbb{P}(X < z)$. Finally we notice that $\{X \leq z\} = \{X < z\} \cup \{X = z\}$

where the two events are disjoint on the right-hand side. Therefore by the additivity of the probability measure $\mathbb{P}$ $\mathbb{P}(X \leq z) = \mathbb{P}(X < z) + \mathbb{P}(X = z)$. Rearranging we obtain $\mathbb{P}(X < z) = \mathbb{P}(X \leq z) - \mathbb{P}(X = z)$. Hence

$$\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = \mathbb{P}(X < z) = \mathbb{P}(X \leq z) - \mathbb{P}(X = z) = F_X(z) - \mathcal{L}_X(\{z\}),$$

where the first equality follows from the continuity of probability from below established above, the second from the additivity of $\mathbb{P}$, the third from the definition of the CDF and of the law of X. Since $z \in \mathbb{R}$ was arbitrary we conclude that $\lim_{\varepsilon\downarrow 0} F_X(z-\varepsilon) = F_X(z) - \mathcal{L}_X(\{z\})$.

Definition — Probability Density Function (pdf)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \to \mathbb{R}$ be a r.v. Recall that cdf $F_X(z) := \mathbb{P}(X \leq z)$, $z \in \mathbb{R}$. A r.v. X admits a probability density function (pdf) if there exists a non-negative measurable function $f_X : \mathbb{R} \to [0, +\infty)$ such that

$$\mathbb{P}(X \in A) = \int_A f_X(z)\, dz \quad \forall A \in \mathcal{B}(\mathbb{R}).$$

In particular taking $A = (-\infty, z]$ we obtain $F_X(z) = \int_{-\infty}^{z} f_X(t)\, dt$. If the CDF is absolutely continuous then it is differentiable almost everywhere and its density is given by $f_X(z) = F_X'(z)$ for almost every $z \in \mathbb{R}$.

Properties of pdf

(a) Non-negativity $f_X(z) \geq 0$ for almost every $z \in \mathbb{R}$. **(b)** Normalization $\int_{-\infty}^{+\infty} f_X(z)\, dz = 1$.

Proof.

(a) Recall that by prop (a) of the CDF, F_X is non-decreasing. Therefore $\forall z \in \mathbb{R}$ and $\forall h > 0$, $F_X(z + h) \geq F_X(z)$. Subtracting $F_X(z)$ and dividing by $h > 0$ we obtain $\frac{F_X(z+h) - F_X(z)}{h} \geq 0$. At every point where F_X is differentiable letting $h \downarrow 0$ gives $F_X'(z) = \lim_{h \downarrow 0} \frac{F_X(z+h) - F_X(z)}{h} \geq 0$. The integral representation $F_X(z) = \int_{-\infty}^{z} f_X(t)\, dt$ from the definition of a density makes F_X absolutely continuous, with $F_X' = f_X$ almost everywhere by the fundamental theorem for the Lebesgue integral. Consequently, $f_X(z) \geq 0$ almost everywhere, consistently with the non-negativity already required in the definition. **(b)** Recall that by def of probability density function $\mathbb{P}(X \in A) = \int_A f_X(z)\, dz$ $\forall A \in \mathcal{B}(\mathbb{R})$. In particular taking $A = \mathbb{R}$ we obtain $\int_{-\infty}^{+\infty} f_X(z)\, dz = \mathbb{P}(X \in \mathbb{R}) = \mathbb{P}(\Omega) = 1$ where the first equality follows from the definition of the pdf, the second from the fact that $X : \Omega \to \mathbb{R}$ and the third from the normalization of the probability measure $\mathbb{P}$.

Theorem – Existence of a r.v. with a prescribed CDF

Let $F : \mathbb{R} \to [0, 1]$ be a function satisfying the following properties:

- **(a)** Range and monotonicity $F : \mathbb{R} \to [0, 1]$, $F(y) \leq F(z)$ $\forall y \leq z$
- **(b)** Limits of infinity $\lim_{z \to +\infty} F(z) = 1$ $\lim_{z \to -\infty} F(z) = 0$
- **(c)** Right continuity $\lim_{\varepsilon \downarrow 0} F(z + \varepsilon) = F(z)$ $\forall z \in \mathbb{R}$

Then there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a r.v. $X : \Omega \to \mathbb{R}$ such that $F_X(c) = F(c)$ $\forall c \in \mathbb{R}$

Proof

First we construct the probability space. Let $\Omega = \mathbb{R}$ and $\mathcal{F} = \mathcal{B}(\mathbb{R})$. We now introduce a measure $\mathcal{L}$ by defining $\forall a, b \in \mathbb{R}$ with $a < b$ $\mathcal{L}((a, b]) := F(b) - F(a)$. Notice that, since F is non-decreasing by property (a) we have $F(a) \leq F(b)$ whenever $a < b$. Therefore $\mathcal{L}((a, b]) = F(b) - F(a) \geq 0$. The monotonicity and right-continuity of F allow this assignment on intervals to be extended to a measure $\mathcal{L}$ on $\mathcal{B}(\mathbb{R})$ by the Lebesgue–Stieltjes measure construction. This construction is taken as a known result here; it is the same existence step used in the handwritten proof. *We now verify that $\mathcal{L}$ is a probability measure.* Since $\mathcal{L}$ is already a measure on $\mathcal{B}(\mathbb{R})$ it remains to prove that $\mathcal{L}(\mathbb{R}) = 1$. Notice that $(-n, n] \subseteq (-(n+1), n+1]$ $\forall n \in \mathbb{N}$ and $\bigcup_{n=1}^{\infty} (-n, n] = \mathbb{R}$. Therefore since $((-n, n])_{n \in \mathbb{N}}$ is an increasing sequence of Borel sets, by continuity of measures from below we obtain

$$\mathcal{L}(\mathbb{R}) = \mathcal{L}\left(\bigcup_{n=1}^{\infty} (-n, n]\right) = \lim_{n \to \infty} \mathcal{L}((-n, n]) = \lim_{n \to \infty} (F(n) - F(-n)) = 1 - 0 = 1$$

Hence $\mathcal{L}$ is a probability measure on $\mathcal{B}(\mathbb{R})$. We can therefore consider the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L})$.

Second we construct the random variable. Define $X : \Omega \to \mathbb{R}$ by $X(\omega) := \omega \ \forall \omega \in \Omega$. Notice that X is identity function on $\mathbb{R}$. *We now verify that X is a r.v. namely that is $\mathcal{F}$-measurable.* Let $A \in \mathcal{B}(\mathbb{R})$. By definition of pre-image

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} = \{\omega \in \mathbb{R} : \omega \in A\} = A$$

where the second equality follows from $\Omega = \mathbb{R}$ and $X(\omega) = \omega$. Since $A \in \mathcal{B}(\mathbb{R}) = \mathcal{F}$ we conclude that $X^{-1}(A) \in \mathcal{F}$. Since $A \in \mathcal{B}(\mathbb{R})$ was arbitrary by Definition 3.1, X is $\mathcal{F}$-measurable and therefore a random variable.

Finally we prove that the CDF of X coincides with F. Fix an arbitrary $c \in \mathbb{R}$. First notice that $(-n, c] \uparrow (-\infty, c]$ as $n \to \infty$ where we consider sufficiently large n so that $-n < c$. Therefore by continuity of measure from below

$$\mathcal{L}((-\infty, c]) = \lim_{n\to\infty} \mathcal{L}((-n, c]) = \lim_{n\to\infty} (F(c) - F(-n)) = F(c)$$

where the first equality follows from continuity of measure from below, the second from the definition of $\mathcal{L}$ and the third from property (b) since $\lim_{n\to\infty} F(-n) = 0$. Now by definition of the CDF

$$\begin{aligned} F_X(c) &= \mathcal{L}_X((-\infty, c]) = \mathbb{P}(X \le c) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \le c\}) \\ &= \mathbb{P}(\{\omega \in \mathbb{R} : \omega \le c\}) = \mathbb{P}((-\infty, c]) = \mathcal{L}((-\infty, c]) = F(c) \end{aligned}$$

where the first and second equalities follow from the definition of CDF and the law of X, the fourth from $X(\omega) = \omega$ and $\Omega = \mathbb{R}$, the sixth from $\mathbb{P} = \mathcal{L}$ and the last from calculation above. Since $c \in \mathbb{R}$ was arbitrary we conclude that $F_X(c) = F(c) \ \forall c \in \mathbb{R}$. Thus we have constructed a r.v. X whose CDF is precisely F.

Construction — Inverse Transform Method

Another way to construct a random variable with a prescribed CDF F is to start from a uniformly distributed random variable $U \sim \text{Unif}(0, 1)$.

Now, define

$$X := F^{-1}(U),$$

where F^{-1} denotes the inverse function when it exists, or the generalized inverse for a general CDF.

The idea is that applying F^{-1} to a uniform random variable transforms its distribution into the prescribed distribution F.

Thus, the inverse transform method allows us to construct a random variable with any prescribed CDF F.

Lemma 3.7 — The σ-Algebra Generated by a Random Variable

Statement

Let $Y : \Omega \to \mathbb{R}$ be a random variable, and consider the class

$$\pi(Y) := \big\{\{\omega \in \Omega : Y(\omega) \le x\} : x \in \mathbb{R}\big\} = Y^{-1}(\pi(\mathbb{R})),$$

where

$$\pi(\mathbb{R}) := \{(-\infty, x] : x \in \mathbb{R}\}.$$

Then

$$\sigma(Y) = \sigma(\pi(Y)).$$

Expanded Proof of Lemma 3.7

As we will see in HW 35, $\pi(Y)$ is a π-system on Ω.

We prove the two inclusions.

First, we prove that

$$\sigma(Y) \subseteq \sigma(\pi(Y)).$$

By definition of the pre-image of a set, for every $x \in \mathbb{R}$,

$$Y^{-1}((-\infty, x]) = \{\omega \in \Omega : Y(\omega) \in (-\infty, x]\} = \{\omega \in \Omega : Y(\omega) \leq x\} = \{Y \leq x\},$$

where the second equality follows from the fact that $Y(\omega) \in (-\infty, x]$ if and only if $Y(\omega) \leq x$.
Since $\{Y \leq x\}$ is precisely one of the sets defining $\pi(Y)$, we obtain

$$Y^{-1}((-\infty, x]) = \{Y \leq x\} \in \pi(Y).$$

Since $\sigma(\pi(Y))$ is the smallest σ-algebra containing $\pi(Y)$, we have

$$\pi(Y) \subseteq \sigma(\pi(Y)).$$

Consequently,

$$Y^{-1}((-\infty, x]) \in \sigma(\pi(Y)) \qquad \forall x \in \mathbb{R}.$$

In other words, the pre-image under Y of every set belonging to $\pi(\mathbb{R})$ belongs to $\sigma(\pi(Y))$.
Moreover, recall that, by Proposition 1.1,

$$\sigma(\pi(\mathbb{R})) = \mathcal{B}(\mathbb{R}).$$

Thus, all the assumptions of Lemma 3.2 are satisfied.
Therefore, by Lemma 3.2, Y is measurable with respect to $\sigma(\pi(Y))$, namely

$$Y \in m\sigma(\pi(Y)).$$

Now, recall that $\sigma(Y)$ is the smallest σ-algebra on Ω with respect to which Y is measurable.
Since $\sigma(\pi(Y))$ is a σ-algebra with respect to which Y is measurable, by the minimality of $\sigma(Y)$, we conclude that

$$\sigma(Y) \subseteq \sigma(\pi(Y)).$$

We now prove the reverse inclusion, namely

$$\sigma(\pi(Y)) \subseteq \sigma(Y).$$

Recall that

$$\sigma(Y) = Y^{-1}(\mathcal{B}(\mathbb{R})).$$

We first prove that

$$\pi(Y) \subseteq \sigma(Y).$$

Let $A \in \pi(Y)$.
By definition of $\pi(Y)$, there exists $x_0 \in \mathbb{R}$ such that

$$A = \{Y \leq x_0\}.$$

As shown above, by definition of pre-image,

$$A = \{Y \leq x_0\} = Y^{-1}((-\infty, x_0]).$$

Notice that $(-\infty, x_0]$ is a Borel set, since it is closed in $\mathbb{R}$. Therefore,

$$(-\infty, x_0] \in \mathcal{B}(\mathbb{R}).$$

Hence, by definition of $\sigma(Y)$,

$$A = Y^{-1}((-\infty, x_0]) \in \sigma(Y).$$

Since $A \in \pi(Y)$ was arbitrary, we conclude that

$$\pi(Y) \subseteq \sigma(Y).$$

Now, since $\sigma(Y)$ is a σ-algebra containing $\pi(Y)$, and $\sigma(\pi(Y))$ is the smallest σ-algebra containing $\pi(Y)$, we obtain

$$\sigma(\pi(Y)) \subseteq \sigma(Y).$$

Combining the two inclusions, we conclude that

$$\sigma(Y) = \sigma(\pi(Y)).$$

HW 35 — Prove That $\pi(Y)$ Is a π-System on Ω

Statement

Let $Y : \Omega \to \mathbb{R}$ be a random variable, and consider

$$\pi(Y) := \{\{Y \leq x\} : x \in \mathbb{R}\}.$$

Prove that $\pi(Y)$ is a π-system on Ω.

Proof

Recall that a π-system is a non-empty collection of sets closed under finite intersections.
It is trivial that $\pi(Y)$ is non-empty.
We now prove that $\pi(Y)$ is closed under finite intersections.
Let $A, B \in \pi(Y)$. By definition, there exist $x, y \in \mathbb{R}$ such that

$$A = \{Y \leq x\}, \qquad B = \{Y \leq y\}.$$

Recall that, as shown in HW 6,

$$(-\infty, x] \cap (-\infty, y] = (-\infty, \min\{x, y\}].$$

Therefore,

$$\begin{aligned} A \cap B &= \{Y \leq x\} \cap \{Y \leq y\} = Y^{-1}((-\infty, x]) \cap Y^{-1}((-\infty, y]) \\ &= Y^{-1}((-\infty, x] \cap (-\infty, y]) = Y^{-1}((-\infty, \min\{x, y\}]) = \{Y \leq \min\{x, y\}\}, \end{aligned}$$

where the third equality follows from HW 28-**(iii)**, and the fourth from HW 6.
Since $\min\{x, y\} \in \mathbb{R}$, by definition of $\pi(Y)$,

$$\{Y \leq \min\{x, y\}\} \in \pi(Y).$$

Hence,

$$A \cap B \in \pi(Y).$$

Since $A, B \in \pi(Y)$ were arbitrary, $\pi(Y)$ is closed under finite intersections.
Therefore,

$$\pi(Y) \text{ is a } \pi\text{-system on } \Omega.$$

The statement is proved.

Properties — Measurable Functions of Random Variables

Here are some useful facts that we state without proof.
(a) Let $Y : \Omega \to \mathbb{R}$ and $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y) \iff \exists f \in m\mathcal{B}(\mathbb{R}) \text{ such that } Z = f(Y),$$

where $f : \mathbb{R} \to \mathbb{R}$.
(b) Let $Y : \Omega \to \mathbb{R}^n$, i.e.,

$$Y(\omega) = (Y_1(\omega), Y_2(\omega), \ldots, Y_n(\omega)),$$

and let $Z : \Omega \to \mathbb{R}$. We have

$$Z \in m\sigma(Y) \iff \exists f \in m\mathcal{B}(\mathbb{R}^n) \text{ such that } Z = f(Y),$$

where $f : \mathbb{R}^n \to \mathbb{R}$.

(c) Let $(Y_\gamma)_{\gamma \in I}$ be a collection of functions $Y_\gamma : \Omega \to \mathbb{R}$, and let $Z : \Omega \to \mathbb{R}$. We have

$$\begin{aligned} Z \in m\sigma(Y_\gamma, \gamma \in I) \iff & \text{ there exist a countable collection} \\ & (\gamma_n)_{n\in\mathbb{N}} \text{ and a function} \\ & f \in m\mathcal{B}(\mathbb{R}^{\mathbb{N}}) \\ & \text{such that } Z = f((Y_{\gamma_n})_{n\in\mathbb{N}}), \end{aligned}$$

where $f : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}$.

Financial Interpretation

Item (b) can be interpreted as follows.

Let $(Y_1, \ldots, Y_n)$ be n assets traded on the market, and let Z be the payoff of a derivative contract written on the vector $(Y_1, \ldots, Y_n)$.

Then there must exist a function f such that

$$Z = f(Y_1, \ldots, Y_n),$$

which describes the random payoff Z of the derivative contract.

4. Independence

Definition 4.1 — Independent Sigma-Algebras

Sub-sigma-algebras $\mathcal{G}$ and $\mathcal{H}$ of $\mathcal{F}$ are independent when

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H) \qquad \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

For a family of sub-sigma-algebras, the corresponding factorisation is required for every finite selection of distinct members and one event from each selected member.

Definition 4.2 — Independent Random Variables

Random variables are independent when their generated sigma-algebras are independent. In particular, the threshold criteria below test this property using the classes $\pi(X)$ introduced in Lemma 3.7.

Definition 4.3 — Independent Events

Events are independent when the sigma-algebras generated by their indicator functions are independent. HW 36 describes each of these sigma-algebras explicitly.

HW 36 — σ-Algebra Generated by an Indicator Function

Statement

Let $E \in \mathcal{F}$. Prove that

$$\sigma(\mathbf{1}_E) = \{\emptyset, \Omega, E, E^c\}.$$

Proof

Recall that, by Lemma 3.7,

$$\sigma(\mathbf{1}_E) = \sigma(\pi(\mathbf{1}_E)),$$

where

$$\pi(\mathbf{1}_E) = \{\{\mathbf{1}_E \leq x\} : x \in \mathbb{R}\}.$$

Since the indicator function takes only the values 0 and 1, we have

$$\{\mathbf{1}_E \leq x\} = \begin{cases} \emptyset, & x < 0, \\ E^c, & 0 \leq x < 1, \\ \Omega, & x \geq 1. \end{cases}$$

Therefore,

$$\pi(\mathbf{1}_E) = \{\emptyset, E^c, \Omega\}.$$

Since every σ-algebra is closed under complements, we must also include

$$E = (E^c)^c.$$

Notice that the collection

$$\mathcal{G} := \{\emptyset, \Omega, E, E^c\}$$

is already a σ-algebra on Ω, since it contains Ω, is closed under complements and countable unions, and consequently also under intersections.
Therefore, no additional sets are needed, and we obtain

$$\sigma(\pi(\mathbf{1}_E)) = \{\emptyset, \Omega, E, E^c\}.$$

Finally, by Lemma 3.7,

$$\sigma(\mathbf{1}_E) = \sigma(\pi(\mathbf{1}_E)) = \{\emptyset, \Omega, E, E^c\}.$$

Therefore, the statement is proved.

Connection with Independence of Events

This tells us how to calculate all elements in $\sigma(\mathbf{1}_E)$. In particular, given two independent events $E, F \in \mathcal{F}$, according to the definition above we have

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F),$$
$$\mathbb{P}(E^c \cap F) = \mathbb{P}(E^c)\mathbb{P}(F),$$
$$\mathbb{P}(E^c \cap F^c) = \mathbb{P}(E^c)\mathbb{P}(F^c),$$
$$\mathbb{P}(E \cap F^c) = \mathbb{P}(E)\mathbb{P}(F^c).$$

Plus analogous relations involving $\emptyset$ and Ω. So here we recognise the "old" notion of independence!
Next comes a lemma that will further allow us to deepen our understanding of independence for random variables.

Lemma 4.1 — Independence of σ-Algebras Generated by π-Systems

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{G}$ and $\mathcal{H}$ be sub-σ-algebras of $\mathcal{F}$, and let $\mathcal{I}$ and $\mathcal{J}$ be π-systems such that

$$\sigma(\mathcal{I}) = \mathcal{G}, \qquad \sigma(\mathcal{J}) = \mathcal{H}.$$

Then the σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent if and only if

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) \qquad \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

Expanded Proof of Lemma 4.1

($\Rightarrow$) Assume that $\mathcal{G}$ and $\mathcal{H}$ are independent.
By Definition 4.1,

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H) \qquad \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

Since

$$\mathcal{I} \subseteq \sigma(\mathcal{I}) = \mathcal{G}, \qquad \mathcal{J} \subseteq \sigma(\mathcal{J}) = \mathcal{H},$$

the equality holds, in particular, for every $I \in \mathcal{I}$ and $J \in \mathcal{J}$. Therefore,

$$\mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) \qquad \forall I \in \mathcal{I},\ J \in \mathcal{J}.$$

($\Leftarrow$) Assume the factorisation in the statement.
First, fix an arbitrary $I \in \mathcal{I}$, and consider the two set functions defined on $(\Omega, \mathcal{H})$ by

$$\mu_1(H) := \mathbb{P}(I \cap H), \qquad \mu_2(H) := \mathbb{P}(I)\mathbb{P}(H), \quad H \in \mathcal{H}.$$

As we will see in HW 37, these two set functions are measures on $(\Omega, \mathcal{H})$.
Notice that they have the same finite total mass. Indeed,

$$\mu_1(\Omega) = \mathbb{P}(I \cap \Omega) = \mathbb{P}(I) = \mathbb{P}(I)\mathbb{P}(\Omega) = \mu_2(\Omega) < \infty,$$

where the second equality follows from $I \cap \Omega = I$, and the third from $\mathbb{P}(\Omega) = 1$.
Moreover, we prove that the two measures coincide on the π-system $\mathcal{J}$.
Let $J \in \mathcal{J}$. By our initial assumption,

$$\mu_1(J) = \mathbb{P}(I \cap J) = \mathbb{P}(I)\mathbb{P}(J) = \mu_2(J),$$

where the second equality follows precisely from the assumption of this implication.
Since $J \in \mathcal{J}$ was arbitrary, we obtain

$$\mu_1(J) = \mu_2(J) \qquad \forall J \in \mathcal{J}.$$

Thus, the two measures have the same finite total mass and coincide on the π-system $\mathcal{J}$.
Since $\sigma(\mathcal{J}) = \mathcal{H}$, all the assumptions of Lemma 1.2 are satisfied. Therefore, by Lemma 1.2, the two measures coincide on the entire σ-algebra $\mathcal{H}$:

$$\mu_1(H) = \mu_2(H) \qquad \forall H \in \mathcal{H}.$$

Substituting their definitions, and recalling that $I \in \mathcal{I}$ was arbitrary, we conclude that

$$\mathbb{P}(I \cap H) = \mathbb{P}(I)\mathbb{P}(H) \qquad \forall I \in \mathcal{I},\ H \in \mathcal{H}.$$

We now fix an arbitrary $H \in \mathcal{H}$ and consider the two measures on $(\Omega, \mathcal{G})$ defined by

$$\nu_1(G) := \mathbb{P}(G \cap H), \qquad \nu_2(G) := \mathbb{P}(G)\mathbb{P}(H), \quad G \in \mathcal{G}.$$

By the same argument as above, these two measures have the same finite total mass:

$$\nu_1(\Omega) = \mathbb{P}(H) = \nu_2(\Omega) < \infty.$$

Moreover, by the equality established in the first part of the proof, they coincide on the π-system $\mathcal{I}$:

$$\nu_1(I) = \nu_2(I) \qquad \forall I \in \mathcal{I}.$$

Therefore, since $\sigma(\mathcal{I}) = \mathcal{G}$, by Lemma 1.2 the two measures also coincide on the entire σ-algebra $\mathcal{G}$:

$$\nu_1(G) = \nu_2(G) \qquad \forall G \in \mathcal{G}.$$

Substituting their definitions, and recalling that $H \in \mathcal{H}$ was arbitrary, we conclude that

$$\mathbb{P}(G \cap H) = \mathbb{P}(G)\mathbb{P}(H) \qquad \forall G \in \mathcal{G},\ H \in \mathcal{H}.$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent.

HW 37 — Prove That Two Set Functions Define Measures

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\mathcal{H} \subseteq \mathcal{F}$ be a σ-algebra, and let $I \in \mathcal{F}$.
Prove that the two set functions

$$H \longmapsto \mathbb{P}(I \cap H), \qquad H \longmapsto \mathbb{P}(I)\mathbb{P}(H)$$

define two measures on $(\Omega, \mathcal{H})$.

Expanded Proof of HW 37

Write μ_1 and μ_2 for the first and second set functions in the statement, respectively.
First, we prove that μ_1 is a measure.
Let $H \in \mathcal{H}$. Since $\mathcal{H} \subseteq \mathcal{F}$, we have $H \in \mathcal{F}$. Moreover, since $I \in \mathcal{F}$ and $\mathcal{F}$ is closed under intersections,

$$I \cap H \in \mathcal{F}.$$

Therefore, $\mu_1(H) = \mathbb{P}(I \cap H)$ is well-defined and non-negative:

$$\mu_1(H) \geq 0 \qquad \forall H \in \mathcal{H}.$$

Furthermore,

$$\mu_1(\emptyset) = \mathbb{P}(I \cap \emptyset) = \mathbb{P}(\emptyset) = 0.$$

We now prove countable additivity.
Let $(H_n)_{n \in \mathbb{N}} \subseteq \mathcal{H}$ be a sequence of pairwise disjoint sets, namely

$$H_i \cap H_j = \emptyset \qquad \forall i \neq j.$$

Notice that the sets $I \cap H_n$ are also pairwise disjoint. Indeed, for every $i \neq j$,

$$(I \cap H_i) \cap (I \cap H_j) = I \cap (H_i \cap H_j) = I \cap \emptyset = \emptyset.$$

Moreover, since $I \in \mathcal{F}$ and $H_n \in \mathcal{H} \subseteq \mathcal{F}$, we have $I \cap H_n \in \mathcal{F}$ for every $n \in \mathbb{N}$.
Therefore,

$$\mu_1\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}\left(I \cap \bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} (I \cap H_n)\right) = \sum_{n=1}^{\infty} \mathbb{P}(I \cap H_n) = \sum_{n=1}^{\infty} \mu_1(H_n),$$

where the second equality follows from the distributive property of intersections over unions, and the third from the countable additivity of $\mathbb{P}$, since the sets $I \cap H_n$ are measurable and pairwise disjoint.
Thus, μ_1 is a measure on $(\Omega, \mathcal{H})$.
We now prove that μ_2 is also a measure.
Since $\mathbb{P}(I) \geq 0$ and $\mathbb{P}$ is a probability measure, we immediately have

$$\mu_2(H) \geq 0 \qquad \forall H \in \mathcal{H},$$

and

$$\mu_2(\emptyset) = \mathbb{P}(I)\mathbb{P}(\emptyset) = 0.$$

Moreover, let $(H_n)_{n\in\mathbb{N}} \subseteq \mathcal{H}$ be a sequence of pairwise disjoint sets. By the countable additivity of $\mathbb{P}$,

$$\mu_2\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}(I)\mathbb{P}\left(\bigcup_{n=1}^{\infty} H_n\right) = \mathbb{P}(I)\sum_{n=1}^{\infty} \mathbb{P}(H_n) = \sum_{n=1}^{\infty} \mathbb{P}(I)\mathbb{P}(H_n) = \sum_{n=1}^{\infty} \mu_2(H_n).$$

Therefore, μ_2 is also a measure on $(\Omega, \mathcal{H})$.
We conclude that both set functions

$$\begin{aligned} \mu_1(H) &= \mathbb{P}(I \cap H), \\ \mu_2(H) &= \mathbb{P}(I)\mathbb{P}(H), \end{aligned} \qquad H \in \mathcal{H}$$

define measures on $(\Omega, \mathcal{H})$.

Corollary — Characterisation of Independence via Threshold Events

Let $X, Y : \Omega \to \mathbb{R}$ be two random variables.
Then X and Y are independent if and only if

$$\mathbb{P}(X \leq x,\ Y \leq y) = \mathbb{P}(X \leq x)\,\mathbb{P}(Y \leq y) \qquad \forall x, y \in \mathbb{R}.$$

Indeed, this is a direct application of Lemma 4.1 with

$$\mathcal{I} := \pi(X) = \big\{\{\omega \in \Omega : X(\omega) \leq x\} : x \in \mathbb{R}\big\},$$

and

$$\mathcal{J} := \pi(Y) = \big\{\{\omega \in \Omega : Y(\omega) \leq y\} : y \in \mathbb{R}\big\}.$$

Recall that, by Lemma 3.7,

$$\sigma(\pi(X)) = \sigma(X), \qquad \sigma(\pi(Y)) = \sigma(Y).$$

Therefore, the independence of the random variables X and Y is equivalent to the factorisation of the probabilities of the events $\{X \leq x\}$ and $\{Y \leq y\}$ for every $x, y \in \mathbb{R}$.

Generalisation — Independence of Finitely Many Random Variables

More generally, random variables $X_1, \dots, X_n$ are independent if and only if, for every $x_1, \dots, x_n \in \mathbb{R}$,

$$\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \le x_i).$$

This is the analogous finite-dimensional characterisation of independence, obtained by applying the same idea to the π-systems generated by the threshold events of the variables $X_1, \dots, X_n$.

HW 38 — Independence of Three σ-Algebras Generated by π-Systems

(Williams, E4.1, p. 225.)

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3$ be three π-systems such that

$$\mathcal{I}_k \subseteq \mathcal{F}, \qquad \Omega \in \mathcal{I}_k, \qquad k = 1, 2, 3.$$

Prove that $\sigma(\mathcal{I}_1)$, $\sigma(\mathcal{I}_2)$ and $\sigma(\mathcal{I}_3)$ are independent if and only if

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k)$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
Why is it necessary to assume that $\Omega \in \mathcal{I}_k$, $k = 1, 2, 3$?

Part I — Expanded Proof of the Equivalence

Let

$$\mathcal{G}_k := \sigma(\mathcal{I}_k), \qquad k = 1, 2, 3.$$

($\Rightarrow$) Assume that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.
By Definition 4.1, for every $G_k \in \mathcal{G}_k$, $k = 1, 2, 3$,

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \prod_{k=1}^{3} \mathbb{P}(G_k).$$

Since

$$\mathcal{I}_k \subseteq \sigma(\mathcal{I}_k) = \mathcal{G}_k, \qquad k = 1, 2, 3,$$

the equality holds, in particular, for every $I_k \in \mathcal{I}_k$. Therefore,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k).$$

($\Leftarrow$) Assume the factorisation in the statement.
First, without loss of generality, we consider $\mathcal{G}_1$ and $\mathcal{G}_2$, since the three indices play symmetric roles. In particular, since $\Omega \in \mathcal{I}_3$, we can choose $I_3 = \Omega$ in our assumption.
Therefore, for every $I_1 \in \mathcal{I}_1$ and $I_2 \in \mathcal{I}_2$,

$$\mathbb{P}(I_1 \cap I_2) = \mathbb{P}(I_1 \cap I_2 \cap \Omega) = \mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(\Omega) = \mathbb{P}(I_1)\mathbb{P}(I_2),$$

where the second equality follows from our assumption, and the third from $\mathbb{P}(\Omega) = 1$.
Since $\mathcal{I}_1$ and $\mathcal{I}_2$ are π-systems generating $\mathcal{G}_1$ and $\mathcal{G}_2$, respectively, by Lemma 4.1 we conclude that

$$\mathcal{G}_1 \text{ and } \mathcal{G}_2 \text{ are independent.}$$

We now introduce the collection

$$\mathcal{I}_{1,2} := \{I_1 \cap I_2 : I_1 \in \mathcal{I}_1,\ I_2 \in \mathcal{I}_2\}.$$

We first prove that $\mathcal{I}_{1,2}$ is a π-system.
Let $A, B \in \mathcal{I}_{1,2}$. By definition, there exist $I_1, J_1 \in \mathcal{I}_1$ and $I_2, J_2 \in \mathcal{I}_2$ such that

$$A = I_1 \cap I_2, \qquad B = J_1 \cap J_2.$$

Therefore,

$$A \cap B = (I_1 \cap I_2) \cap (J_1 \cap J_2) = (I_1 \cap J_1) \cap (I_2 \cap J_2),$$

where the second equality follows from the associative and commutative properties of intersections.
Since $\mathcal{I}_1$ and $\mathcal{I}_2$ are π-systems,

$$I_1 \cap J_1 \in \mathcal{I}_1, \qquad I_2 \cap J_2 \in \mathcal{I}_2.$$

Hence,

$$A \cap B \in \mathcal{I}_{1,2}.$$

Moreover, $\Omega = \Omega \cap \Omega \in \mathcal{I}_{1,2}$, since $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$.
Thus, $\mathcal{I}_{1,2}$ is a non-empty collection closed under intersections, and therefore a π-system.
Notice also that, since $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$,

$$I_1 = I_1 \cap \Omega \in \mathcal{I}_{1,2} \qquad \forall I_1 \in \mathcal{I}_1,$$

and

$$I_2 = \Omega \cap I_2 \in \mathcal{I}_{1,2} \qquad \forall I_2 \in \mathcal{I}_2.$$

Therefore,

$$\mathcal{I}_1 \subseteq \mathcal{I}_{1,2}, \qquad \mathcal{I}_2 \subseteq \mathcal{I}_{1,2}.$$

Since $\sigma(\mathcal{I}_{1,2})$ is a σ-algebra containing both $\mathcal{I}_1$ and $\mathcal{I}_2$, by the minimality of generated σ-algebras we obtain

$$\mathcal{G}_1 \subseteq \sigma(\mathcal{I}_{1,2}), \qquad \mathcal{G}_2 \subseteq \sigma(\mathcal{I}_{1,2}).$$

We now prove that $\sigma(\mathcal{I}_{1,2})$ and $\mathcal{G}_3$ are independent.
Let $A \in \mathcal{I}_{1,2}$ and $I_3 \in \mathcal{I}_3$. By definition, there exist $I_1 \in \mathcal{I}_1$ and $I_2 \in \mathcal{I}_2$ such that

$$A = I_1 \cap I_2.$$

Therefore,

$$\mathbb{P}(A \cap I_3) = \mathbb{P}(I_1 \cap I_2 \cap I_3) = \mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(I_3) = \mathbb{P}(I_1 \cap I_2)\mathbb{P}(I_3) = \mathbb{P}(A)\mathbb{P}(I_3),$$

where the second equality follows from our initial assumption, and the third from the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$, established above.
Since $A \in \mathcal{I}_{1,2}$ and $I_3 \in \mathcal{I}_3$ were arbitrary,

$$\mathbb{P}(A \cap I_3) = \mathbb{P}(A)\mathbb{P}(I_3),$$
$$\forall A \in \mathcal{I}_{1,2},\ I_3 \in \mathcal{I}_3.$$

Since $\mathcal{I}_{1,2}$ and $\mathcal{I}_3$ are π-systems, by Lemma 4.1 we conclude that

$$\sigma(\mathcal{I}_{1,2}) \text{ and } \mathcal{G}_3 \text{ are independent.}$$

Finally, let

$$G_k \in \mathcal{G}_k, \qquad k = 1, 2, 3.$$

Since $\mathcal{G}_1, \mathcal{G}_2 \subseteq \sigma(\mathcal{I}_{1,2})$, and $\sigma(\mathcal{I}_{1,2})$ is closed under intersections, we have

$$G_1 \cap G_2 \in \sigma(\mathcal{I}_{1,2}).$$

Therefore,

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \mathbb{P}((G_1 \cap G_2) \cap G_3) = \mathbb{P}(G_1 \cap G_2)\mathbb{P}(G_3) = \mathbb{P}(G_1)\mathbb{P}(G_2)\mathbb{P}(G_3),$$

where the second equality follows from the independence of $\sigma(\mathcal{I}_{1,2})$ and $\mathcal{G}_3$, and the third from the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$.
Since $G_k \in \mathcal{G}_k$, $k = 1, 2, 3$, were arbitrary, we conclude that

$$\mathbb{P}(G_1 \cap G_2 \cap G_3) = \prod_{k=1}^{3} \mathbb{P}(G_k),$$
$$\forall G_k \in \mathcal{G}_k.$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are independent.
Therefore, the equivalence is proved.

HW 38 — Part II: Why Is the Assumption $\Omega \in \mathcal{I}_k$ Necessary?

In Part I, the assumption $\Omega \in \mathcal{I}_3$ allowed us to choose $I_3 = \Omega$ and establish the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$.
However, pairwise independence alone does not imply joint independence. We first illustrate this fact and then construct a counterexample showing why the assumption $\Omega \in \mathcal{I}_k$ is needed.

Counterexample 1 — Pairwise Independence Does Not Imply Joint Independence

Let X_1 and X_2 be two independent and identically distributed random variables such that

$$\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}, \qquad i = 1, 2.$$

Define

$$X_3 := X_1 X_2.$$

We first prove that X_1, X_2, X_3 are pairwise independent.
By assumption, X_1 and X_2 are independent. Therefore, it remains to consider the pairs (X_1, X_3) and (X_2, X_3).
Notice that $X_3 = 1$ if and only if X_1 and X_2 have the same sign. Thus,

$$\{X_3 = 1\} = \big(\{X_1 = 1\} \cap \{X_2 = 1\}\big) \cup \big(\{X_1 = -1\} \cap \{X_2 = -1\}\big).$$

Since these two events are disjoint, by additivity and the independence of X_1 and X_2,

$$\begin{aligned}\mathbb{P}(X_3 = 1) &= \mathbb{P}(X_1 = 1, X_2 = 1) \quad + \mathbb{P}(X_1 = -1, X_2 = -1) \\ &= \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) \quad + \mathbb{P}(X_1 = -1)\mathbb{P}(X_2 = -1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.\end{aligned}$$

We now prove that X_1 and X_3 are independent.
Since each X_i takes only the values -1 and 1, we have

$$\sigma(X_i) = \{\emptyset, \Omega, \{X_i = 1\}, \{X_i = -1\}\}, \qquad i = 1, 2, 3.$$

Therefore, the collection

$$\mathcal{J}_i := \big\{\{X_i = 1\}\big\}$$

is a π-system generating $\sigma(X_i)$. Indeed, it is closed under intersections, and its generated σ-algebra also contains

$$\{X_i = 1\}^c = \{X_i = -1\}.$$

Thus, by Lemma 4.1, it is sufficient to prove that

$$\mathbb{P}(X_1 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1)\mathbb{P}(X_3 = 1).$$

Notice that, since $X_3 = X_1 X_2$,

$$\{X_1 = 1, X_3 = 1\} = \{X_1 = 1, X_2 = 1\}.$$

Indeed, when $X_1 = 1$, the equality $X_3 = 1$ holds if and only if $X_2 = 1$.
Therefore,

$$\mathbb{P}(X_1 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_1 = 1)\mathbb{P}(X_3 = 1),$$

where the second equality follows from independence, and the last from $\mathbb{P}(X_3 = 1) = 1/2$, established above.
Hence, by Lemma 4.1, $\sigma(X_1)$ and $\sigma(X_3)$ are independent.
Similarly, exchanging X_1 and X_2, we obtain

$$\mathbb{P}(X_2 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4} = \mathbb{P}(X_2 = 1)\mathbb{P}(X_3 = 1).$$

Therefore, again by Lemma 4.1, $\sigma(X_2)$ and $\sigma(X_3)$ are independent.
We conclude that

$$X_1, X_2, X_3 \text{ are pairwise independent.}$$

We now prove that they are not jointly independent.
Since $X_3 = X_1X_2$, we have

$$\{X_1 = 1, X_2 = 1, X_3 = 1\} = \{X_1 = 1, X_2 = 1\}.$$

Therefore,

$$\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) = \frac{1}{4}.$$

On the other hand,

$$\prod_{i=1}^{3} \mathbb{P}(X_i = 1) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Since $\frac{1}{4} \neq \frac{1}{8}$, we obtain

$$\mathbb{P}(X_1 = 1, X_2 = 1, X_3 = 1) \neq \prod_{i=1}^{3} \mathbb{P}(X_i = 1).$$

Thus, by Definition 4.1, the three σ-algebras $\sigma(X_1)$, $\sigma(X_2)$ and $\sigma(X_3)$ are not jointly independent.
Therefore, this counterexample shows that pairwise independence does not imply joint independence.
In particular, it explains why, in Part I of HW 38, establishing the independence of $\mathcal{G}_1$ and $\mathcal{G}_2$ was not sufficient.

Counterexample 2 — Necessity of $\Omega \in \mathcal{I}_k$

We now prove that, without the assumption $\Omega \in \mathcal{I}_k$, the factorisation on the three π-systems does not necessarily imply the independence of the generated σ-algebras.
Consider the probability space

$$\Omega = \{0, 1\}, \qquad \mathcal{F} = \mathcal{P}(\Omega),$$

with

$$\mathbb{P}(\{0\}) = \mathbb{P}(\{1\}) = \frac{1}{2}.$$

Let $A := \{1\}$, so that

$$\mathbb{P}(A) = \frac{1}{2}.$$

Define

$$\mathcal{I}_1 = \mathcal{I}_2 = \{\Omega, A\}, \qquad \mathcal{I}_3 = \{\emptyset\}.$$

First, notice that these collections are π-systems.
Indeed, $\{\Omega, A\}$ is closed under intersections, since

$$\Omega \cap A = A, \qquad A \cap A = A, \qquad \Omega \cap \Omega = \Omega.$$

Moreover, $\{\emptyset\}$ is a non-empty collection closed under intersections, because

$$\emptyset \cap \emptyset = \emptyset.$$

However, although $\Omega \in \mathcal{I}_1$ and $\Omega \in \mathcal{I}_2$,

$$\Omega \notin \mathcal{I}_3.$$

We first prove that the factorisation in HW 38 holds on these three π-systems.
Let $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
Since $\mathcal{I}_3 = \{\emptyset\}$, necessarily $I_3 = \emptyset$.
Therefore,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \mathbb{P}(I_1 \cap I_2 \cap \emptyset) = \mathbb{P}(\emptyset) = 0,$$

where the second equality follows from the fact that the intersection of any set with the empty set is empty.
On the other hand,

$$\mathbb{P}(I_1)\mathbb{P}(I_2)\mathbb{P}(I_3) = \mathbb{P}(I_1)\mathbb{P}(I_2) \cdot 0 = 0.$$

Consequently,

$$\mathbb{P}(I_1 \cap I_2 \cap I_3) = \prod_{k=1}^{3} \mathbb{P}(I_k).$$

for every $I_k \in \mathcal{I}_k$, $k = 1, 2, 3$.
We now prove that the generated σ-algebras are not independent.
Let

$$\mathcal{G}_k := \sigma(\mathcal{I}_k), \qquad k = 1, 2, 3.$$

Since $A \in \mathcal{I}_1$ and $A \in \mathcal{I}_2$, we have

$$A \in \mathcal{G}_1, \qquad A \in \mathcal{G}_2.$$

Moreover, although $\Omega \notin \mathcal{I}_3$, we necessarily have

$$\Omega \in \mathcal{G}_3,$$

because every σ-algebra on Ω contains Ω.
Therefore, we can choose

$$G_1 = A, \qquad G_2 = A, \qquad G_3 = \Omega.$$

If the three σ-algebras were independent, by Definition 4.1 we would have

$$\mathbb{P}(A \cap A \cap \Omega) = \mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega).$$

However,

$$\mathbb{P}(A \cap A \cap \Omega) = \mathbb{P}(A) = \frac{1}{2},$$

where the first equality follows from $A \cap A \cap \Omega = A$.
On the other hand,

$$\mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega) = \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{1}{4}.$$

Since $\frac{1}{2} \neq \frac{1}{4}$, we obtain

$$\mathbb{P}(A \cap A \cap \Omega) \neq \mathbb{P}(A)\mathbb{P}(A)\mathbb{P}(\Omega).$$

Thus, by Definition 4.1, the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are not independent.

Notice that this does not contradict the factorisation established above. Indeed, that factorisation holds only for events belonging to the original π-systems. Since $\Omega \notin \mathcal{I}_3$, we could not choose $I_3 = \Omega$ there, whereas Ω necessarily belongs to the generated σ-algebra $\mathcal{G}_3$.

Therefore, the factorisation holds on the three π-systems but fails on their generated σ-algebras.

Hence, the assumption $\Omega \in \mathcal{I}_k$, $k = 1, 2, 3$, is necessary to guarantee the equivalence in HW 38.

Therefore, the statement is proved.

Lemma 4.2 — Second Borel–Cantelli's Lemma

Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be a sequence of independent events such that

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) = +\infty.$$

Then

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 1.$$

In other words, with probability one, infinitely many of the events E_n occur.

Expanded Proof of Lemma 4.2

Set

$$E := \limsup_{n\to\infty} E_n.$$

By definition of the limit superior of a sequence of sets and De Morgan's laws, we have

$$E^c = \left(\bigcap_{n=1}^{\infty} \bigcup_{m\geq n} E_m\right)^c = \bigcup_{n=1}^{\infty} \left(\bigcup_{m\geq n} E_m\right)^c = \bigcup_{n=1}^{\infty} \bigcap_{m\geq n} E_m^c = \liminf_{n\to\infty} E_n^c,$$

where the second and third equalities follow from De Morgan's laws, and the last from the definition of the limit inferior.

We want to prove that $\mathbb{P}(E^c) = 0$, which will imply $\mathbb{P}(E) = 1$.

Set

$$p_n := \mathbb{P}(E_n), \qquad n \in \mathbb{N}.$$

By assumption,

$$\sum_{n=1}^{\infty} p_n = +\infty.$$

Fix an arbitrary $n \in \mathbb{N}$. We first calculate

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right).$$

For every $k \geq 0$, define

$$A_k := \bigcap_{m=n}^{n+k} E_m^c.$$

Notice that $(A_k)_{k\geq 0}$ is a decreasing sequence of measurable sets. Indeed,

$$A_{k+1} = A_k \cap E_{n+k+1}^c \subseteq A_k.$$

Moreover,

$$\bigcap_{k=0}^{\infty} A_k = \bigcap_{m\geq n} E_m^c.$$

Since $\mathbb{P}$ is a probability measure, we also have

$$\mathbb{P}(A_0) \leq \mathbb{P}(\Omega) = 1 < \infty.$$

Thus, all the assumptions of Lemma 1.6 are satisfied. Therefore, by continuity of measures from above,

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) = \lim_{k\to\infty} \mathbb{P}\left(\bigcap_{m=n}^{n+k} E_m^c\right).$$

Now, since the events $(E_n)_{n\in\mathbb{N}}$ are independent, their complements are also independent. Indeed, by Definition 4.3 and HW 36,

$$E_m^c \in \sigma(\mathbf{1}_{E_m}) \qquad \forall m \in \mathbb{N},$$

and the σ-algebras generated by the indicator functions of the independent events are themselves independent.

Consequently, for every finite $k \geq 0$,

$$\mathbb{P}\left(\bigcap_{m=n}^{n+k} E_m^c\right) = \prod_{m=n}^{n+k} \mathbb{P}(E_m^c) = \prod_{m=n}^{n+k} (1 - p_m),$$

where the first equality follows from the independence of the complementary events, and the second from

$$\mathbb{P}(E_m^c) = 1 - \mathbb{P}(E_m) = 1 - p_m.$$

Combining this equality with continuity from above, we obtain

$$\mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) = \lim_{k\to\infty} \prod_{m=n}^{n+k} (1 - p_m).$$

We now use the inequality

$$1 - x \leq e^{-x}, \qquad x \geq 0.$$

Since $p_m = \mathbb{P}(E_m) \in [0, 1]$, applying this inequality with $x = p_m$ gives

$$0 \leq 1 - p_m \leq e^{-p_m}.$$

Therefore, multiplying these inequalities for $m = n, \ldots, n + k$, we obtain

$$\prod_{m=n}^{n+k} (1 - p_m) \leq \prod_{m=n}^{n+k} e^{-p_m} = \exp\left(-\sum_{m=n}^{n+k} p_m\right),$$

where the last equality follows from the properties of the exponential function.

Consequently,

$$0 \leq \mathbb{P}\left(\bigcap_{m\geq n} E_m^c\right) \leq \lim_{k\to\infty} \exp\left(-\sum_{m=n}^{n+k} p_m\right).$$

Notice that, since

$$\sum_{m=1}^{\infty} p_m = +\infty$$

and each $p_m \in [0,1]$, removing finitely many terms does not change the divergence of the series. Hence, for every fixed $n \in \mathbb{N}$,

$$\sum_{m=n}^{\infty} p_m = +\infty.$$

Therefore,

$$\lim_{k\to\infty} \exp\left(-\sum_{m=n}^{n+k} p_m\right) = 0,$$

since $e^{-t} \to 0$ as $t \to +\infty$.

Thus, by the inequality established above,

$$0 \le \mathbb{P}\left(\bigcap_{m\ge n} E_m^c\right) \le 0.$$

We conclude that

$$\mathbb{P}\left(\bigcap_{m\ge n} E_m^c\right) = 0 \qquad \forall n \in \mathbb{N}.$$

Finally, recall that we established at the beginning of the proof that

$$E^c = \bigcup_{n=1}^{\infty} \bigcap_{m\ge n} E_m^c.$$

Since each event in this countable union has probability zero, by Lemma 1.7 (a countable union of null sets is a null set), we obtain

$$\mathbb{P}(E^c) = 0.$$

Therefore,

$$\mathbb{P}(E) = 1 - \mathbb{P}(E^c) = 1.$$

Recalling that $E = \limsup_{n\to\infty} E_n$, we conclude that

$$\mathbb{P}\left(\limsup_{n\to\infty} E_n\right) = 1.$$

Therefore, the statement is proved.

Example 2 (Williams, Example 4.4) — Application of the Borel–Cantelli Lemmas Statement

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables such that

$$X_n \sim \mathrm{Exp}(1), \qquad n \in \mathbb{N}.$$

For $\alpha > 0$, determine the probability of the event

$$E_\alpha := \{\omega \in \Omega : X_n(\omega) > \alpha \log n, \text{ i.o.}\},$$

where i.o. means infinitely often. Then study the almost sure behaviour of

$$L(\omega) := \limsup_{n\to\infty} \frac{X_n(\omega)}{\log n}.$$

Expanded Solution of Example 2

Fix $\alpha > 0$, and define

$$E_n^\alpha := \{\omega \in \Omega : X_n(\omega) > \alpha \log n\}, \qquad n \in \mathbb{N}.$$

The survival function of X_n satisfies

$$\mathbb{P}(X_n > t) = e^{-t}, \qquad t \geq 0.$$

This follows from $X_n \sim \mathrm{Exp}(1)$. Consequently,

$$\mathbb{P}(E_n^\alpha) = \mathbb{P}(X_n > \alpha \log n) = e^{-\alpha \log n} = n^{-\alpha}.$$

The first equality uses the definition of E_n^α, the second the exponential survival function (since $\alpha \log n \geq 0$), and the last the identity $e^{a \log n} = n^a$.

Moreover,

$$E_n^\alpha = X_n^{-1}((\alpha \log n, \infty)) \in \sigma(X_n) \subseteq \mathcal{F}.$$

The equality is the definition of a pre-image; the membership follows from the definition of $\sigma(X_n)$, and the final inclusion from the $\mathcal{F}$-measurability of X_n. Since the X_n are independent, the events $(E_n^\alpha)_{n\in\mathbb{N}}$ are independent by Definition 4.2.

We now express E_α as a limit superior:

$$\begin{aligned} E_\alpha &= \{\omega \in \Omega : X_n(\omega) > \alpha \log n \text{ for infinitely many } n\} \\ &= \{\omega \in \Omega : \forall N \in \mathbb{N},\ \exists m \geq N \text{ such that } X_m(\omega) > \alpha \log m\} \\ &= \bigcap_{N=1}^{\infty} \bigcup_{m \geq N} \{\omega \in \Omega : X_m(\omega) > \alpha \log m\} = \bigcap_{N=1}^{\infty} \bigcup_{m \geq N} E_m^\alpha = \limsup_{n\to\infty} E_n^\alpha. \end{aligned}$$

The first equality states the definition of E_α. The second expresses infinitely often: after every N there must be some $m \geq N$ for which the inequality holds. The third translates "for every N" into an intersection and "there exists $m \geq N$" into a union. The fourth substitutes the definition of E_m^α, and the last is the definition of the limit superior of sets.

We will use formula (4.3):

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

As we will prove in HW 39, this follows from the two Borel–Cantelli lemmas and the convergence criterion for the p-series $\sum_{n=1}^{\infty} n^{-\alpha}$.

We now consider

$$L(\omega) := \limsup_{n\to\infty} \frac{X_n(\omega)}{\log n},$$

where the ratios are defined for $n \geq 2$, because $\log 1 = 0$. For each $n \geq 2$, the function $X_n/\log n$ is measurable, since X_n is measurable and $\log n$ is a positive constant. Moreover,

$$L = \inf_{N\geq 2} \sup_{n \geq N} \frac{X_n}{\log n}.$$

This is the definition of the numerical limit superior; since countable suprema and infima of measurable functions are measurable, L is an extended-real-valued measurable random variable.

We first prove that $\mathbb{P}(L \geq 1) = 1$.

Let $\omega_0 \in E_1$. We have

$$\frac{X_n(\omega_0)}{\log n} > 1 \qquad \text{for infinitely many } n \geq 2.$$

Indeed, the definition of E_1 gives $X_n(\omega_0) > \log n$ infinitely often, and division by $\log n > 0$ preserves the inequality. Consequently,

$$L(\omega_0) \geq 1.$$

This follows from the definition of the limit superior: a sequence exceeding 1 infinitely often cannot have limit superior below 1. Since $\omega_0 \in E_1$ was arbitrary,

$$E_1 \subseteq \{L \geq 1\}.$$

We therefore obtain

$$1 \geq \mathbb{P}(L \geq 1) \geq \mathbb{P}(E_1) = 1.$$

The first bound holds for every probability, the second inequality follows from the inclusion just proved and monotonicity of probability, and the last equality uses formula (4.3) with $\alpha = 1$. Hence $\mathbb{P}(L \geq 1) = 1$.

We now prove that $\mathbb{P}(L > 1) = 0$.

First, notice that

$$\{L > 1\} = \bigcup_{k=1}^{\infty} \left\{ L \geq 1 + \frac{2}{k} \right\} = \bigcup_{k=1}^{\infty} A_k,$$

where, for each fixed $k \in \mathbb{N}$, we define

$$A_k := \left\{ L \geq 1 + \frac{2}{k} \right\}.$$

The first equality follows from $2/k \to 0$ and the Archimedean property of the real numbers (including the case $L = +\infty$); the second follows from the definition of A_k.

Now fix $k \in \mathbb{N}$, and define

$$B_k := \left\{ X_n > \left(1 + \frac{1}{k}\right) \log n, \text{ i.o.} \right\}.$$

We want to show that

$$A_k \subseteq B_k \quad \Longleftrightarrow \quad A_k \cap B_k^c = \emptyset.$$

The equivalence holds because an inclusion $A_k \subseteq B_k$ means that no outcome can belong to A_k without belonging to B_k.

Suppose, for a contradiction, that there exists

$$\omega_0 \in A_k \cap B_k^c.$$

From the membership $\omega_0 \in A_k$, we obtain

$$L(\omega_0) \geq 1 + \frac{2}{k}.$$

This is the definition of A_k. On the other hand,

$$X_n(\omega_0) > \left(1 + \frac{1}{k}\right) \log n \qquad \text{only finitely often.}$$

This follows from $\omega_0 \in B_k^c$: the outcome does not belong to the event where the inequality holds infinitely often. Thus, for some $N \geq 2$,

$$X_n(\omega_0) \leq \left(1 + \frac{1}{k}\right) \log n \qquad \forall n \geq N.$$

Indeed, after the last index (if any) at which the strict inequality holds, its negation must hold for every subsequent index. Dividing yields

$$\frac{X_n(\omega_0)}{\log n} \leq 1 + \frac{1}{k} \qquad \forall n \geq N.$$

The inequality is preserved because $\log n > 0$ for $n \geq 2$. Therefore,

$$L(\omega_0) \leq 1 + \frac{1}{k}.$$

This follows from the definition of limit superior, since every sufficiently late term is bounded by $1 + 1/k$, and contradicts the membership $\omega_0 \in A_k$ established above. Thus,

$$A_k \cap B_k^c = \emptyset, \qquad A_k \subseteq B_k.$$

The contradiction rules out any outcome in $A_k \cap B_k^c$, which is equivalent to the desired inclusion.

Consequently,

$$\mathbb{P}(A_k) \le \mathbb{P}(B_k) = \mathbb{P}(E_{1+1/k}) = 0.$$

The inequality follows from $A_k \subseteq B_k$ and monotonicity of probability, the first equality from the definition of E_α, and the last from formula (4.3), since $1 + 1/k > 1$.

Since k was arbitrary, we obtain

$$\mathbb{P}(L > 1) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = 0.$$

The first equality uses the identity for $\{L > 1\}$, and the second follows from Lemma 1.7 because each A_k has probability zero.

Finally,

$$\mathbb{P}(L = 1) = \mathbb{P}(L \ge 1) - \mathbb{P}(L > 1) = 1 - 0 = 1.$$

The first equality follows from the disjoint decomposition $\{L \ge 1\} = \{L = 1\} \cup \{L > 1\}$ and additivity of probability; the other equalities use the two probabilities already established. Recalling the definition of L, we conclude that

$$\mathbb{P}\left(\limsup_{n\to\infty} \frac{X_n}{\log n} = 1\right) = 1.$$

HW 39 — Proof of Formula (4.3)

Statement

Let $(X_n)_{n\in\mathbb{N}}$ be independent and identically distributed random variables with

$$X_n \sim \text{Exp}(1).$$

For $\alpha > 0$, let

$$E_\alpha := \{X_n > \alpha \log n, \text{ i.o.}\}.$$

Prove formula (4.3):

$$\mathbb{P}(E_\alpha) = \begin{cases} 0, & \alpha > 1, \\ 1, & \alpha \in (0, 1]. \end{cases}$$

Expanded Proof of HW 39

Fix $\alpha > 0$, and consider

$$E_n^\alpha := \{X_n > \alpha \log n\}, \qquad n \in \mathbb{N}.$$

As established in Example 2, these events are measurable and independent, and

$$\mathbb{P}(E_n^\alpha) = n^{-\alpha}, \qquad E_\alpha = \limsup_{n\to\infty} E_n^\alpha.$$

The first identity was obtained from the exponential survival function; the second from the meaning of infinitely often and the definition of the limit superior. Hence,

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = \sum_{n=1}^{\infty} \frac{1}{n^\alpha}.$$

The equality follows by substituting the probability of each E_n^α.

If $\alpha > 1$, the p-series criterion gives

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) < \infty.$$

Consequently,

$$\mathbb{P}(E_\alpha) = \mathbb{P}\left(\limsup_{n\to\infty} E_n^\alpha\right) = 0.$$

The first equality uses the representation of E_α, and the second follows from Lemma 2.3 (First Borel–Cantelli), whose assumptions hold because the events are measurable and their probabilities have finite sum. Independence is not required here.

If $0 < \alpha \leq 1$, the same p-series diverges, so

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n^\alpha) = +\infty.$$

At $\alpha = 1$ this is the harmonic series. Therefore,

$$\mathbb{P}(E_\alpha) = \mathbb{P}\left(\limsup_{n\to\infty} E_n^\alpha\right) = 1.$$

The first equality uses the same representation of E_α, and the second follows from Lemma 4.2 (Second Borel–Cantelli), because the events are independent and the sum of their probabilities diverges.

Both cases of (4.3) follow.

Fundamental Question for Modelling

So far, we have considered sequences of random variables with arbitrary distributions. A natural question is whether we can actually construct a sequence of independent random variables $(X_n)_{n\in\mathbb{N}}$, each with a prescribed CDF F_n.

The answer is yes. Indeed, let $(U_n)_{n\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that

$$U_n \sim \text{Unif}(0,1).$$

Define

$$X_n := F_n^{-1}(U_n), \qquad n \in \mathbb{N},$$

where F_n^{-1} denotes the generalised inverse of the CDF F_n.

Then

$$X_n \sim F_n,$$

by the inverse transform method. Moreover, the random variables $(X_n)_{n\in\mathbb{N}}$ are independent, since each X_n is a measurable function of U_n, and measurable transformations of independent random variables preserve independence.

Stochastic Processes

A stochastic process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a family of random variables

$$(Y_t)_{t\in\mathcal{I}},$$

indexed by a parameter t, usually interpreted as time.

For example, the index set $\mathcal{I}$ may be $\mathbb{N}$, corresponding to discrete time, or $[0,\infty)$, corresponding to continuous time.

Existence of Stochastic Processes

A natural question is whether a stochastic process can be constructed with a prescribed distributional structure.

Under suitable consistency conditions, the answer is yes, as established by the Kolmogorov Existence Theorem.

The technical details of this theorem are beyond the scope of the present discussion.

Example 3 — Markov Chains

Statement

Let E be a finite or countable set, and let $P = (p_{ij})_{i,j\in E}$ be a square matrix with positive entries satisfying

$$\sum_{j\in E} p_{ij} = 1, \qquad \forall i \in E.$$

Such a matrix is called a stochastic matrix.

Let μ be a probability measure on E, and set

$$\mu_i := \mu(\{i\}), \qquad i \in E.$$

A time-homogeneous Markov chain $Z = (Z_n)_{n\in\mathbb{N}\cup\{0\}}$ on E, with initial distribution μ and transition matrix P, is a stochastic process satisfying

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}$$

for every $n \in \mathbb{N}$ and every choice of states $i_0, \dots, i_n \in E$.

We want to construct such a process using independent random variables.

Expanded Construction and Proof of Example 3

First, let Z_0 be an E-valued random variable with distribution μ, so that

$$\mathbb{P}(Z_0 = i) = \mu_i, \qquad \forall i \in E.$$

Next, consider a family of E-valued random variables

$$(Y_{i,n})_{i\in E,\ n\in\mathbb{N}}$$

such that

$$\mathbb{P}(Y_{i,n} = j) = p_{ij}, \qquad \forall i, j \in E,\ n \in \mathbb{N}.$$

We assume that all the random variables $Y_{i,n}$ are mutually independent and also independent of Z_0.

Notice that this independence is an assumption of our initial construction: **the random variables $Y_{i,n}$ are chosen independently before defining the process Z.**

For each fixed state i and time n, the random variable $Y_{i,n}$ takes values in E, according to the probabilities p_{ij}. Thus, the index i identifies which random variable we are considering, whereas j represents one of its possible values.

We now define the process recursively by setting

$$Z_n(\omega) := Y_{Z_{n-1}(\omega),n}(\omega), \qquad n \geq 1.$$

This means that the value of $Z_{n-1}(\omega)$ determines which random variable $Y_{i,n}$ is selected to obtain $Z_n(\omega)$.

In particular, if $Z_0(\omega) = i_0$, then

$$Z_1(\omega) = Y_{i_0,1}(\omega).$$

The equality follows from the recursive definition with $n = 1$, after substituting $Z_0(\omega) = i_0$.

Notice that knowing $Z_0(\omega) = i_0$ identifies the random variable $Y_{i_0,1}$, **but does not determine its realised value**. We only know that

$$\mathbb{P}(Y_{i_0,1} = j) = p_{i_0 j}, \qquad j \in E.$$

If we additionally consider an outcome ω for which $Z_1(\omega) = i_1$, then

$$Z_2(\omega) = Y_{i_1,2}(\omega).$$

This follows from the same recursive definition with $n = 2$. Again, the condition $Z_1(\omega) = i_1$ identifies the variable used at the second step, without specifying its realised value.

Notice also that, although the variables $Y_{i,n}$ are independent, the resulting variables Z_n are generally not independent, since the selection of the variable used at time n depends on the previous state Z_{n-1}.

Nevertheless, the construction defines measurable random variables. Indeed, for every $j \in E$ and $n \geq 1$,

$$\{Z_n = j\} = \bigcup_{i \in E} \left(\{Z_{n-1} = i\} \cap \{Y_{i,n} = j\}\right).$$

The equality follows from the recursive definition: $Z_n = j$ precisely when the previous state equals some $i \in E$ and the corresponding variable $Y_{i,n}$ takes the value j. Since E is finite or countable, the right-hand side is a countable union of measurable events. Starting from Z_0, this proves the measurability of each Z_n by induction.

We now verify that the constructed process has the required joint distribution.

We first consider the first two transitions.

Fix arbitrary states $i_0, i_1, i_2 \in E$. We obtain

$$\begin{aligned}
&\{Z_0 = i_0, Z_1 = i_1, Z_2 = i_2\} \\
&\quad = \{Z_0 = i_0, Y_{i_0,1} = i_1, Z_2 = i_2\} \\
&\quad = \{Z_0 = i_0, Y_{i_0,1} = i_1, Y_{i_1,2} = i_2\}.
\end{aligned}$$

The first equality follows from the recursive definition: on the event $\{Z_0 = i_0\}$, we have $Z_1 = Y_{i_0,1}$. Therefore, requiring $Z_1 = i_1$ is equivalent to requiring $Y_{i_0,1} = i_1$.

The second equality follows by applying the same reasoning to the next step. Once the first two conditions hold, we have $Z_1 = i_1$, and therefore $Z_2 = Y_{i_1,2}$. Hence, requiring $Z_2 = i_2$ is equivalent to requiring $Y_{i_1,2} = i_2$.

The important point is that i_0, i_1, i_2 are fixed states. We are considering the event in which the process follows the particular path

$$i_0 \longrightarrow i_1 \longrightarrow i_2.$$

Consequently, the event on the right-hand side involves the three specific random variables

$$Z_0, \qquad Y_{i_0,1}, \qquad Y_{i_1,2},$$

which are independent by our initial assumption. We are not using independence of Z_0, Z_1, Z_2.

Therefore,

$$\begin{aligned}
&\mathbb{P}(Z_0 = i_0, Z_1 = i_1, Z_2 = i_2) \\
&\quad = \mathbb{P}(Z_0 = i_0, Y_{i_0,1} = i_1, Y_{i_1,2} = i_2) \\
&\quad = \mathbb{P}(Z_0 = i_0)\mathbb{P}(Y_{i_0,1} = i_1)\mathbb{P}(Y_{i_1,2} = i_2) \\
&\quad = \mu_{i_0} p_{i_0 i_1} p_{i_1 i_2}.
\end{aligned}$$

The first equality follows from the equality of events established above. The second follows from the independence of $Z_0, Y_{i_0,1}, Y_{i_1,2}$, and the last from their prescribed distributions.

We now extend the argument to an arbitrary number of transitions.

Fix $n \in \mathbb{N}$ and arbitrary states $i_0, \ldots, i_n \in E$.

Repeatedly applying the recursive definition gives

$$\begin{aligned}
&\{Z_0 = i_0, Z_1 = i_1, \ldots, Z_n = i_n\} \\
&\quad = \{Z_0 = i_0\} \cap \bigcap_{r=1}^{n} \{Y_{i_{r-1},r} = i_r\}.
\end{aligned}$$

This follows from the same argument used for the first two transitions. For a fixed path, the condition $Z_{r-1} = i_{r-1}$ implies that the variable selected at time r is $Y_{i_{r-1},r}$. Therefore, requiring $Z_r = i_r$ is equivalent to requiring $Y_{i_{r-1},r} = i_r$.

Since the states $i_0, \dots, i_n$ are fixed, the event on the right-hand side involves the random variables

$$Z_0,\, Y_{i_0,1},\, Y_{i_1,2},\, \dots,\, Y_{i_{n-1},n}.$$

These are independent by construction. In particular, the selected Y-variables are distinct because their time indices are different, even if some of the states coincide.

Consequently,

$$\begin{aligned}
&\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) \\
&\quad = \mathbb{P}\left(\{Z_0 = i_0\} \cap \bigcap_{r=1}^{n} \{Y_{i_{r-1},r} = i_r\}\right) \\
&\quad = \mathbb{P}(Z_0 = i_0) \prod_{r=1}^{n} \mathbb{P}(Y_{i_{r-1},r} = i_r) \\
&\quad = \mu_{i_0} \prod_{r=1}^{n} p_{i_{r-1} i_r} \\
&\quad = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.
\end{aligned}$$

The first equality follows from the representation of the event just established. The second follows from the independence of the selected random variables, the third from their prescribed distributions, and the last from expanding the finite product.

Since $n \in \mathbb{N}$ and the states $i_0, \dots, i_n \in E$ were arbitrary, we have proved that

$$\mathbb{P}(Z_0 = i_0, Z_1 = i_1, \dots, Z_n = i_n) = \mu_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.$$

Thus, the constructed process satisfies the required joint distribution and is a time-homogeneous Markov chain with initial distribution μ and transition matrix P.

Therefore, the statement is proved.

A Monkey Types Shakespeare

Statement and construction

Consider a fixed text, such as the complete works of William Shakespeare (WS), consisting of $N \in \mathbb{N}$ symbols in a specified order. Write the target sequence as

$$(w_1, w_2, \dots, w_N).$$

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of independent random variables, where X_n is the symbol typed by the monkey at the n-th keystroke. We assume that each symbol needed to type WS has a probability uniformly bounded below by a constant $\varepsilon > 0$ at every keystroke:

$$\mathbb{P}(X_n = x) \geq \varepsilon > 0 \qquad \text{for every } n \in \mathbb{N} \text{ and every symbol } x \text{ appearing in WS.}$$

This makes explicit the uniform-in-time meaning of the positive lower bound used in the lecture notes. No assumption that the X_n have identical distributions is needed.

For $s \in \mathbb{N} \cup \{0\}$, let

$$C_s := \bigcap_{r=1}^{N} \{X_{s+r} = w_r\}.$$

The event C_s means that a complete copy of WS starts at keystroke $s+1$ and ends at keystroke $s+N$. Different occurrences may overlap; each occurrence is identified by its starting position.

Following the lecture notes, introduce the events

$$\begin{aligned} H &:= \{\text{the monkey types infinitely many copies of WS}\},\\ H_k &:= \{\text{the monkey types at least } k \text{ copies of WS}\},\\ H_{m,k} &:= \{\text{the monkey types at least } k \text{ copies of WS by time } m\},\\ H^{(m)} &:= \{\text{the monkey types infinitely many copies of WS after time } m\}. \end{aligned}$$

Here $H_{m,k}$ concerns copies completed no later than the m-th keystroke, whereas $H^{(m)}$ concerns copies whose starting positions are after that keystroke. In particular, $H_{m,k}$ depends only on $X_1, \dots, X_m$, while $H^{(m)}$ depends only on $X_{m+1}, X_{m+2}, \dots$. These are measurable events: each occurrence C_s is a finite intersection of events determined by random variables, and the conditions involving finitely or infinitely many occurrences can be expressed through countable unions and intersections.

We want to prove

$$\mathbb{P}(H) = 1.$$

Expanded proof

We first show that $\mathbb{P}(H)$ *is either* 0 *or* 1.

Fix $m, k \in \mathbb{N}$. We obtain

$$\mathbb{P}\big(H_{m,k} \cap H^{(m)}\big) = \mathbb{P}(H_{m,k})\mathbb{P}\big(H^{(m)}\big).$$

The equality follows from independence: $H_{m,k}$ is determined by the first m symbols, whereas $H^{(m)}$ is determined by symbols typed strictly after time m; the corresponding families of random variables are independent.

As we will see in HW 40, deleting finitely many possible starting positions does not change whether infinitely many copies occur, so $H^{(m)} = H$. Consequently,

$$\mathbb{P}(H_{m,k} \cap H) = \mathbb{P}(H_{m,k})\mathbb{P}(H).$$

This equality follows by substituting $H^{(m)} = H$ into the preceding identity.

As we will also see in HW 40,

$$H_{m,k} \uparrow H_k \quad \text{as } m \to \infty, \qquad H_k \cap H = H.$$

Therefore,

$$\mathbb{P}(H) = \mathbb{P}(H_k \cap H) = \lim_{m\to\infty} \mathbb{P}(H_{m,k} \cap H) = \lim_{m\to\infty} \big[\mathbb{P}(H_{m,k})\mathbb{P}(H)\big] = \mathbb{P}(H_k)\mathbb{P}(H).$$

The first equality uses $H_k \cap H = H$. The second follows from Lemma 1.5 (continuity from below), since $H_{m,k} \cap H \uparrow H_k \cap H$. The third uses the independence identity already established, and the last follows from continuity from below applied to $H_{m,k} \uparrow H_k$.

As we will prove in HW 40,

$$H_k \downarrow H \quad \text{as } k \to \infty.$$

Consequently,

$$\mathbb{P}(H) = \mathbb{P}(H)^2.$$

The equality follows by taking $k \to \infty$ in $\mathbb{P}(H) = \mathbb{P}(H_k)\mathbb{P}(H)$ and applying Lemma 1.6 (continuity from above). Since $\mathbb{P}(H) \in [0, 1]$, we conclude that

$$\mathbb{P}(H) \in \{0, 1\}.$$

Indeed, the equation $p = p^2$ is equivalent to $p(1 - p) = 0$.

We now prove that $\mathbb{P}(H) = 1$.

Divide the successive keystrokes into disjoint blocks of length N. For every $k \in \mathbb{N}$, define

$$E_k := \bigcap_{r=1}^{N} \{X_{(k-1)N+r} = w_r\}.$$

Thus E_k is the event that the monkey types an entire copy of WS in the k-th block, consisting precisely of keystrokes $(k-1)N+1, \ldots, kN$.

We obtain

$$\mathbb{P}(E_k) = \prod_{r=1}^{N} \mathbb{P}(X_{(k-1)N+r} = w_r) \geq \varepsilon^N > 0.$$

The equality follows from independence of the N random variables used in the block, while the inequality follows from the uniform lower bound on the probability of each required symbol. In particular, the same positive bound holds for every k.

Consequently,

$$\sum_{k=1}^{\infty} \mathbb{P}(E_k) \geq \sum_{k=1}^{\infty} \varepsilon^N = +\infty.$$

The inequality follows from the preceding bound, and the equality holds because the sum contains infinitely many copies of the fixed positive constant ε^N.

Moreover, the events $(E_k)_{k\in\mathbb{N}}$ are independent, because distinct blocks involve disjoint sets of the independent random variables $(X_n)_{n\in\mathbb{N}}$. Thus all the assumptions of Lemma 4.2, the Second Borel–Cantelli Lemma, are satisfied, and we obtain

$$\mathbb{P}\left(\limsup_{k\to\infty} E_k\right) = 1.$$

As we will see in HW 40,

$$\limsup_{k\to\infty} E_k \subseteq H.$$

Therefore,

$$1 = \mathbb{P}\left(\limsup_{k\to\infty} E_k\right) \leq \mathbb{P}(H) \leq 1.$$

The first equality is the conclusion of the Second Borel–Cantelli Lemma, the first inequality follows from the inclusion and monotonicity of probability, and the last inequality follows from $H \in \mathcal{F}$.

Hence,

$$\mathbb{P}(H) = 1.$$

Therefore, the statement is proved.

HW 40 — Set-Theoretic Verification of the Events in the Monkey Example

Statement

With the notation introduced in A Monkey Types Shakespeare, prove that

- **(i)** $H = H^{(m)}$;
- **(ii)** $H_k \cap H = H$;
- **(iii)** $\limsup_{k\to\infty} E_k \subseteq H$;
- **(iv)** $H_k = \bigcup_{m=1}^{\infty} H_{m,k}$;
- **(v)** $H = \bigcap_{k=1}^{\infty} H_k$.

Expanded proof of HW 40

Recall that $C_s := \bigcap_{r=1}^{N} \{X_{s+r} = w_r\}$, $s \in \mathbb{N}\cup\{0\}$, is the event that the monkey types one complete copy of WS starting at keystroke $s+1$.

(i) Recall that H is the event of infinitely many occurrences of WS in the entire sequence, while $H^{(m)}$ concerns infinitely many occurrences whose first keystrokes are strictly after time m. We also allow $m = 0$.

For every fixed threshold $q \geq 0$, consider the event $\bigcup_{s \geq q} C_s$. This is the event that at least one complete copy of WS starts at a keystroke $s+1$ with $s \geq q$. Here s labels a possible starting position and q is a variable threshold; all indices are nonnegative integers.

For fixed $m \geq 0$, we have

$$H^{(m)} = \bigcap_{q=m}^{\infty} \bigcup_{s \geq q} C_s,$$

$$\omega \in H^{(m)} \iff \forall q \geq m,\ \exists s \geq q :\ \omega \in C_s.$$

The outer intersection requires an occurrence beyond every threshold $q \geq m$, which is precisely the meaning of infinitely many occurrences after time m.

In particular,

$$H = H^{(0)} = \limsup_{s \to \infty} C_s.$$

The first equality uses the case $m = 0$, when the entire sequence is considered; the second is the definition of the limit superior of events.

Fix $m \in \mathbb{N}$. *First, we prove that $H^{(m)} \subseteq H$.*

For every $0 \leq q < m$, we have

$$H^{(m)} \subseteq \bigcup_{s \geq m} C_s \subseteq \bigcup_{s \geq q} C_s.$$

The first inclusion follows by taking the threshold m in the intersection defining $H^{(m)}$; the second follows because $s \geq m$ implies $s \geq q$. Hence,

$$H^{(m)} \subseteq H^{(0)} = H.$$

The inclusion holds because the conditions with $q < m$ follow from the preceding inclusions, while those with $q \geq m$ already enter the definition of $H^{(m)}$; the equality was established above.

Conversely, we prove that $H \subseteq H^{(m)}$ by contradiction.

Recall that

$$H \subseteq H^{(m)} \quad \iff \quad H \cap \left(H^{(m)}\right)^c = \emptyset.$$

The equivalence follows because the inclusion means that no outcome can belong to H without belonging to $H^{(m)}$.

Suppose, for a contradiction, that there exists

$$\omega_0 \in H \cap \left(H^{(m)}\right)^c.$$

Since $\omega_0 \in H$, the monkey types infinitely many copies of WS in the entire sequence. On the other hand, since $\omega_0 \in \left(H^{(m)}\right)^c$, only finitely many copies start strictly after time m.

However, at most m distinct copies can start at or before time m, corresponding to the starting indices $s = 0, \ldots, m-1$. Therefore, infinitely many copies in total and only finitely many copies starting after time m are incompatible.

We now express this contradiction formally.

We obtain

$$\left(H^{(m)}\right)^c = \left(\bigcap_{q=m}^{\infty} \bigcup_{s \geq q} C_s \right)^c = \bigcup_{q=m}^{\infty} \bigcap_{s \geq q} C_s^c.$$

The first equality uses the representation of $H^{(m)}$, while the second follows from De Morgan's laws.

Consequently, since $\omega_0 \in \left(H^{(m)}\right)^c$, there exists $q_0 \geq m$ such that

$$\omega_0 \notin C_s \qquad \forall s \geq q_0.$$

By the definitions of union and intersection, membership in the preceding event gives such a threshold q_0. This is the meaning of finitely often: no further copy starts beyond that threshold.

On the other hand, since $\omega_0 \in H$, we obtain

$$\omega_0 \in C_{s_0} \qquad \text{for some } s_0 \geq q_0.$$

This follows by taking $q = q_0$ in the membership characterisation of H, since $q_0 \geq m \geq 0$. It contradicts the fact that $\omega_0 \notin C_s$ for every $s \geq q_0$.
Hence,

$$H \cap \left(H^{(m)}\right)^c = \emptyset,$$

and therefore

$$H \subseteq H^{(m)}.$$

Combining the two inclusions, we conclude that

$$H = H^{(m)}.$$

□

(iv) We first express the two events in terms of the C_s:

$$H_k = \bigcup_{0 \leq s_1 < \cdots < s_k} \bigcap_{j=1}^{k} C_{s_j}$$

and

$$H_{m,k} = \bigcup_{\substack{0 \leq s_1 < \cdots < s_k \\ s_k + N \leq m}} \bigcap_{j=1}^{k} C_{s_j}.$$

In the first expression, the intersection requires k distinct occurrences of WS, while the union considers every possible selection of their starting positions. In the second, the additional restriction $s_k + N \leq m$ requires all selected copies to be completed by time m.
Consequently,

$$\bigcup_{m=1}^{\infty} H_{m,k} = \bigcup_{m=1}^{\infty} \bigcup_{\substack{0 \leq s_1 < \cdots < s_k \\ s_k + N \leq m}} \bigcap_{j=1}^{k} C_{s_j} = \bigcup_{0 \leq s_1 < \cdots < s_k} \bigcap_{j=1}^{k} C_{s_j} = H_k.$$

The first equality substitutes the definition of $H_{m,k}$. The second follows because, for every finite selection of starting positions, we can choose $m \geq s_k + N$, so that selection eventually appears in the union. The last equality uses the representation of H_k.
Moreover, $H_{m,k} \subseteq H_{m+1,k}$, because increasing the deadline from m to $m+1$ does not exclude any previously completed copies.
Therefore,

$$H_{m,k} \uparrow H_k \text{ as } m \to \infty.$$

(v) Using complements, we obtain

$$\left(\bigcap_{k=1}^{\infty} H_k\right)^c = \bigcup_{k=1}^{\infty} H_k^c = \bigcup_{q=0}^{\infty} \bigcap_{s \geq q} C_s^c = \left(\bigcap_{q=0}^{\infty} \bigcup_{s \geq q} C_s\right)^c = H^c.$$

The first equality follows from De Morgan's law. The second expresses the equivalence between having fewer than k occurrences for some k and having only finitely many occurrences, which means that from some position q onward no further C_s occurs. The third equality follows again from De Morgan's law, while the last uses the representation of H established above.
Taking complements, we conclude that

$$H = \bigcap_{k=1}^{\infty} H_k.$$

Moreover, $H_{k+1} \subseteq H_k$, because having at least $k+1$ copies implies having at least k copies. Hence,

$$H_k \downarrow H \text{ as } k \to \infty.$$

(ii) In particular,

$$H \subseteq H_k,\ H_k \cap H = H.$$

The inclusion follows because an intersection is contained in each of its constituent sets, while the equality follows from the identity $A \cap B = B$ whenever $B \subseteq A$.

(iii) Recall that $E_k = C_{(k-1)N}$. The equality follows because E_k requires one complete copy of WS in the k-th block of N keystrokes.

For every $r \in \mathbb{N}$,

$$\bigcup_{k \geq r} E_k = \bigcup_{k \geq r} C_{(k-1)N} \subseteq \bigcup_{s \geq (r-1)N} C_s.$$

The equality substitutes the definition of E_k, while the inclusion follows because the positions $(k-1)N$, for $k \geq r$, form a subset of all starting positions $s \geq (r-1)N$.

Therefore,

$$\limsup_{k \to \infty} E_k = \bigcap_{r=1}^{\infty} \bigcup_{k \geq r} E_k \subseteq \bigcap_{r=1}^{\infty} \bigcup_{s \geq (r-1)N} C_s = \bigcap_{q=0}^{\infty} \bigcup_{s \geq q} C_s = H.$$

The first equality is the definition of the limit superior, and the inclusion follows from the preceding inclusion of unions. The next equality holds because every omitted threshold q is below some retained threshold $(r-1)N$, whose union is smaller and hence imposes a stronger condition. The last equality follows from the representation of H.

HW 41 — Independence of Products of Random Variables

Statement

Let $(Y_n)_{n \in \mathbb{N} \cup \{0\}}$ be a sequence of i.i.d. random variables such that

$$\mathbb{P}(Y_n = 1) = \mathbb{P}(Y_n = -1) = \frac{1}{2}.$$

Define

$$X_n := \prod_{i=0}^{n} Y_i, \qquad n \in \mathbb{N} \cup \{0\}.$$

Prove that $(X_n)_{n \in \mathbb{N} \cup \{0\}}$ is a sequence of i.i.d. random variables.

Expanded Proof of HW 41

We first notice that

$$X_0 = Y_0, \qquad X_n = X_{n-1} Y_n, \quad n \geq 1.$$

The first equality follows from the definition of X_0, while the second follows by separating the last factor in the product defining X_n.

Moreover, $X_n \in \{-1, 1\}$ for every n, since each Y_i takes values in $\{-1, 1\}$.

We now use the same idea as in Example 3 on Markov chains: we fix a possible path of the process and calculate its probability.

Fix $n \in \mathbb{N} \cup \{0\}$ and arbitrary states $x_0, \ldots, x_n \in \{-1, 1\}$. We obtain

$$\begin{aligned} &\{X_0 = x_0, \ldots, X_n = x_n\} \\ &\quad = \{Y_0 = x_0, Y_1 = x_0 x_1, \ldots, Y_n = x_{n-1} x_n\}. \end{aligned}$$

The equality follows from the recursive relation $X_k = X_{k-1} Y_k$. Indeed, since $X_{k-1}^2 = 1$, the conditions $X_{k-1} = x_{k-1}$ and $X_k = x_k$ are equivalent to requiring $Y_k = x_{k-1} x_k$, once the previous states have been fixed.

Therefore,

$$\begin{aligned}
&\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) \\
&\quad = \mathbb{P}(Y_0 = x_0, Y_1 = x_0x_1, \dots, Y_n = x_{n-1}x_n) \\
&\quad = \mathbb{P}(Y_0 = x_0) \prod_{k=1}^{n} \mathbb{P}(Y_k = x_{k-1}x_k) \\
&\quad = \left(\frac{1}{2}\right)^{n+1}.
\end{aligned}$$

The first equality follows from the identity of events just established, the second from independence of the Y_k, and the last from their common distribution, since every $x_{k-1}x_k$ belongs to $\{-1, 1\}$.

We now relate this result to Example 3.

Consider the state space $E = \{-1, 1\}$, with initial distribution and transition matrix given by

$$\mu_{-1} = \mu_1 = \frac{1}{2}, \qquad P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The joint distribution obtained above can be written as

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mu_{x_0} \prod_{k=1}^{n} p_{x_{k-1}x_k}.$$

This is precisely the joint-distribution formula for a Markov chain introduced in Example 3, with $\mu_i = 1/2$ and $p_{ij} = 1/2$ for every $i, j \in E$.

In this particular case, all rows of the transition matrix are identical: the probability of the next state does not depend on the current state.

We now determine the marginal distributions. For every $m \geq 0$ and $a \in \{-1, 1\}$,

$$\mathbb{P}(X_m = a) = \sum_{x_0, \dots, x_{m-1} \in E} \mathbb{P}(X_0 = x_0, \dots, X_{m-1} = x_{m-1}, X_m = a) = 2^m \left(\frac{1}{2}\right)^{m+1} = \frac{1}{2}.$$

The first equality follows by summing over all possible values of the preceding variables, while the second uses the joint distribution already established and the fact that there are 2^m possible choices of $x_0, \dots, x_{m-1}$.

Consequently,

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \left(\frac{1}{2}\right)^{n+1} = \prod_{k=0}^{n} \mathbb{P}(X_k = x_k).$$

The first equality was proved above, and the second follows from the marginal distributions. The factorisation also holds for any finite selection of the X_n, by summing over the remaining coordinates. Since the variables take values only in $\{-1, 1\}$, this proves their independence according to Definition 4.2.

Moreover, all the X_n have the same distribution. Therefore,

$$(X_n)_{n \in \mathbb{N} \cup \{0\}} \text{ is an i.i.d. sequence.}$$

Therefore, the statement is proved.

5. Integration

Motivation — Expectation as an Integral

We seek a general meaning for the expected value $\mathbb{E}[X]$ of a random variable X.
If X takes countably many possible values $x_1, x_2, \ldots$, then

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} x_i \, \mathbb{P}(X = x_i).$$

The sum is understood whenever it is well-defined, in particular when X is integrable.
If X is real-valued and admits a probability density function f_X, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} z f_X(z) \, dz.$$

This formula also requires a well-defined expectation.
These two expressions do not cover every random variable: its distribution need not be discrete or admit a density. We therefore seek a general notion of integration that allows us to write

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) \, d\mathbb{P}(\omega).$$

Here $\mathbb{P}$ is the probability measure on $(\Omega, \mathcal{F})$; Lebesgue integration gives meaning to this expression whenever the integral is defined.

Recall — Riemann Integral

Let $g \in C([a,b])$, and for each $n \in \mathbb{N}$ consider the dyadic partition

$$z_i^{(n)} = a + \frac{i}{2^n}(b-a), \qquad i = 0, 1, \ldots, 2^n.$$

The Riemann integral can be expressed as

$$\int_a^b g(z) \, dz = \lim_{n\to\infty} \sum_{i=0}^{2^n-1} g\big(z_i^{(n)}\big)\big(z_{i+1}^{(n)} - z_i^{(n)}\big).$$

Each term is the left-endpoint value times the subinterval length. The 2^n subintervals give the last index $2^n - 1$.
A random variable $X : \Omega \to \mathbb{R}$ need only be $\mathcal{F}$-measurable, which does not ensure Riemann integrability; moreover, Ω need not be a real interval. We need integration on general measure spaces, agreeing with the Riemann integral for continuous functions on real intervals. Such functions are Borel-measurable.

Notation — Lebesgue Integral with Respect to a Measure

Let (S, Σ, μ) be a measure space. For a Σ-measurable function f for which the integral is well-defined (in particular, for any non-negative measurable f), we use the notation

$$\mu(f) := \int_S f(s) \, d\mu(s) = \int_S f \, d\mu.$$

The measure μ supplies the weights; the integral is constructed below.
For any $A \in \Sigma$, we also write

$$\mu(f; A) := \mu(f\mathbf{1}_A) = \int_S f\mathbf{1}_A \, d\mu = \int_A f \, d\mu.$$

The indicator $\mathbf{1}_A$ keeps f on A and sets it to zero outside A, giving integration restricted to A.

Example — Counting Measure and Infinite Series

Consider the measure space

$$(S, \Sigma, \mu) = \big(\mathbb{N}, \mathcal{P}(\mathbb{N}), \#\big),$$

where $\#$ is the counting measure, so that $\mu(\{k\}) = 1$ for every $k \in \mathbb{N}$.
Let $f : \mathbb{N} \to [0, \infty)$ be given by $f(k) = a_k$. Our construction must recover

$$\int_S f \, d\mu = \sum_{k=1}^{\infty} f(k)\, \mu(\{k\}) = \sum_{k=1}^{\infty} a_k.$$

Each singleton has measure one, so its contribution should be $f(k) = a_k$. For signed a_k, absolute convergence also ensures the identity.
Thus integration must recover series as well as the continuous-function Riemann integral.

Integrals of Positive Simple Functions

Definition — Integral of an Indicator Function

Let (S, Σ, μ) be a measure space. For any $A \in \Sigma$, recall that the indicator function of A is

$$\mathbf{1}_A(s) = \begin{cases} 1, & s \in A, \\ 0, & s \notin A. \end{cases}$$

We first define its integral by

$$\mu_0(\mathbf{1}_A) := \int_S \mathbf{1}_A \, d\mu := \mu(A).$$

Thus, integrating an indicator function gives the measure of the set on which it equals 1. The notation μ_0 emphasizes that, at this stage, the integral has been defined only for indicator functions. The value $\mu(A)$ may be infinite.

Definition — Positive Simple Function

A non-negative function $f : S \to [0, \infty)$ is called a simple function if it can be written as

$$f(s) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(s), \qquad s \in S,$$

for some $m \in \mathbb{N}$, coefficients $a_k \in [0, \infty)$ and sets $A_k \in \Sigma$.
For each $s \in S$, only the terms with $s \in A_k$ contribute to the sum. The sets A_k need not be disjoint, so their contributions may add up. Since there are only finitely many indicators, f can take only finitely many distinct values; this explains the word simple.

HW 42 — Measurability of a Simple Function

Statement. Let

$$f(s) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(s), \qquad a_k \in [0, \infty), \quad A_k \in \Sigma.$$

Check that $f \in m\Sigma$.

Expanded Proof.

Fix $k \in \{1, \dots, m\}$. For any Borel set $B \in \mathcal{B}(\mathbb{R})$, we have

$$(\mathbf{1}_{A_k})^{-1}(B) \in \{\emptyset, A_k, A_k^c, S\} \subseteq \Sigma.$$

The pre-image depends on whether 0 and 1 belong to B. All four sets are measurable since $A_k \in \Sigma$ and Σ is a σ-algebra. Thus $\mathbf{1}_{A_k} \in m\Sigma$ by Definition 3.1.
We now obtain

$$a_k \mathbf{1}_{A_k} \in m\Sigma, \qquad k = 1, \dots, m.$$

Lemma 3.4**(iii)** applies to each real constant a_k.

Consequently,

$$f = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k} \in m\Sigma.$$

Repeated application of Lemma 3.4**(i)** gives the membership. Nonnegative coefficients and indicators give $f \geq 0$, hence $f \in (m\Sigma)^+$.

Therefore, the statement is proved.

Different Representations and Canonical Form of Simple Functions

A simple function may admit different representations, corresponding to different choices of coefficients a_k and measurable sets A_k.

For example, let $S = [0, 1]$ and $\Sigma = \mathcal{B}([0, 1])$. Then

$$f(s) = \mathbf{1}_{[1/2,1]}(s) = \mathbf{1}_{[1/2,2/3)}(s) + \mathbf{1}_{[2/3,1]}(s).$$

The equality follows because the two intervals on the right are disjoint and their union is $[1/2, 1]$. Thus, the same function can be represented using different measurable sets.

A simple function is expressed in canonical form if

$$f(s) = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}(s),$$

where

$$a_1 < a_2 < \cdots < a_m, \qquad A_k = f^{-1}(\{a_k\}), \quad k = 1, \ldots, m.$$

In canonical form, the coefficients are precisely the distinct values attained by f, including zero when attained. Their measurable level sets are pairwise disjoint and cover S, so they form a finite partition. Every nonnegative simple function admits this representation, obtained by collecting points with the same value.

Definition — Integral of a Positive Simple Function

We denote the collection of non-negative simple functions by SF^+. For $f \in SF^+$, represented as

$$f = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k},$$

we define

$$\mu_0(f) := \int_S f \, d\mu := \sum_{k=1}^{m} a_k \mu(A_k).$$

This extends the definition of the integral of an indicator function, since for $f = \mathbf{1}_A$ we recover $\mu_0(f) = \mu(A)$.

Observation — The Convention $0 \cdot \infty = 0$

Let $S = \mathbb{R}$, $\Sigma = \mathcal{B}(\mathbb{R})$, $\mu = \text{Leb}$, and consider the zero function $f = 0$.

We obtain

$$\int_S f \, d\mu = 0 \cdot \mu(\mathbb{R}) = 0 \cdot \infty.$$

Since the integral of the zero function must equal zero, we adopt the convention

$$0 \cdot \infty := 0.$$

This ensures that zero coefficients contribute nothing to the integral, even when the corresponding sets have infinite measure.

Finally, since a simple function may admit different representations, we must verify that the value of $\mu_0(f)$ does not depend on the chosen representation.

This is the well-definedness of the integral of simple functions, which will be established in the next homework.

HW 43 — Well-Definedness of the Integral of Simple Functions

Statement. Fix (S, Σ, μ) and let $f \in SF^+$ have the following two representations:

$$f = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i} \qquad \text{and} \qquad f = \sum_{j=1}^{m} b_j \mathbf{1}_{B_j},$$

for numbers $(a_i)_{i=1}^n, (b_j)_{j=1}^m \subseteq [0, \infty)$ and sets $(A_i)_{i=1}^n, (B_j)_{j=1}^m \subseteq \Sigma$. Show that

$$\mu_0(f) = \sum_{i=1}^{n} a_i \mu(A_i) = \sum_{j=1}^{m} b_j \mu(B_j).$$

Proof.

Reduction to disjoint partitions.

A finite measurable partition of S is a family of nonempty measurable sets that are pairwise disjoint and whose union is S. For instance, two sets $A, B \in \Sigma$ give the pieces

$$A \cap B, \qquad A \cap B^c, \qquad A^c \cap B, \qquad A^c \cap B^c,$$

with empty pieces omitted. In particular,

$$A = (A \cap B) \cup (A \cap B^c), \qquad B = (A \cap B) \cup (A^c \cap B).$$

Thus each original set can be recovered as a union of disjoint pieces, while the remaining piece completes the partition of S.

For the first representation, start with S and consider $A_1, \ldots, A_n$ one at a time. At the ith step, split every current piece E into the nonempty sets among

$$E \cap A_i, \qquad E \cap A_i^c.$$

Indeed,

$$E = (E \cap A_i) \cup (E \cap A_i^c), \qquad (E \cap A_i) \cap (E \cap A_i^c) = \emptyset.$$

Each step preserves disjointness and the union S; measurability follows from closure of Σ under complements and finite intersections. After finitely many steps we obtain a partition $C_1, \ldots, C_r$ of S. Each C_k lies entirely inside or outside every A_i, so each A_i is a union of some C_k and f is constant on each C_k.

Applying the same construction to $B_1, \ldots, B_m$ gives another partition $D_1, \ldots, D_t$ of S. Each B_j is a union of some D_ℓ, and f is constant on each D_ℓ.

Choose $s_k \in C_k$ and $t_\ell \in D_\ell$, and set

$$c_k = f(s_k), \qquad d_\ell = f(t_\ell), \qquad k = 1, \ldots, r, \quad \ell = 1, \ldots, t.$$

From the two representations of f,

$$c_k = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}(s_k), \qquad d_\ell = \sum_{j=1}^{m} b_j \mathbf{1}_{B_j}(t_\ell),$$

for every $k = 1, \ldots, r$ and $\ell = 1, \ldots, t$.

Since the C_k partition S, we obtain

$$\sum_{i=1}^{n} a_i \mu(A_i) = \sum_{i=1}^{n} a_i \mu\left(A_i \cap \bigcup_{k=1}^{r} C_k\right) = \sum_{k=1}^{r} \sum_{i=1}^{n} a_i \mu(A_i \cap C_k)$$
$$= \sum_{k=1}^{r} \left(\sum_{i=1}^{n} a_i \mathbf{1}_{A_i}(s_k)\right) \mu(C_k) = \sum_{k=1}^{r} c_k \mu(C_k).$$

The second equality uses finite additivity on the disjoint pieces $A_i \cap C_k$ and rearranges finite nonnegative sums. The third follows because each C_k lies entirely inside or outside A_i, and the last uses the expression for c_k. Likewise,

$$\sum_{j=1}^{m} b_j \mu(B_j) = \sum_{\ell=1}^{t} d_\ell \mu(D_\ell).$$

These identities show that we may assume, without loss of generality, that the representing families are finite measurable partitions of S, since the replacement preserves both f and the weighted sums. Moreover, by merging partition elements with equal coefficients, any such representation can be reduced to canonical form, again without changing the weighted sum.

Proof for partition representations.

Assume now that $(A_i)_{i=1}^n$ and $(B_j)_{j=1}^m$ are finite measurable partitions of S, with the corresponding coefficients a_i and b_j. As proved above, this entails no loss of generality.

The nonempty intersections $A_i \cap B_j$ form a common finite measurable partition of S. Indeed, they cover S because every $s \in S$ belongs to some A_i and some B_j. Moreover, for $(i,j) \neq (k,\ell)$,

$$(A_i \cap B_j) \cap (A_k \cap B_\ell) = (A_i \cap A_k) \cap (B_j \cap B_\ell) = \emptyset.$$

The last equality follows because either $i \neq k$ or $j \neq \ell$, so at least one intersection is empty. Finally, $A_i \cap B_j \in \Sigma$ by closure under finite intersections.

For every $i = 1, \ldots, n$ and $j = 1, \ldots, m$, we distinguish two cases.

Case 1 — Empty intersection.

If $A_i \cap B_j = \emptyset$, then $\mu(A_i \cap B_j) = 0$.

Case 2 — Nonempty intersection.

If $A_i \cap B_j \neq \emptyset$, choose $s \in A_i \cap B_j$. Since the two families are partitions, s belongs to exactly one set in each family. Hence,

$$a_i = \sum_{k=1}^{n} a_k \mathbf{1}_{A_k}(s) = f(s) = \sum_{\ell=1}^{m} b_\ell \mathbf{1}_{B_\ell}(s) = b_j.$$

The first and last equalities follow because only the indicators of A_i and B_j equal 1 in their respective sums; the middle equalities use the two representations of f.

Thus $a_i = b_j$ holds for pairs whose intersection is nonempty. For the remaining pairs the coefficients need not agree, but the intersection has measure zero. In both cases, therefore,

$$a_i \mu(A_i \cap B_j) = b_j \mu(A_i \cap B_j), \qquad \forall\, i = 1, \ldots, n, \quad j = 1, \ldots, m.$$

Using $S = \bigcup_{j=1}^m B_j = \bigcup_{i=1}^n A_i$, we obtain

$$\sum_{i=1}^{n} a_i \mu(A_i) = \sum_{i=1}^{n} a_i \mu(A_i \cap S) = \sum_{i=1}^{n} a_i \mu\left(A_i \cap \bigcup_{j=1}^{m} B_j\right) = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \mu(A_i \cap B_j)$$
$$= \sum_{j=1}^{m} \sum_{i=1}^{n} b_j \mu(A_i \cap B_j) = \sum_{j=1}^{m} b_j \mu\left(B_j \cap \bigcup_{i=1}^{n} A_i\right) = \sum_{j=1}^{m} b_j \mu(B_j).$$

The third and fifth equalities use finite additivity on disjoint intersections. The fourth follows from the equality of the contributions in both cases and rearrangement of finite sums. All expressions remain valid when some measures are infinite, under the convention $0 \cdot \infty = 0$.

Thus the weighted sum is independent of the chosen representation of f, and $\mu_0(f)$ is well-defined. Therefore, the statement is proved.

HW 44 — Properties of Integrals of Simple Functions

Statement. Fix a measure space (S, Σ, μ). Prove the following properties.

(i) If $f, g \in SF^+$ and $\mu(\{s \in S : f(s) \neq g(s)\}) = 0$, then $\mu_0(f) = \mu_0(g)$.

(ii) If $f, g \in SF^+$ and $c \geq 0$, then $f + g, cf \in SF^+$; moreover,

$$\mu_0(f + g) = \mu_0(f) + \mu_0(g), \qquad \mu_0(cf) = c\mu_0(f).$$

(iii) If $f, g \in SF^+$ and $f \leq g$, then $\mu_0(f) \leq \mu_0(g)$.

(iv) If $f, g \in SF^+$, then $f \vee g := \max\{f, g\}$ and $f \wedge g := \min\{f, g\}$ also belong to SF^+.

Recall — μ-almost everywhere.

A property $P(s)$ holds μ-almost everywhere (μ-a.e.) if it fails only on a set of μ-measure zero. In particular, for measurable f, g,

$$f = g \quad \mu\text{-a.e.} \quad \iff \quad \mu\big(\{s \in S : f(s) \neq g(s)\}\big) = 0.$$

Thus f and g need not agree at every point.

When constructing a common partition as in HW 43, we discard only empty intersections, since partition elements must be nonempty. We do not discard nonempty sets of measure zero: for example, on $S = [0, 1]$ with $\mu = \text{Leb}$,

$$D = \{\tfrac{1}{2}\} \neq \emptyset, \qquad \mu(D) = 0.$$

Such a set remains a partition element, although its contribution to the integral is zero.

Proof.

By HW 43, both f and g admit canonical representations, whose level sets form finite measurable partitions of S. Intersecting these partitions and omitting empty intersections gives a common finite measurable partition $D_1, \ldots, D_r$, on which both functions are constant.

$$f = \sum_{k=1}^{r} a_k \mathbf{1}_{D_k}, \qquad g = \sum_{k=1}^{r} b_k \mathbf{1}_{D_k},$$

where $a_k, b_k \geq 0$. By HW 43, the integral is independent of the representation, so

$$\mu_0(f) = \sum_{k=1}^{r} a_k \mu(D_k), \qquad \mu_0(g) = \sum_{k=1}^{r} b_k \mu(D_k).$$

(i) Equality almost everywhere. Suppose $f = g$ μ-a.e., and let

$$N := \{s \in S : f(s) \neq g(s)\}, \qquad \mu(N) = 0.$$

Fix k such that $a_k \neq b_k$. For any $s \in D_k$, the partition representations give

$$f(s) = a_k \neq b_k = g(s) \quad \Longrightarrow \quad s \in N.$$

Since $s \in D_k$ was arbitrary, $D_k \subseteq N$. By monotonicity of μ,

$$0 \leq \mu(D_k) \leq \mu(N) = 0 \quad \Longrightarrow \quad \mu(D_k) = 0.$$

Consequently, for every $k = 1, \ldots, r$,

$$a_k \mu(D_k) = b_k \mu(D_k).$$

Indeed, if $a_k = b_k$ the equality is immediate; otherwise $\mu(D_k) = 0$ and both sides vanish. Summing gives

$$\mu_0(f) = \sum_{k=1}^{r} a_k \mu(D_k) = \sum_{k=1}^{r} b_k \mu(D_k) = \mu_0(g).$$

(ii) Additivity and positive homogeneity. Using the common partition,

$$f + g = \sum_{k=1}^{r} (a_k + b_k) \mathbf{1}_{D_k}, \qquad cf = \sum_{k=1}^{r} (ca_k) \mathbf{1}_{D_k}.$$

Thus $f + g, cf \in SF^+$, since their coefficients are nonnegative and the D_k are measurable. Moreover,

$$\mu_0(f + g) = \sum_{k=1}^{r} (a_k + b_k) \mu(D_k) = \sum_{k=1}^{r} a_k \mu(D_k) + \sum_{k=1}^{r} b_k \mu(D_k) = \mu_0(f) + \mu_0(g),$$

and

$$\mu_0(cf) = \sum_{k=1}^{r} (ca_k) \mu(D_k) = c \sum_{k=1}^{r} a_k \mu(D_k) = c\mu_0(f).$$

The equalities follow from the definition of μ_0 and distributivity of finite nonnegative sums, with $0 \cdot \infty = 0$ when necessary.

(iii) Monotonicity. Suppose $f \leq g$. Since $D_k \neq \emptyset$ and both functions are constant on it, $a_k \leq b_k$ for every $k = 1, \ldots, r$. Hence

$$\mu_0(f) = \sum_{k=1}^{r} a_k \mu(D_k) \leq \sum_{k=1}^{r} b_k \mu(D_k) = \mu_0(g).$$

The inequality follows from $a_k \leq b_k$ and $\mu(D_k) \geq 0$ for every k.

(iv) Maximum and minimum. Since $f = a_k$ and $g = b_k$ on D_k,

$$f \vee g = \sum_{k=1}^{r} \max\{a_k, b_k\} \mathbf{1}_{D_k},$$

$$f \wedge g = \sum_{k=1}^{r} \min\{a_k, b_k\} \mathbf{1}_{D_k}.$$

Both functions belong to SF^+, because their coefficients are nonnegative and their partition elements are measurable.

Therefore, the statement is proved.

Example — Equality Lebesgue-Almost Everywhere

Let $S = [0, 1]$, $\Sigma = \mathcal{B}([0, 1])$, $\mu = \text{Leb}$, and consider

$$f = \mathbf{1}_{[\frac{1}{2}, 1]}, \qquad g = \mathbf{1}_{(\frac{1}{2}, 1]}.$$

The functions differ only at $s = \frac{1}{2}$, so

$$\{s \in S : f(s) \neq g(s)\} = \{\tfrac{1}{2}\}, \qquad \text{Leb}(\{\tfrac{1}{2}\}) = 0.$$

Hence $f = g$ Leb-a.e. and, by HW 44(i),

$$\mu_0(f) = \text{Leb}([\tfrac{1}{2}, 1]) = \tfrac{1}{2} = \text{Leb}((\tfrac{1}{2}, 1]) = \mu_0(g).$$

The equalities follow from the definition of the integral of an indicator function and the length of the two intervals.

Definition — Integral of a Positive Measurable Function

In this chapter, $(m\Sigma)^+$ includes all measurable maps $f : S \to [0, +\infty]$, where measurability is understood through the real threshold tests. Simple functions in SF^+ still have only finitely many finite values.

For signed functions, $m\Sigma$ retains the real-valued meaning of Definition 3.1 unless stated otherwise. The convention $0 \cdot \infty = 0$ also applies pointwise to products with indicators.

Let (S, Σ, μ) be a measure space. For any $f \in (m\Sigma)^+$, define

$$\mu(f) := \int_S f \, d\mu := \sup\{\mu_0(h) : h \in SF^+, \ h \leq f\}.$$

Thus the integral of f is the supremum of the integrals of all nonnegative simple functions lying below f.

Observation — Agreement with the Simple Integral

If $f \in SF^+$, then

$$\mu(f) = \mu_0(f).$$

Indeed, f itself belongs to the family over which the supremum is taken, so $\mu(f) \geq \mu_0(f)$. Conversely, for every $h \in SF^+$ with $h \leq f$, monotonicity of μ_0 (HW 44(iii)) gives $\mu_0(h) \leq \mu_0(f)$. Taking the supremum yields $\mu(f) \leq \mu_0(f)$, proving the equality.

HW 45 — Monotonicity of the Positive Lebesgue Integral

Statement. Prove that if $f, g \in (m\Sigma)^+$ satisfy $f \leq g$, then

$$\mu(f) \leq \mu(g).$$

Proof.

We have

$$\{h \in SF^+ : h \leq f\} \subseteq \{h \in SF^+ : h \leq g\}.$$

Indeed, $h \leq f \leq g$ implies $h \leq g$. Both families contain the zero function. Taking suprema gives

$$\mu(f) = \sup\{\mu_0(h) : h \in SF^+, \ h \leq f\} \leq \sup\{\mu_0(h) : h \in SF^+, \ h \leq g\} = \mu(g).$$

The equalities are the definition of the positive integral; the inequality follows from the family inclusion. Therefore, the statement is proved.

Lemma 5.1 — Zero Integral Implies Zero Almost Everywhere

Statement. Let (S, Σ, μ) be a measure space and let $f \in (m\Sigma)^+$ satisfy $\mu(f) = 0$. Then

$$\mu(\{f > 0\}) = 0.$$

Expanded Proof.

We argue by contradiction. Suppose $\mu(\{f > 0\}) > 0$ and define

$$A_n := \left\{ s \in S : f(s) > \frac{1}{n} \right\} \in \Sigma, \qquad n \in \mathbb{N}.$$

These sets are measurable since f is measurable. Moreover, $A_n \subseteq A_{n+1}$ because $1/(n+1) < 1/n$. As we will see in HW 46,

$$\{f > 0\} = \bigcup_{n=1}^{\infty} A_n = \lim_{n \to \infty} A_n.$$

Hence, by continuity from below (Lemma 1.5),

$$0 < \mu(\{f > 0\}) = \lim_{n \to \infty} \mu(A_n).$$

Since $\mu(A_n)$ is nondecreasing and its limit is strictly positive, there exists $\bar{n} \in \mathbb{N}$ such that $\mu(A_n) > 0$ for every $n \geq \bar{n}$.

For any $n \geq \bar{n}$, the sets A_n and A_n^c partition S, so $f = f\mathbf{1}_{A_n} + f\mathbf{1}_{A_n^c}$ pointwise. Moreover, $f\mathbf{1}_{A_n^c} \geq 0$ and

$$f\mathbf{1}_{A_n} \geq \frac{1}{n}\mathbf{1}_{A_n}.$$

Indeed, if $s \in A_n$, then $f(s) > 1/n$ by definition; if $s \notin A_n$, both sides of the latter inequality vanish. Hence, using monotonicity of the positive integral (HW 45),

$$\begin{aligned} 0 = \mu(f) &= \int_S \left(f\mathbf{1}_{A_n} + f\mathbf{1}_{A_n^c}\right) d\mu \\ &\geq \int_S f\mathbf{1}_{A_n}\, d\mu \geq \int_S \frac{1}{n}\mathbf{1}_{A_n}\, d\mu = \frac{1}{n}\mu(A_n) > 0. \end{aligned}$$

The first equality is the hypothesis $\mu(f) = 0$; the second uses the pointwise decomposition of f, without linearity of the integral. The next two inequalities follow from the pointwise bounds above and HW 45. The final equality is the integral of the nonnegative simple function $(1/n)\mathbf{1}_{A_n}$, and the strict inequality holds since $\mu(A_n) > 0$ and $1/n > 0$, even if $\mu(A_n) = \infty$.

This contradicts $\mu(f) = 0$. Therefore $\mu(\{f > 0\}) = 0$.

Therefore, the statement is proved.

HW 46 — Level-Set Representation of $\{f > 0\}$

Statement. Let $f \in (m\Sigma)^+$. Prove that

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \left\{f > \frac{1}{n}\right\} = \lim_{n\to\infty} \left\{f > \frac{1}{n}\right\}.$$

Proof.

($\subseteq$) Let $s_0 \in \{f > 0\}$. Then $f(s_0) > 0$, so there exists $n_0 \in \mathbb{N}$ such that

$$\frac{1}{n_0} < f(s_0).$$

This follows from the Archimedean property of $\mathbb{R}$, and is immediate if $f(s_0) = \infty$. Therefore,

$$s_0 \in \left\{f > \frac{1}{n_0}\right\} \subseteq \bigcup_{n=1}^{\infty} \left\{f > \frac{1}{n}\right\}.$$

Since s_0 was arbitrary, the first inclusion follows.

($\supseteq$) Conversely, let

$$s_0 \in \bigcup_{n=1}^{\infty} \left\{f > \frac{1}{n}\right\}.$$

By definition of union, there exists $n_0 \in \mathbb{N}$ such that

$$f(s_0) > \frac{1}{n_0} > 0.$$

Hence $s_0 \in \{f > 0\}$, proving the reverse inclusion.

Finally, define

$$A_n := \left\{f > \frac{1}{n}\right\}, \qquad n \in \mathbb{N}.$$

Since

$$\frac{1}{n+1} < \frac{1}{n} \quad \Longrightarrow \quad A_n \subseteq A_{n+1},$$

the sequence $(A_n)_{n\in\mathbb{N}}$ is increasing. Therefore, by the definition of the limit of an increasing sequence of sets,

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n = \{f > 0\}.$$

Therefore, the statement is proved.

Example — Application of Lemma 5.1

Let $(S, \Sigma, \mu) = ([0,1], \mathcal{B}([0,1]), \text{Leb})$ and

$$f(s) = \mathbf{1}_{\{\frac{1}{2}\}}(s).$$

Since $f \in SF^+$,

$$\mu(f) = \mu_0(f) = \text{Leb}\left(\left\{\frac{1}{2}\right\}\right) = 0.$$

The equalities follow from the agreement of μ with μ_0 on simple functions, the integral of an indicator, and the fact that a singleton has zero Lebesgue measure. By Lemma 5.1,

$$\mu(\{f > 0\}) = \text{Leb}\left(\left\{\frac{1}{2}\right\}\right) = 0.$$

Thus $f = 0$, Leb-a.e., although $f(\frac{1}{2}) = 1$.

The next theorem is probably the most useful single tool in the whole course.

Observation — Reduction from Almost-Everywhere to Pointwise Convergence

Statement. Let $N \in \Sigma$ satisfy $\mu(N) = 0$. For any $u \in (m\Sigma)^+$, define

$$u^N(s) := \begin{cases} u(s), & s \in N^c, \\ 0, & s \in N. \end{cases}$$

Then $u^N \in (m\Sigma)^+$ and

$$\mu(u^N) = \mu(u).$$

Consequently, if $f_n \uparrow f$ μ-almost everywhere, we may modify all f_n and f on a common null set so that the convergence holds everywhere, without changing their integrals.

Expanded Proof.

Fix $u \in (m\Sigma)^+$. For $a \in \mathbb{R}$,

$$\{u^N > a\} = \begin{cases} N^c \cap \{u > a\}, & a \geq 0, \\ S, & a < 0. \end{cases}$$

Both sets are measurable since $N \in \Sigma$ and u is measurable. Thus $u^N \in (m\Sigma)^+$, including when u takes the value ∞.

Let $h \in SF^+$ satisfy $h \leq u$. Consider $h\mathbf{1}_{N^c}$. We have $h\mathbf{1}_{N^c} \in SF^+$, $h\mathbf{1}_{N^c} \leq u^N$. Indeed, outside N, the inequality follows from $h \leq u$, while on N both functions are zero.

Moreover,

$$\mu_0(h) = \mu_0(h\mathbf{1}_{N^c}) \leq \mu(u^N).$$

The equality follows from HW 44(i), since the two simple functions agree outside the null set N. The inequality follows from the definition of the positive integral, because $h\mathbf{1}_{N^c}$ is an admissible simple function below u^N.

Taking the supremum over all $h \in SF^+$ with $h \leq u$, we obtain

$$\mu(u) = \sup\{\mu_0(h) : h \in SF^+,\ h \leq u\} \leq \mu(u^N).$$

Conversely,

$$u^N \leq u \quad \Longrightarrow \quad \mu(u^N) \leq \mu(u).$$

The implication follows from HW 45. Combining the inequalities gives

$$\mu(u^N) = \mu(u).$$

Now suppose $f_n \uparrow f$ μ-almost everywhere. There exists a single measurable set N, with $\mu(N) = 0$, outside which both monotonicity and convergence hold. Define

$$\widehat{f_n} := f_n \mathbf{1}_{N^c}, \qquad \widehat{f} := f\mathbf{1}_{N^c}.$$

Then

$$\widehat{f_n}(s) \uparrow \widehat{f}(s) \qquad \forall s \in S.$$

On N^c, this is the original hypothesis; on N, all the modified functions are identically zero. By the result just proved,

$$\mu(\widehat{f_n}) = \mu(f_n) \quad \forall n \in \mathbb{N}, \qquad \mu(\widehat{f}) = \mu(f).$$

Thus, when proving a statement involving these integrals, we may work with the modified functions and assume pointwise monotone convergence.

This reduction is important in MON. Without it, the sets $E_n = \{ch \le f_n\}$ need not increase to S. For example, on $S = [0, 1]$ with Lebesgue measure, take $f = h = 1$, $c = 1/2$, and $f_n = \mathbf{1}_{(0,1]}$ for every n. Then $f_n \uparrow f$ almost everywhere, but $E_n = (0, 1]$ for every n. Monotonicity of the original E_n can also fail on the exceptional set. The reduction above eliminates both difficulties before E_n is defined.

Therefore, the statement is proved.

Theorem 5.2 — Monotone Convergence Theorem (MON)

Statement.

Let $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$ and let $f \in (m\Sigma)^+$ satisfy $f_n(s) \uparrow f(s)$ for μ-almost every $s \in S$. Then

$$\int_S f_n \, d\mu \uparrow \int_S f \, d\mu.$$

Expanded Proof.

By the preceding Observation, we may assume that $f_n \uparrow f$ everywhere, without changing the integrals.

Recall that $\mu(f) = \sup\{\mu_0(h) : h \in SF^+,\ h \le f\}$. We first obtain

$$\mu(f_n) \le \mu(f_{n+1}) \le \mu(f).$$

Both inequalities follow from HW 45, since $f_n \le f_{n+1} \le f$ pointwise. Thus $(\mu(f_n))_{n\in\mathbb{N}}$ is nondecreasing, and

$$\alpha := \lim_{n\to\infty} \mu(f_n) \le \mu(f).$$

It remains to prove the reverse inequality.

Fix $c \in (0, 1)$ and an arbitrary $h \in SF^+$ such that $h \le f$. Define

$$E_n := \{s \in S : ch(s) \le f_n(s)\} \in \Sigma.$$

The sets are measurable because h and f_n are measurable. As we will see in HW 47, $E_n \uparrow S$. Writing

$$h = \sum_{k=1}^{m} a_k \mathbf{1}_{A_k}, \qquad a_k \ge 0, \quad A_k \in \Sigma,$$

we obtain

$$\mu(f_n) \ge \int_S f_n \mathbf{1}_{E_n} \, d\mu \ge c \int_S h\mathbf{1}_{E_n} \, d\mu = c \sum_{k=1}^{m} a_k \mu(E_n \cap A_k).$$

The inequalities follow from $f_n \ge f_n\mathbf{1}_{E_n}$ and $f_n \ge ch$ on E_n, using HW 45. The scalar factor is extracted by HW 44(ii), since $h\mathbf{1}_{E_n}$ is simple; the last equality is its simple-function integral.

Since $E_n \uparrow S$, we have $E_n \cap A_k \uparrow A_k$. By continuity from below (Lemma 1.5),

$$\mu(E_n \cap A_k) \uparrow \mu(A_k).$$

Letting $n \to \infty$ in the previous inequality yields

$$\alpha \geq \lim_{n\to\infty} c \sum_{k=1}^{m} a_k \mu(E_n \cap A_k) = c \sum_{k=1}^{m} a_k \mu(A_k) = c\mu_0(h).$$

The first equality follows because the sum has finitely many nonnegative terms; the final equality follows from the definition of $\mu_0(h)$. The calculation also allows $\mu(A_k) = \infty$, with the convention $0 \cdot \infty = 0$.

Since $h \in SF^+$ with $h \leq f$ was arbitrary, taking the supremum gives

$$\alpha \geq c \sup\{\mu_0(h) : h \in SF^+,\ h \leq f\} = c\mu(f).$$

The equality follows from the definition of $\mu(f)$.

Finally, letting $c \uparrow 1$, we obtain $\alpha \geq \mu(f)$. If $\mu(f) = \infty$, the same conclusion already follows from $\alpha \geq c\mu(f) = \infty$ for any $c \in (0, 1)$.

Combining this with $\alpha \leq \mu(f)$, we conclude that

$$\mu(f_n) \uparrow \mu(f).$$

Therefore, the statement is proved.

HW 47 — Increasing Sets in MON

Statement. In the proof of Theorem 5.2, after the reduction to pointwise convergence, let $f_n \uparrow f$ everywhere. Fix $c \in (0, 1)$, $h \in SF^+$ with $h \leq f$, and define

$$E_n := \{s \in S : ch(s) \leq f_n(s)\} \in \Sigma.$$

Prove that

$$E_n \subseteq E_{n+1}, \qquad E_n \uparrow S.$$

Expanded Proof.

We first show that $E_n \subseteq E_{n+1}$. Let $s_0 \in E_n$. Then

$$ch(s_0) \leq f_n(s_0) \leq f_{n+1}(s_0).$$

The first inequality follows from the definition of E_n, while the second follows from the monotonicity of (f_n). Hence $s_0 \in E_{n+1}$, so $E_n \subseteq E_{n+1}$ $\forall n \in \mathbb{N}$.

We now prove that $\bigcup_{n=1}^{\infty} E_n = S$. Fix $s_0 \in S$.

If $h(s_0) = 0$, then

$$ch(s_0) = 0 \leq f_n(s_0) \qquad \forall n \in \mathbb{N},$$

since $f_n \geq 0$. Thus $s_0 \in E_n$ for every n.

If $h(s_0) > 0$, then

$$f(s_0) \geq h(s_0) > ch(s_0).$$

The first inequality follows from $h \leq f$, and the second from $c \in (0, 1)$. Since $f_n(s_0) \uparrow f(s_0)$, there exists $n_0 \in \mathbb{N}$ such that

$$f_n(s_0) \geq ch(s_0) \qquad \forall n \geq n_0.$$

This also holds when $f(s_0) = \infty$, because $ch(s_0)$ is finite and $f_n(s_0) \to \infty$.

In either case, $s_0 \in E_n$ for some n. Since $s_0 \in S$ was arbitrary,

$$\bigcup_{n=1}^{\infty} E_n = S.$$

Together with $E_n \subseteq E_{n+1}$, this gives

$$E_n \uparrow S.$$

Therefore, the statement is proved.

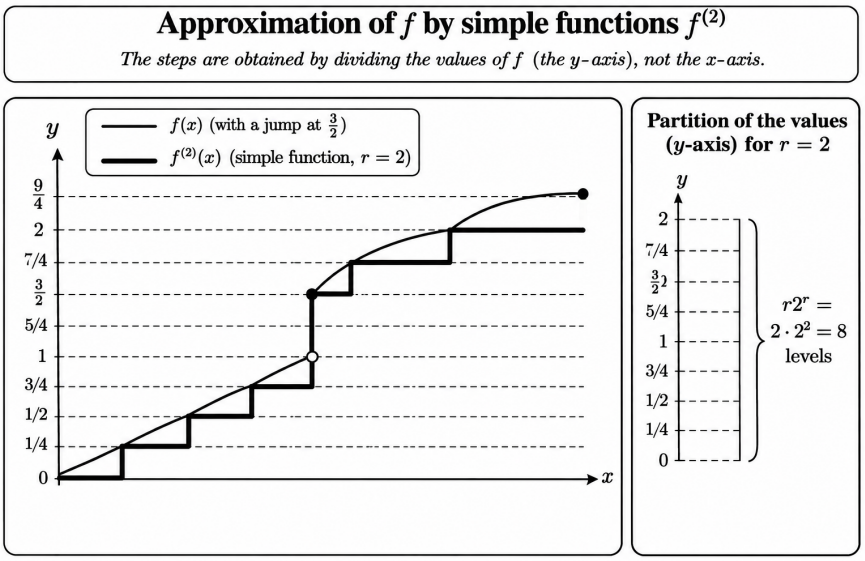
Approximation by Positive Simple Functions

From the monotone convergence theorem (Theorem 5.2), the next question arises: given $f \in (m\Sigma)^+$, how can we construct an increasing sequence $(f^{(r)})_{r\in\mathbb{N}} \subseteq SF^+$ that approximates f?
For $k, r \in \mathbb{N}$, define

$$A_k^r := \left\{ s \in S : \frac{k-1}{2^r} \le f(s) < \frac{k}{2^r} \right\}.$$

Set

$$f^{(r)}(s) = \sum_{k=1}^{r2^r} \frac{k-1}{2^r} \mathbf{1}_{A_k^r}(s) + r\mathbf{1}_{\{f\ge r\}}(s). \tag{5.1}$$



Verification.

First, $A_k^r, \{f \ge r\} \in \Sigma$ by measurability of f. The sets $A_1^r, \ldots, A_{r2^r}^r, \{f \ge r\}$ are pairwise disjoint and cover S: if $0 \le f(s) < r$, its value belongs to exactly one half-open interval $[(k-1)/2^r, k/2^r)$; if $f(s) \ge r$, it belongs to the last set. Hence $f^{(r)} \in SF^+$, since (5.1) is a finite sum of measurable indicators with nonnegative coefficients.
We now prove that $f^{(r)} \le f^{(r+1)} \le f$. For any $s \in S$ with $x := f(s) < \infty$, formula (5.1) gives

$$f^{(r)}(s) = \min\left\{ r, \frac{\lfloor 2^r x \rfloor}{2^r} \right\}.$$

Indeed, below r the coefficient is the left endpoint of the interval containing x, while at and above r the value is truncated at r. For every $x \ge 0$,

$$\frac{\lfloor 2^r x \rfloor}{2^r} \le \frac{\lfloor 2^{r+1} x \rfloor}{2^{r+1}} \le x.$$

The first inequality follows from $\lfloor 2^{r+1}x \rfloor \ge 2\lfloor 2^r x \rfloor$; the second is the defining property of the integer part. Since the truncation level also increases from r to $r+1$, we obtain $f^{(r)}(s) \le f^{(r+1)}(s) \le f(s)$. If $f(s) = \infty$, then $f^{(r)}(s) = r$, so the same inequalities hold.
Finally, for $f(s) < \infty$, choose $r > f(s)$. The truncation no longer applies and

$$0 \le f(s) - f^{(r)}(s) < \frac{1}{2^r},$$

by the defining bounds of the interval A_k^r containing s. Thus $f^{(r)}(s) \to f(s)$. If $f(s) = \infty$, then $f^{(r)}(s) = r \to \infty$. Consequently,

$$f^{(r)}(s) \uparrow f(s), \qquad \forall s \in S.$$

In particular, the monotone convergence theorem applies to this sequence.

Lemma 5.3 — Equality Almost Everywhere

Statement. If $f, g \in (m\Sigma)^+$ and $\mu(\{f \neq g\}) = 0$, then $\mu(f) = \mu(g)$.

Proof.

Let $(f^{(r)})_{r\in\mathbb{N}}$ and $(g^{(r)})_{r\in\mathbb{N}}$ be the simple-function approximations given by (5.1). By the construction above,

$$f^{(r)} \uparrow f, \qquad g^{(r)} \uparrow g.$$

Set $N := \{s \in S : f(s) \neq g(s)\} \in \Sigma$, so that $\mu(N) = 0$. For every $r \in \mathbb{N}$ and $s \notin N$,

$$f(s) = g(s) \quad \Longrightarrow \quad f^{(r)}(s) = g^{(r)}(s),$$

since (5.1) applies the same dyadic approximation to both values. Therefore,

$$\{f^{(r)} \neq g^{(r)}\} \subseteq N \quad \Longrightarrow \quad 0 \leq \mu(\{f^{(r)} \neq g^{(r)}\}) \leq \mu(N) = 0.$$

The implication follows from monotonicity of μ. Thus $f^{(r)} = g^{(r)}$ μ-a.e., and HW 44(i) gives

$$\mu(f^{(r)}) = \mu_0(f^{(r)}) = \mu_0(g^{(r)}) = \mu(g^{(r)}), \qquad \forall r \in \mathbb{N}.$$

The first and last equalities use the agreement of the positive Lebesgue integral with the simple integral.

Applying the monotone convergence theorem (Theorem 5.2), we conclude that

$$\mu(f) = \lim_{r\to\infty} \mu(f^{(r)}) = \lim_{r\to\infty} \mu(g^{(r)}) = \mu(g).$$

The first and last equalities follow from monotone convergence, and the middle one from equality of the integrals for every r.

Therefore, the statement is proved.

Lemma 5.4 — Fatou's Lemma for Functions

Statement. Let (S, Σ, μ) be a measure space. For any $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$, it holds

$$\liminf_{n\to\infty} \int_S f_n \, d\mu \geq \int_S \liminf_{n\to\infty} f_n \, d\mu.$$

Proof.

Recall that

$$\liminf_{n\to\infty} f_n = \sup_{n\geq 1} \inf_{m\geq n} f_m.$$

Define

$$g_n(s) := \inf_{m\geq n} f_m(s), \qquad n \in \mathbb{N}.$$

First, $g_n \in (m\Sigma)^+$, since $g_n \geq 0$ and, for every $a \in \mathbb{R}$,

$$\{g_n < a\} = \bigcup_{m=n}^{\infty} \{f_m < a\} \in \Sigma.$$

The equality follows from the definition of infimum; the membership follows from measurability of the f_m and closure of Σ under countable unions.

Moreover,

$$g_n \leq g_{n+1}, \qquad g_n \leq f_m \quad \forall m \geq n.$$

The first inequality follows because the infimum defining g_{n+1} is taken over a smaller family; the second follows from the definition of g_n. Consequently,

$$g_n(s) \uparrow \sup_{n\geq 1} g_n(s) = \liminf_{n\to\infty} f_n(s), \qquad \forall s \in S.$$

Since $g_n \leq f_m$ for every $m \geq n$, monotonicity of the positive Lebesgue integral (HW 45) gives

$$\int_S g_n \, d\mu \leq \int_S f_m \, d\mu, \qquad \forall m \geq n.$$

Taking the infimum over $m \geq n$, we obtain

$$\int_S g_n \, d\mu \leq \inf_{m\geq n} \int_S f_m \, d\mu.$$

Therefore,

$$\int_S \liminf_{n\to\infty} f_n \, d\mu = \lim_{n\to\infty} \int_S g_n \, d\mu \leq \lim_{n\to\infty} \inf_{m\geq n} \int_S f_m \, d\mu = \liminf_{n\to\infty} \int_S f_n \, d\mu.$$

The first equality follows from MON, the inequality from the preceding bound, and the last equality from the definition of the numerical lim inf.

Therefore, the statement is proved.

Lemma 5.5 — Reverse Fatou's Lemma for Functions

Statement. Let $(f_n)_{n\in\mathbb{N}} \subseteq (m\Sigma)^+$. Suppose there exists $g \in (m\Sigma)^+$ such that $f_n \leq g$ for all but finitely many $n \in \mathbb{N}$ and $\mu(g) < \infty$. Then

$$\limsup_{n\to\infty} \int_S f_n \, d\mu \leq \int_S \limsup_{n\to\infty} f_n \, d\mu.$$

Observation.

The hypothesis "$f_n \leq g$ for all but finitely many n" means that there is $n_0 \in \mathbb{N}$ such that

$$0 \leq f_n(s) \leq g(s), \qquad n \geq n_0, \ s \in S.$$

By the tail definitions recalled before HW 19, removing finitely many initial terms changes neither lim inf nor lim sup, pointwise or for the sequence of integrals: the tails eventually exclude every removed term.

Let $N := \{s \in S : g(s) = \infty\} \in \Sigma$; its measurability follows from that of g. For $M > 0$, if $s \in N$, then $\mathbf{1}_N(s) = 1$ and $M\mathbf{1}_N(s) = M \leq \infty = g(s)$. If $s \notin N$, then $\mathbf{1}_N(s) = 0$ and $M\mathbf{1}_N(s) = 0 \leq g(s)$, since $g \geq 0$. Thus $M\mathbf{1}_N \leq g$ on all of S, and

$$M\mu(N) = \mu_0(M\mathbf{1}_N) \leq \mu(g) < \infty.$$

The equality is the definition of the simple-function integral; the inequality is HW 45. If $\mu(N) > 0$, the left-hand side becomes arbitrarily large as $M \to \infty$ (and is already infinite if $\mu(N) = \infty$), contradicting $\mu(g) < \infty$. Hence $\mu(N) = 0$, so $g < \infty$ μ-almost everywhere.

We now modify g and every f_n by setting them equal to zero on N; we do not assert that the original functions vanished there. By Lemma 5.3, these modifications change none of their integrals, nor the integral of $\limsup_n f_n$, since the latter also agrees outside N. We retain the same notation for the modified functions below.

HW 48 — Proof of Reverse Fatou's Lemma for Functions

Statement. Prove Lemma 5.5 by applying Fatou's lemma to $g - f_n$.

Expanded Proof.

Use n_0 and the null-set modifications from the preceding Observation. For $n \in \mathbb{N}$, set

$$h_n := g - f_{n_0+n} \in (m\Sigma)^+.$$

The difference is well-defined, measurable and nonnegative because $0 \le f_{n_0+n} \le g < \infty$. Fatou's lemma (Lemma 5.4) gives

$$\mu\Big(\liminf_{n\to\infty} h_n\Big) \le \liminf_{n\to\infty} \mu(h_n).$$

Set $L := \limsup_{n\to\infty} f_n$. Pointwise, we have

$$\begin{aligned}\liminf_{n\to\infty} h_n(s) &= \liminf_{n\to\infty} \big(g(s) - f_{n_0+n}(s)\big) \\ &= g(s) - \limsup_{n\to\infty} f_{n_0+n}(s) = g(s) - L(s).\end{aligned}$$

The middle equality is HW 22: subtraction from finite $g(s)$ reverses infima and suprema. The last uses tail invariance from the Observation. Consequently, $0 \le L \le g$.

For integral subtraction without the later Lemma 5.6, take $u \in (m\Sigma)^+$ with $0 \le u \le g$ and set

$$v := g - u, \qquad \text{so that} \qquad g = u + v.$$

Since g is finite, $v \in (m\Sigma)^+$. For the increasing simple approximations $u^{(r)} \uparrow u$ and $v^{(r)} \uparrow v$ from formula (5.1), their simple sums satisfy

$$u^{(r)} + v^{(r)} \uparrow u + v = g.$$

Hence

$$\begin{aligned}\mu(g) &= \lim_{r\to\infty} \mu_0\big(u^{(r)} + v^{(r)}\big) \\ &= \lim_{r\to\infty} \big(\mu_0(u^{(r)}) + \mu_0(v^{(r)})\big) = \mu(u) + \mu(v).\end{aligned}$$

MON (Theorem 5.2), agreement of μ and μ_0 on simple functions, and addition of limits give the first and last equalities; HW 44(ii) gives the second. Since $\mu(g) < \infty$ and $v = g - u$, we obtain

$$\mu(g - u) = \mu(g) - \mu(u), \qquad 0 \le u \le g.$$

Applying this identity with $u = L$ and $u = f_{n_0+n}$ gives

$$\begin{aligned}\mu(g) - \mu(L) = \mu(g - L) &= \mu\Big(\liminf_{n\to\infty} h_n\Big) \\ &\le \liminf_{n\to\infty} \mu(h_n) = \liminf_{n\to\infty} \big(\mu(g) - \mu(f_{n_0+n})\big) \\ &= \mu(g) - \limsup_{n\to\infty} \mu(f_n).\end{aligned}$$

The first and third equalities use integral subtraction; the second, the pointwise identity. The inequality is the earlier Fatou application; the last equality is HW 22 with finite $\mu(g)$ and tail invariance. Subtracting $\mu(g)$ and multiplying by -1 gives

$$\limsup_{n\to\infty} \int_S f_n \, d\mu \le \int_S \limsup_{n\to\infty} f_n \, d\mu.$$

Therefore, the statement is proved.

Lemma 5.6 — Linearity of the Positive Lebesgue Integral

Statement. Let $\alpha, \beta \in [0, \infty)$ and $f, g \in (m\Sigma)^+$. Then

$$\int_S (\alpha f + \beta g) \, d\mu = \alpha \int_S f \, d\mu + \beta \int_S g \, d\mu.$$

Expanded Proof.

Let $(f^{(r)})_{r\in\mathbb{N}}$ and $(g^{(r)})_{r\in\mathbb{N}}$ be the sequences of nonnegative simple functions constructed in (5.1), so that

$$f^{(r)} \uparrow f, \qquad g^{(r)} \uparrow g.$$

Since $\alpha, \beta \ge 0$, we have

$$\alpha f^{(r)} + \beta g^{(r)} \uparrow \alpha f + \beta g.$$

Moreover, $\alpha f^{(r)} + \beta g^{(r)} \in SF^+$ by HW 44(ii).
We obtain

$$\int_S (\alpha f + \beta g)\, d\mu = \lim_{r\to\infty} \int_S (\alpha f^{(r)} + \beta g^{(r)})\, d\mu = \lim_{r\to\infty} \left(\alpha \int_S f^{(r)}\, d\mu + \beta \int_S g^{(r)}\, d\mu \right)$$
$$= \alpha \lim_{r\to\infty} \int_S f^{(r)}\, d\mu + \beta \lim_{r\to\infty} \int_S g^{(r)}\, d\mu = \alpha \int_S f\, d\mu + \beta \int_S g\, d\mu.$$

The first and last equalities follow from MON. The second uses HW 44(ii), since the approximating functions are simple, and the third follows from the limit rules for nonnegative sequences. The convention $0 \cdot \infty = 0$ applies when necessary.
Therefore, the statement is proved.

Positive and Negative Parts of a Measurable Function

We now extend the definition of the Lebesgue integral to measurable functions that can change sign.
Given $f \in m\Sigma$, define its positive and negative parts by

$$f^+ := \max\{f, 0\}, \qquad f^- := \max\{-f, 0\}.$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Both functions are nonnegative. Their measurability and the preceding identities are verified in HW 49.

HW 49 — Positive and Negative Parts

Statement. Let $f \in m\Sigma$. Prove that

$$f^+, f^- \in (m\Sigma)^+, \qquad |f| = f^+ + f^-.$$

Expanded Proof.
Both parts are nonnegative by definition.
We first verify measurability. For every $a \in \mathbb{R}$,

$$\{f^+ > a\} = \begin{cases} S, & a < 0, \\ \{f > a\}, & a \geq 0, \end{cases} \qquad \{f^- > a\} = \begin{cases} S, & a < 0, \\ \{f < -a\}, & a \geq 0. \end{cases}$$

All these sets belong to Σ, since f is measurable. Thus

$$f^+, f^- \in (m\Sigma)^+.$$

Finally, for every $s \in S$,

$$(f^+(s), f^-(s)) = \begin{cases} (f(s), 0), & f(s) \geq 0, \\ (0, -f(s)), & f(s) < 0. \end{cases}$$

In both cases,

$$f(s) = f^+(s) - f^-(s), \qquad |f(s)| = f^+(s) + f^-(s).$$

Therefore, the statement is proved.

Definition — Lebesgue Integral of a Signed Measurable Function

Let $f \in m\Sigma$. If

$$\min\{\mu(f^+), \mu(f^-)\} < \infty,$$

the Lebesgue integral of f is defined by

$$\mu(f) := \int_S f\, d\mu = \int_S f^+\, d\mu - \int_S f^-\, d\mu.$$

The condition ensures that at least one of the two nonnegative integrals is finite, so the subtraction is well-defined in the extended real numbers.

If both integrals are infinite, the expression would be $\infty - \infty$, which is undefined. If both are finite, then $\mu(f) \in \mathbb{R}$.

Definition 5.1 — L^1-Space

Let (S, Σ, μ) be a measure space. We denote by $L^1(S, \Sigma, \mu)$ the collection of measurable functions $f \in m\Sigma$ such that

$$\mu(|f|) = \int_S f^+ \, d\mu + \int_S f^- \, d\mu < \infty.$$

The equality follows from $|f| = f^+ + f^-$ (HW 49) and Lemma 5.6.

In particular, if $f \in L^1(S, \Sigma, \mu)$, both $\mu(f^+)$ and $\mu(f^-)$ are finite. Therefore,

$$\mu(f) = \mu(f^+) - \mu(f^-) \in \mathbb{R}.$$

Such functions are called Lebesgue integrable.

We work with measurable representatives. When $L^1(S, \Sigma, \mu)$ is viewed as a normed space, functions equal μ-almost everywhere are identified; the norm of their common equivalence class is $\|f\|_1 := \mu(|f|)$. Before this identification, the same formula defines a seminorm on the collection of integrable functions.

HW 50 — Absolute Value of the Integral

Statement. For $f \in L^1(S, \Sigma, \mu)$, prove that

$$|\mu(f)| \leq \mu(|f|).$$

Expanded Proof.

Both $\mu(f^+)$ and $\mu(f^-)$ are finite by Definition 5.1. We obtain

$$|\mu(f)| = |\mu(f^+) - \mu(f^-)| \leq \mu(f^+) + \mu(f^-) = \mu(|f|).$$

The first equality is the signed-integral definition; the inequality is the triangle inequality for finite nonnegative integrals, and the last equality is Definition 5.1.

Therefore, the statement is proved.

HW 51 — Closure of L^1 under Linear Combinations

Statement. Let $f, g \in L^1(S, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$. Prove that

$$\alpha f + \beta g \in L^1(S, \Sigma, \mu).$$

Expanded Proof.

The function $\alpha f + \beta g$ is measurable by Lemma 3.4. Moreover,

$$|\alpha f + \beta g| \leq |\alpha| \, |f| + |\beta| \, |g|.$$

The inequality is the pointwise triangle inequality. Therefore,

$$\begin{aligned} \mu(|\alpha f + \beta g|) &\leq \mu(|\alpha| \, |f| + |\beta| \, |g|) \\ &= |\alpha| \, \mu(|f|) + |\beta| \, \mu(|g|) \\ &< \infty. \end{aligned}$$

The inequality follows from HW 45, the equality from Lemma 5.6, and the last inequality from $f, g \in L^1(S, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$. Hence $\alpha f + \beta g \in L^1(S, \Sigma, \mu)$ by Definition 5.1. Therefore, the statement is proved.

Linearity of the Integral on L^1

Linearity extends to signed integrable functions. If $f, g \in L^1(S, \Sigma, \mu)$ and $\alpha, \beta \in \mathbb{R}$, then

$$\int_S (\alpha f + \beta g)\, d\mu = \alpha \int_S f\, d\mu + \beta \int_S g\, d\mu.$$

Expanded Proof.

By HW 51, all linear combinations below are integrable.

We first prove additivity. Set $u := f + g$. We have

$$u = u^+ - u^- \qquad \text{and} \qquad f + g = f^+ - f^- + g^+ - g^-,$$

and hence

$$u^+ + f^- + g^- = u^- + f^+ + g^+.$$

These are the signed decompositions rearranged pointwise. Integrating gives

$$\mu(u^+) + \mu(f^-) + \mu(g^-) = \mu(u^-) + \mu(f^+) + \mu(g^+).$$

Lemma 5.6 applies to the nonnegative measurable parts. Thus

$$\begin{aligned} \mu(f+g) &= \mu(u^+) - \mu(u^-) \\ &= \big(\mu(f^+) - \mu(f^-)\big) + \big(\mu(g^+) - \mu(g^-)\big) = \mu(f) + \mu(g). \end{aligned}$$

The outer equalities are the signed-integral definition; the middle equality rearranges the preceding identity. All terms are finite by integrability.

We now prove homogeneity. Let $c \in \mathbb{R}$. For $c \geq 0$, $(cf)^+ = cf^+$ and $(cf)^- = cf^-$. We obtain

$$\mu(cf) = \mu(cf^+) - \mu(cf^-) = c\mu(f^+) - c\mu(f^-) = c\mu(f).$$

The outer equalities use the signed-integral definition and these parts; the middle equality is Lemma 5.6.

For $c < 0$, we have

$$(-f)^+ = f^-, \qquad (-f)^- = f^+.$$

Hence,

$$\mu(-f) = \mu(f^-) - \mu(f^+) = -\mu(f).$$

The parts interchange under negation, and the signed-integral definition gives the integral identity. Writing $c = -|c|$ yields

$$\mu(cf) = \mu\big(|c|(-f)\big) = |c|\mu(-f) = -|c|\mu(f) = c\mu(f).$$

The second equality uses the nonnegative-scalar case; the others use $c = -|c|$ and the identity for $\mu(-f)$. Finally,

$$\mu(\alpha f + \beta g) = \mu(\alpha f) + \mu(\beta g) = \alpha\mu(f) + \beta\mu(g).$$

The equalities use additivity and homogeneity, respectively. Therefore, the statement is proved.

We will often need to interchange limits and integrals. One result of this kind is MON; the next theorem provides another important tool for doing so.

Theorem 5.7 — Dominated Convergence (DOM)

Statement. Suppose $(f_n)_{n\in\mathbb{N}} \subseteq m\Sigma$ and $f \in m\Sigma$ satisfy $f_n(s) \to f(s)$ for μ-almost every $s \in S$. If

$$|f_n| \leq g, \qquad \forall n \in \mathbb{N},$$

for some $g \in L^1(S, \Sigma, \mu)$, then

$$\lim_{n\to\infty} \int_S |f_n - f|\, d\mu = 0, \qquad \lim_{n\to\infty} \int_S f_n\, d\mu = \int_S f\, d\mu.$$

Proof.

Since $|f_n| \le g$ and $g \in L^1(S, \Sigma, \mu)$, monotonicity (HW 45) gives

$$0 \le \mu(|f_n|) \le \mu(g) < \infty, \qquad \forall n \in \mathbb{N}.$$

Thus every f_n is integrable. Let $N \in \Sigma$ be a null set outside which $f_n(s) \to f(s)$. Passing to the pointwise limit in $|f_n(s)| \le g(s)$ yields

$$|f(s)| \le g(s), \qquad \forall s \in N^c.$$

Replace f_n, f by $f_n \mathbf{1}_{N^c}, f\mathbf{1}_{N^c}$. Lemma 5.3, applied to their positive and negative parts, preserves their integrals. The modified functions satisfy $f_n \to f$ and $|f_n|, |f| \le g$ everywhere. Thus

$$\mu(|f|) \le \mu(g) < \infty,$$

so $f \in L^1(S, \Sigma, \mu)$ as well.

By the triangle inequality,

$$0 \le |f_n - f| \le |f_n| + |f| \le 2g, \qquad \forall n \in \mathbb{N}.$$

Since $g \ge 0$ and $g \in L^1$, Lemma 5.6 gives $\mu(2g) = 2\mu(g) < \infty$. We may therefore apply reverse Fatou's lemma (Lemma 5.5) to the sequence $|f_n - f|$:

$$0 \le \limsup_{n\to\infty} \int_S |f_n - f|\, d\mu \le \int_S \limsup_{n\to\infty} |f_n - f|\, d\mu = \int_S 0\, d\mu = 0.$$

The first inequality is nonnegativity of the integrals, the second follows from reverse Fatou and the bound by $2g$, and the equality uses $f_n \to f$ pointwise. Thus

$$\lim_{n\to\infty} \int_S |f_n - f|\, d\mu = 0.$$

Finally, by linearity of the integral on L^1 and HW 50,

$$\left| \int_S f_n\, d\mu - \int_S f\, d\mu \right| = \left| \int_S (f_n - f)\, d\mu \right| \le \int_S |f_n - f|\, d\mu \longrightarrow 0.$$

The inequality is valid since $f_n - f \in L^1$ by HW 51. This proves the second assertion.

Therefore, the statement is proved.

HW 52 — Why the Domination Hypothesis Is Necessary

Statement. Let $(S, \Sigma, \mu) = ([0,1], \mathcal{B}([0,1]), \mathrm{Leb})$ and define

$$f_n(s) = n\mathbf{1}_{(0,\,1/n)}(s), \qquad n \in \mathbb{N}.$$

Show that the conclusion of DOM fails without an integrable dominating function.

Proof.

Fix $s \in (0,1]$. For every $n \ge 1/s$, we have $s \notin (0, 1/n)$, so $f_n(s) = 0$. Moreover, $f_n(0) = 0$ for every n. Hence

$$f_n(s) \longrightarrow f(s) := 0, \qquad \forall s \in [0,1].$$

Nevertheless,

$$\int_0^1 f_n\, d\mathrm{Leb} = n\,\mathrm{Leb}\big((0, 1/n)\big) = 1, \qquad \forall n \in \mathbb{N},$$
$$\int_0^1 f\, d\mathrm{Leb} = 0.$$

The first line uses the simple-function integral and $\mathrm{Leb}((0,1/n)) = 1/n$; the second uses the integral of the zero function. Consequently,

$$\lim_{n\to\infty} \int_0^1 f_n\, d\mathrm{Leb} = 1 \ne 0 = \int_0^1 f\, d\mathrm{Leb}.$$

We now verify directly which hypothesis of DOM fails. Suppose that a function

$$g \in L^1([0,1], \mathcal{B}([0,1]), \mathrm{Leb})$$

satisfies $f_n \le g$ for every n. For each $k \in \mathbb{N}$, set

$$I_k := \left(\frac{1}{k+1}, \frac{1}{k}\right).$$

On I_k we have $f_k = k$, so $g \ge k$. Since the intervals I_k are pairwise disjoint, for every $N \in \mathbb{N}$,

$$\mu(g) \ge \mu\left(\sum_{k=1}^{N} k\mathbf{1}_{I_k}\right) = \sum_{k=1}^{N} k\,\mathrm{Leb}(I_k) = \sum_{k=1}^{N} k\left(\frac{1}{k} - \frac{1}{k+1}\right) = \sum_{k=1}^{N} \frac{1}{k+1}.$$

The inequality follows from HW 45, and the first equality uses the definition of the simple integral. As $N \to \infty$, the last sum diverges, contradicting $\mu(g) < \infty$.

Thus no integrable function dominates all f_n: the domination hypothesis cannot be omitted in general.

Therefore, the statement is proved.

The Standard Machinery

Many properties of Lebesgue integrals can be established through the following four steps:

1. Prove the property for indicators $\mathbf{1}_A$, where $A \in \Sigma$.
2. Use linearity to extend it to nonnegative simple functions.
3. Approximate nonnegative measurable functions by increasing simple functions and apply the monotone convergence theorem (MON, Theorem 5.2).
4. Use $f = f^+ - f^-$ to extend the result to signed measurable functions whenever the relevant integrals are well-defined.

This is the standard machinery used throughout the course.

A Measure Defined by a Nonnegative Density

Different measures on the same measurable space yield different Lebesgue integrals. Fix $f \in (m\Sigma)^+$ and define, for every $A \in \Sigma$,

$$\lambda(A) := \int_S f\mathbf{1}_A\, d\mu = \int_A f\, d\mu.$$

The following homework proves that $\lambda : \Sigma \to [0,\infty]$ is a measure. In addition, λ is absolutely continuous with respect to μ, written $\lambda \ll \mu$, meaning

$$\mu(N) = 0 \quad \Longrightarrow \quad \lambda(N) = 0, \qquad \forall N \in \Sigma.$$

HW 53 — Countable Additivity of λ

Statement. Let $f \in (m\Sigma)^+$ and define $\lambda(A) = \int_S f\mathbf{1}_A\, d\mu$ for every $A \in \Sigma$. Prove that λ is countably additive and hence is a measure on (S, Σ).

Proof.

First, $\lambda(A) \ge 0$ for every $A \in \Sigma$, since $f\mathbf{1}_A \ge 0$, and

$$\lambda(\emptyset) = \int_S f\mathbf{1}_\emptyset\, d\mu = \int_S 0\, d\mu = 0.$$

The equality follows from $\mathbf{1}_\emptyset = 0$ and the definition of the positive integral.

Let $(A_n)_{n\in\mathbb{N}} \subseteq \Sigma$ be pairwise disjoint. Set

$$U_r := \bigcup_{n=1}^{r} A_n, \qquad U := \bigcup_{n=1}^{\infty} A_n.$$

Since the sets are pairwise disjoint and $f \geq 0$,

$$f\mathbf{1}_{U_r} = \sum_{n=1}^{r} f\mathbf{1}_{A_n} \uparrow f\mathbf{1}_U \qquad \text{pointwise on } S.$$

Equivalently, for every $s \in S$,

$$f(s)\mathbf{1}_{U_r}(s) = \sum_{n=1}^{r} f(s)\mathbf{1}_{A_n}(s) \uparrow f(s)\mathbf{1}_U(s).$$

The equality follows from $\mathbf{1}_{U_r} = \sum_{n=1}^{r} \mathbf{1}_{A_n}$; the monotone convergence follows from $U_r \uparrow U$ and $f \geq 0$.

Therefore,

$$\lambda\left(\bigcup_{n=1}^{\infty} A_n\right) = \int_S f\mathbf{1}_U \, d\mu = \lim_{r\to\infty} \int_S f\mathbf{1}_{U_r} \, d\mu = \lim_{r\to\infty} \sum_{n=1}^{r} \int_S f\mathbf{1}_{A_n} \, d\mu = \sum_{n=1}^{\infty} \lambda(A_n).$$

The second equality follows from MON (Theorem 5.2), the third from nonnegative linearity (Lemma 5.6), and the last from the definition of an infinite nonnegative sum. Hence λ is countably additive and therefore is a measure on (S, Σ).

Moreover, let $N \in \Sigma$ satisfy $\mu(N) = 0$. Define

$$h_r := (f \wedge r)\mathbf{1}_N, \qquad \text{where } f \wedge r = \min\{f, r\}, \qquad r \in \mathbb{N}.$$

Then $h_r \in (m\Sigma)^+$, $h_r \uparrow f\mathbf{1}_N$, and $0 \leq h_r \leq r\mathbf{1}_N$. By monotonicity (HW 45),

$$0 \leq \mu(h_r) \leq \mu(r\mathbf{1}_N) = r\mu(N) = 0.$$

The equality uses the integral of a simple indicator. Applying MON once more gives

$$\lambda(N) = \int_S f\mathbf{1}_N \, d\mu = \lim_{r\to\infty} \mu(h_r) = 0.$$

Thus $\lambda \ll \mu$.

Therefore, the statement is proved.

Theorem 5.8 — Radon–Nikodym Theorem

Statement. Let μ and λ be σ-finite measures on (S, Σ) such that $\lambda \ll \mu$. Then there exists $f \in (m\Sigma)^+$ satisfying

$$\lambda(A) = \int_S f\mathbf{1}_A \, d\mu, \qquad \forall A \in \Sigma.$$

The function

$$f =: \frac{d\lambda}{d\mu}$$

is called the Radon–Nikodym derivative of λ with respect to μ.

The density is unique up to μ-almost-everywhere equality.

Recall.

A measure ν is σ-finite if there exist measurable sets $(E_n)_{n\in\mathbb{N}}$ such that

$$S = \bigcup_{n=1}^{\infty} E_n, \qquad \nu(E_n) < \infty \quad \forall n \in \mathbb{N}.$$

HW 53 constructs a measure λ from a given nonnegative density f; the Radon–Nikodym theorem gives the converse under σ-finiteness and absolute continuity. As specified in the lecture notes, the theorem is stated without proof.

Summary

A source-ordered summary of the construction of the Lebesgue integral and its main results, with additional clarifications motivated by our study discussions.

- **Why we need Lebesgue integration.** We began with the question of what $\mathbb{E}[X]$ means when a random variable is neither discrete nor described by a density. The Riemann integral is not designed for arbitrary measurable functions, so we built an integral with respect to a measure, general enough to define expectations.
- **Notation and counting measure.** We wrote $\mu(f) = \int_S f \, d\mu$ and $\mu(f; A) = \int_A f \, d\mu$. The counting-measure example tests the definition: on $S = \mathbb{N}$, integrating a nonnegative function should reproduce the corresponding series, connecting measure theory with familiar sums.
- **Indicators as building blocks.** For a measurable set A, we defined the integral of its indicator by $\mu_0(\mathbf{1}_A) = \mu(A)$. This turns integration into measurement of sets at the simplest level and supplies the elementary pieces from which the integral of a simple function is constructed.
- **Simple functions and HW 42.** We introduced nonnegative simple functions as finite sums $\sum_k a_k \mathbf{1}_{A_k}$ and checked their measurability in HW 42. Each takes only finitely many values, which allows its integral to be computed through finitely many measurable sets rather than a limiting procedure.
- **Canonical representation and zero times infinity.** A simple function can be written using different indicator sums; its canonical form groups points by distinct attained values. We also adopted $0 \cdot \infty = 0$, so the zero function has integral zero even on a space of infinite measure. Both choices make the definition coherent.
- **HW 43 — Well-defined simple integral.** Because a simple function may have two different representations, we needed to show that $\sum_i a_i \mu(A_i)$ gives the same answer in either form. HW 43 establishes this independence, using a common measurable refinement so that the proposed integral is genuinely a function of f.
- **HW 44 — Fundamental simple-function rules.** For nonnegative simple functions we established invariance under equality almost everywhere, positive linearity, monotonicity, and closure under minima and maxima. These are initially properties of μ_0, not yet of the integral of every nonnegative measurable function.
- **Extension to nonnegative measurable functions.** We defined $\mu(f)$ for any $f \in (m\Sigma)^+$ as the supremum of $\mu_0(h)$ over simple h satisfying $0 \le h \le f$. Rather than integrating a complicated function directly, we approach it from below by objects whose integrals are already defined.
- **HW 45 — Monotonicity of the extended integral.** If $0 \le f \le g$, every simple function lying below f also lies below g. Therefore the supremum defining $\mu(f)$ cannot exceed the one defining $\mu(g)$. HW 45 proves this basic inequality, which repeatedly supports later integral comparisons.
- **Lemma 5.1 and HW 46 — Zero integral.** We proved that $f \ge 0$ and $\mu(f) = 0$ force $f = 0$ almost everywhere. HW 46 identifies $\{f > 0\}$ with the increasing union of $\{f > 1/n\}$; if that union had positive measure, the lecturer's integral inequality would contradict $\mu(f) = 0$. This connects level sets directly to integral values.
- **MON — From almost everywhere to everywhere.** Before Theorem 5.2, we clarified that an exceptional null set may still contain points where the functions are nonzero. Mod-

ifying all f_n and f on one common null set preserves their integrals and lets us work with monotone pointwise convergence everywhere, avoiding a gap in the set argument.

- **Theorem 5.2 — Monotone convergence.** MON says that if nonnegative measurable $f_n \uparrow f$ almost everywhere, then $\mu(f_n) \uparrow \mu(f)$. The proof starts with $\mu(f_n) \leq \mu(f)$ and recovers the opposite inequality by testing against every simple $h \leq f$, making this the chapter's main limit tool.

- **HW 47 — Why the auxiliary sets fill the space.** After the common-null-set modification, for $0 < c < 1$ and simple $h \leq f$, we used $E_n = \{ch \leq f_n\}$. These sets increase, and every point enters at some stage: when $h(s) = 0$ it belongs immediately, while for $h(s) > 0$ convergence eventually gives the inequality. Thus $E_n \uparrow S$; no point must lie in every E_n.

- **Formula (5.1) — Simple approximation.** We built a specific increasing sequence $f^{(r)} \in SF^+$ using intervals of width 2^{-r} below the truncation level r. The sum has $r2^r$ terms because it partitions $[0, r)$ into that many intervals; the final indicator handles $f \geq r$. The approximations increase to f.

- **Lemma 5.3 versus HW 44(i).** HW 44(i) covered only two simple functions equal almost everywhere. Lemma 5.3 extends the same conclusion to arbitrary nonnegative measurable f, g: their corresponding simple approximations agree almost everywhere, so HW 44(i) applies at each stage and MON passes equality to the limit.

- **Lemma 5.4 — Fatou.** Fatou does not assume f_n is increasing. Instead we construct the increasing tail infima $g_n = \inf_{m \geq n} f_m$, which converge to $\liminf f_n$. Applying MON to g_n and using $g_n \leq f_m$ for all $m \geq n$ yields the comparison between the integral and the lim inf of integrals.

- **Lemma 5.5 — Reverse Fatou hypotheses.** Reverse Fatou asks for a nonnegative majorant g with finite integral and $f_n \leq g$ for all but finitely many indices. This means there is an n_0 after which domination always holds; deleting earlier terms changes neither lim inf nor lim sup, so we can work only with the dominated tail.

- **Reverse Fatou observation — Finiteness almost everywhere.** We also clarified why $\mu(g) < \infty$ implies $g < \infty$ almost everywhere. For $N = \{g = \infty\}$ and every $M > 0$, monotonicity gives $M\mu(N) \leq \mu(g)$; hence $\mu(N) = 0$. Lemma 5.3 then permits harmless modifications on N before writing $g - f_n$.

- **HW 48 — Bovo's proof and HW 22.** Following Bovo, we apply Fatou to $h_n = g - f_{n_0+n}$ after discarding the initial undominated terms. HW 22 explains $\liminf(g - f_n) = g - \limsup f_n$: subtraction from a finite constant reverses infima and suprema. We justify the required integral subtraction using HW 44(ii) and MON, not the later Lemma 5.6.

- **Lemma 5.6 — Positive linearity.** Now the general positive-linearity theorem is proved in its proper place. Approximate f and g by increasing simple functions, apply HW 44(ii) to their sums and nonnegative scalar multiples, and use MON. This extends finite-sum linearity to every nonnegative measurable integrand.

- **Signed functions and HW 49.** To integrate functions that may be negative, we split $f = f^+ - f^-$, where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. HW 49 verifies measurability and $|f| = f^+ + f^-$. The decomposition reduces signed integration to the nonnegative theory just developed.

- **Signed integral and the L^1 definition.** We define the signed integral as $\mu(f^+) - \mu(f^-)$ only when this does not produce the undefined quantity $\infty - \infty$. Definition 5.1 then singles

out L^1: measurable functions with $\mu(|f|) < \infty$, for which both positive and negative parts have finite integrals.

- **HW 50 — Absolute-value bound.** For $f \in L^1$, we verified $|\mu(f)| \le \mu(|f|)$. This follows from the positive/negative decomposition and the ordinary triangle inequality for finite real numbers. The estimate later controls differences of integrals and helps turn convergence in L^1 into convergence of expectations.
- **HW 51 — A short closure proof.** We simplified HW 51: Lemma 3.4 makes $\alpha f + \beta g$ measurable, and $|\alpha f + \beta g| \le |\alpha||f| + |\beta||g|$. HW 45 and Lemma 5.6 bound its absolute integral by $|\alpha|\mu(|f|) + |\beta|\mu(|g|) < \infty$, proving closure of L^1 without extra decompositions.
- **Linearity for integrable signed functions.** Linearity develops in three stages: HW 44(ii) treats nonnegative simple functions and coefficients; Lemma 5.6 uses MON to extend it to all nonnegative measurable functions, with nonnegative coefficients; linearity on L^1 then permits functions and real coefficients of either sign. HW 51 proves only L^1 closure; additivity and homogeneity then establish integral linearity, with finite integrals avoiding $\infty - \infty$.
- **Theorem 5.7 — Dominated convergence.** DOM replaces the monotonicity requirement with a single integrable bound $|f_n| \le g \in L^1$ and almost-everywhere convergence $f_n \to f$. For the functions modified on the common null set, $|f_n - f| \le 2g$ everywhere; Reverse Fatou gives $\mu(|f_n - f|) \to 0$, and the integral inequality from HW 50 yields convergence of the integrals.
- **HW 52 — Pointwise convergence alone fails.** On $[0, 1]$, the functions $f_n = n\mathbf{1}_{(0,1/n)}$ tend to zero everywhere, yet each has Lebesgue integral one. HW 52 shows that convergence alone cannot justify exchanging limit and integral; the missing hypothesis is a common integrable dominating function.
- **The standard machinery.** The lecturer then names the recurring four-stage strategy: prove the claim for indicators, extend to simple functions by linearity, pass to general nonnegative measurable functions by MON, and finally treat signed functions through $f^+ - f^-$. This explains the architecture of the chapter's proofs.
- **HW 53 — Building a measure from a density.** Given nonnegative measurable f, define $\lambda(A) = \int_S f\mathbf{1}_A \, d\mu$. HW 53 verifies countable additivity, so λ is a measure. It also satisfies $\lambda \ll \mu$: any μ-null set is λ-null, connecting integration with absolute continuity of measures.
- **Theorem 5.8 — Radon–Nikodym.** The chapter ends with the converse perspective: under the stated σ-finiteness assumptions, a measure $\lambda \ll \mu$ can be represented through a nonnegative density $d\lambda/d\mu$. The lecturer states the theorem without proof, linking changes of measure to integrals and setting up later probability applications.

6. Expectation

The integral developed in Chapter 5 now gives a rigorous definition of the expectation of a random variable. Taking $(S, \Sigma, \mu) = (\Omega, \mathcal{F}, \mathbb{P})$, we write

$$\mathbb{E}[X] = \int_\Omega X(\omega)\, d\mathbb{P}(\omega).$$

Here $X \in m\mathcal{F}$ is measurable. As in Chapter 5, the integral of a nonnegative random variable may equal $+\infty$, while the integral of a signed random variable is defined only when its positive and negative parts do not give the indeterminate expression $\infty - \infty$.

We denote the class of integrable random variables by

$$L^1(\Omega, \mathcal{F}, \mathbb{P}) := \{X \in m\mathcal{F} : \mathbb{E}[|X|] < \infty\}.$$

This is Definition 5.1 expressed with the probability measure $\mathbb{P}$. In particular, $X \in L^1$ has a finite expectation.

Recap — Properties of Expectation

The following properties have already been proved in Chapter 5. For $X, Y \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $\alpha, \beta \in \mathbb{R}$, linearity of the integral on L^1 gives

$$\mathbb{E}[\alpha X + \beta Y] = \alpha\mathbb{E}[X] + \beta\mathbb{E}[Y].$$

Moreover, if $X \le Y$, monotonicity of the integral gives

$$\mathbb{E}[X] \le \mathbb{E}[Y].$$

The first identity is the signed-integral linearity established after HW 51; the second is the corresponding order property for integrable functions.

Now suppose $(X_n)_{n\in\mathbb{N}} \subseteq m\mathcal{F}$. The convergence results from Chapter 5 become the following statements in expectation notation.

Monotone Convergence (MON, Theorem 5.2). If $0 \le X_n \uparrow X$ (almost surely), then

$$\mathbb{E}[X_n] \uparrow \mathbb{E}[X].$$

Fatou's Lemma (Lemma 5.4). If $X_n \ge 0$ for every $n \in \mathbb{N}$, then

$$\liminf_{n\to\infty} \mathbb{E}[X_n] \ge \mathbb{E}\left[\liminf_{n\to\infty} X_n\right].$$

Dominated Convergence (DOM, Theorem 5.7). If $|X_n(\omega)| \le Y(\omega)$ for every $(n, \omega) \in \mathbb{N} \times \Omega$, with $\mathbb{E}[Y] < \infty$, and $\mathbb{P}(X_n \to X) = 1$ for some $X \in m\mathcal{F}$, then

$$\lim_{n\to\infty} \mathbb{E}[|X_n - X|] = 0.$$

The condition $|X_n| \le Y$ ensures $Y \ge 0$; its finite expectation makes Y an integrable dominating random variable. Thus the preceding results are exactly MON, Fatou and DOM from Chapter 5, with μ replaced by $\mathbb{P}$.

Lemma 6.1 — Markov's Inequality

Statement. Let $Z \in m\mathcal{F}$ and let $g : \mathbb{R} \to [0, \infty]$ be Borel-measurable and non-decreasing. Then $g(Z) \in m\mathcal{F}$ is nonnegative and, for every $c \in \mathbb{R}$,

$$\mathbb{E}[g(Z)] \ge \mathbb{E}\big[g(Z)\mathbf{1}_{\{Z\ge c\}}\big] \ge g(c)\mathbb{P}(Z \ge c).$$

Expanded Proof.

The composition $g(Z)$ is measurable by Lemma 3.3, since Z is $\mathcal{F}$-measurable and g is Borel-measurable. Also, $g(Z) \ge 0$. Fix $c \in \mathbb{R}$. Because Z is measurable, the events $\{Z \ge c\}$ and $\{Z < c\}$ belong to $\mathcal{F}$. We obtain

$$\mathbb{E}[g(Z)] = \mathbb{E}\big[g(Z)\mathbf{1}_{\{Z\ge c\}} + g(Z)\mathbf{1}_{\{Z<c\}}\big] \ge \mathbb{E}\big[g(Z)\mathbf{1}_{\{Z\ge c\}}\big] \ge \mathbb{E}\big[g(c)\mathbf{1}_{\{Z\ge c\}}\big] = g(c)\mathbb{P}(Z \ge c).$$

The first equality holds because the two events partition Ω. The first inequality follows from nonnegativity and monotonicity of the integral. For the second inequality, $Z(\omega) \geq c$ implies $g(Z(\omega)) \geq g(c)$ because g is non-decreasing; multiplying by $\mathbf{1}_{\{Z \geq c\}}$ gives the required pointwise inequality. For the final equality, set $A := \{Z \geq c\}$. If $g(c) < \infty$, then $\mathbb{E}[g(c)\mathbf{1}_A] = g(c)\mathbb{P}(A)$ by the integral of a simple function. If $g(c) = \infty$, the function $g(c)\mathbf{1}_A$ is understood to equal ∞ on A and zero on A^c. For $M \in \mathbb{N}$, we have $M\mathbf{1}_A \uparrow g(c)\mathbf{1}_A$; therefore, by MON (Theorem 5.2),

$$\begin{aligned}\mathbb{E}[g(c)\mathbf{1}_A] &= \lim_{M\to\infty} \mathbb{E}[M\mathbf{1}_A] \\ &= \lim_{M\to\infty} M\mathbb{P}(A) \\ &= \begin{cases} 0, & \mathbb{P}(A) = 0, \\ +\infty, & \mathbb{P}(A) > 0. \end{cases}\end{aligned}$$

This is exactly the meaning of $g(c)\mathbb{P}(A)$ when $g(c) = \infty$, using the convention $0 \cdot \infty = 0$ only if $\mathbb{P}(A) = 0$. Therefore, the statement is proved.

Example 4 — Applications of Markov's Inequality

Statements. The following consequences of Lemma 6.1 hold.

(1) For $c > 0$, if $Z \in (m\mathcal{F})^+$, then

$$\mathbb{P}(Z \geq c) \leq \frac{1}{c}\mathbb{E}[Z].$$

(2) If $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $c > 0$, then

$$\mathbb{P}(|X| \geq c) \leq \frac{1}{c}\mathbb{E}[|X|].$$

(3) For $Z \in m\mathcal{F}$, $\theta > 0$, and any $c \in \mathbb{R}$, taking $g(z) = e^{\theta z}$ gives

$$\mathbb{P}(Z \geq c) \leq e^{-\theta c}\mathbb{E}[e^{\theta Z}].$$

Derivation of (1).

Apply Lemma 6.1 to Z with $g(z) := z \vee 0 = \max\{z, 0\}$. Since $Z \geq 0$, we have $g(Z) = Z$ and $g(c) = c$ for $c > 0$. Hence

$$\mathbb{E}[Z] \geq c\,\mathbb{P}(Z \geq c),$$

which yields (1) after division by c. The same inequality holds if Z may take the value $+\infty$, because $c\,\mathbf{1}_{\{Z \geq c\}} \leq Z$ and the integral is monotone.

Derivation of (2).

Apply (1) to $Z := |X|$. The variable $|X|$ is measurable and nonnegative, while $X \in L^1$ ensures $\mathbb{E}[|X|] < \infty$.

Derivation of (3).

The function $g(z) = e^{\theta z}$ is Borel-measurable, positive, and non-decreasing because $\theta > 0$. Lemma 6.1 gives

$$\mathbb{E}[e^{\theta Z}] \geq e^{\theta c}\mathbb{P}(Z \geq c).$$

Multiplying by $e^{-\theta c} > 0$ gives (3); the result remains valid when the expectation is infinite.

Properties — Nonnegative Random Variables

We state four useful properties. Properties (a)–(c) are proved in HW 54; property (d) follows from (c).

(a) Finite expectation implies almost-sure finiteness. If $X \in (m\mathcal{F})^+$ and $\mathbb{E}[X] < \infty$, then

$$\mathbb{P}(X < \infty) = 1.$$

(b) Expectation of a nonnegative series. If $(Z_k)_{k\in\mathbb{N}} \subseteq (m\mathcal{F})^+$, then

$$\mathbb{E}\left[\sum_{k=1}^{\infty} Z_k\right] = \sum_{k=1}^{\infty} \mathbb{E}[Z_k].$$

Both sides may equal $+\infty$.

(c) Summable expectations imply almost-sure convergence. If $(Z_k)_{k\in\mathbb{N}} \subseteq (m\mathcal{F})^+$ and $\sum_{k=1}^{\infty} \mathbb{E}[Z_k] < \infty$, then

$$\sum_{k=1}^{\infty} Z_k < \infty \qquad \mathbb{P}\text{-a.s.},$$

and in particular $Z_k \to 0$ $\mathbb{P}$-almost surely.

(d) Application to indicators — Borel–Cantelli I. If $(F_k)_{k\in\mathbb{N}} \subseteq \mathcal{F}$ and $\sum_{k=1}^{\infty} \mathbb{P}(F_k) < \infty$, take $Z_k = \mathbf{1}_{F_k}$. Then (c) gives

$$\sum_{k=1}^{\infty} \mathbf{1}_{F_k} < \infty \qquad \mathbb{P}\text{-a.s.}$$

For each $\omega \in \Omega$, the sum counts how many sets F_k contain ω. Thus the conclusion means that almost every ω belongs to only finitely many F_k. This is the first Borel–Cantelli lemma (BC1), using property (c) as we will prove in HW 54, because

$$\left\{\sum_{k=1}^{\infty} \mathbf{1}_{F_k} = \infty\right\} = \limsup_{k\to\infty} F_k, \qquad \mathbb{P}\left(\limsup_{k\to\infty} F_k\right) = 0.$$

The event equality follows from the definition of $\limsup F_k$: it contains precisely the outcomes belonging to infinitely many F_k.

HW 54 — Proofs of Properties (a)–(c)

Statement. Prove properties (a)–(c) above.

Proof of (a).

Let $X \in (m\mathcal{F})^+$ with $\mathbb{E}[X] < \infty$. For each $n \in \mathbb{N}$, we obtain

$$0 \le \mathbb{P}(X = \infty) \le \mathbb{P}(X \ge n) \le \frac{1}{n}\mathbb{E}[X] \xrightarrow[n\to\infty]{} 0.$$

The first inequality follows from $\{X = \infty\} \subseteq \{X \ge n\}$; the second is Example 4(1). Since $\mathbb{E}[X]$ is finite, $\mathbb{P}(X = \infty) = 0$, hence $\mathbb{P}(X < \infty) = 1$.

Proof of (b).

Let $(Z_k)_{k\in\mathbb{N}} \subseteq (m\mathcal{F})^+$, and set

$$S_n := \sum_{k=1}^{n} Z_k, \qquad S := \sum_{k=1}^{\infty} Z_k.$$

Every S_n is nonnegative and measurable, and $S_n \uparrow S$ pointwise. Therefore,

$$\mathbb{E}\left[\sum_{k=1}^{\infty} Z_k\right] = \lim_{n\to\infty} \mathbb{E}[S_n] = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{E}[Z_k] = \sum_{k=1}^{\infty} \mathbb{E}[Z_k].$$

The first equality follows from MON (Theorem 5.2), the second from positive linearity (Lemma 5.6), and the last from the definition of an infinite series of nonnegative numbers.

Proof of (c).

Under the assumption $\sum_{k=1}^{\infty} \mathbb{E}[Z_k] < \infty$, property (b) gives

$$\mathbb{E}\left[\sum_{k=1}^{\infty} Z_k\right] = \sum_{k=1}^{\infty} \mathbb{E}[Z_k] < \infty.$$

The random variable $S = \sum_{k=1}^{\infty} Z_k$ is measurable and nonnegative as the increasing limit of measurable nonnegative partial sums. Applying (a) to S, we obtain

$$\mathbb{P}\left(\sum_{k=1}^{\infty} Z_k < \infty\right) = 1.$$

Fix ω in this probability-one event. The partial sums $S_n(\omega) = \sum_{k=1}^{n} Z_k(\omega)$ converge to a finite limit, so

$$Z_n(\omega) = S_n(\omega) - S_{n-1}(\omega) \xrightarrow[n\to\infty]{} 0.$$

The last limit follows because both successive partial sums converge to the same finite number. Hence $Z_n \to 0$ $\mathbb{P}$-almost surely, as required.

Theorem 6.2 — Jensen's Inequality

Statement. Let $c : \mathbb{R} \to \mathbb{R}$ be a convex function. Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ satisfy $\mathbb{E}[|c(X)|] < \infty$. Then

$$\mathbb{E}[c(X)] \geq c(\mathbb{E}[X]).$$

Expanded Proof.

Fix $x_0 := \mathbb{E}[X] \in \mathbb{R}$. The geometric idea is that a convex function admits a supporting line through $(x_0, c(x_0))$. We establish this property from convexity. For any $a < x_0 < b$, convexity gives

$$\frac{c(x_0) - c(a)}{x_0 - a} \leq \frac{c(b) - c(x_0)}{b - x_0}.$$

Indeed, $x_0 = \frac{b-x_0}{b-a}a + \frac{x_0-a}{b-a}b$, and rearranging the resulting convexity inequality gives the displayed result. Consequently, the supremum of the secant slopes to the left of x_0 does not exceed the infimum of those to its right. Both are finite: the left supremum is bounded above by the slope to $x_0 + 1$, and the right infimum is bounded below by the slope from $x_0 - 1$. Choose a finite number m between them. For every $x \in \mathbb{R}$,

$$c(x) \geq c(x_0) + m(x - x_0).$$

For $x < x_0$ this follows by multiplying the left-slope inequality by the negative number $x - x_0$; for $x > x_0$ it follows directly from the right-slope inequality. Equality holds at $x = x_0$. Applying this supporting-line inequality to X and taking expectations, we obtain

$$\mathbb{E}[c(X)] \geq \mathbb{E}[c(x_0) + m(X - x_0)] = c(x_0) + m\big(\mathbb{E}[X] - x_0\big) = c(\mathbb{E}[X]).$$

The inequality follows from monotonicity of expectation; the first equality uses linearity on L^1, applicable since X and $c(X)$ are integrable and m is finite. The last equality follows from the definition of x_0. Therefore, the statement is proved.

Definition — L^p Spaces of Random Variables

For $1 \leq p < \infty$, we say that $X \in L^p(\Omega, \mathcal{F}, \mathbb{P})$ if $X \in m\mathcal{F}$ and

$$\|X\|_p := \big(\mathbb{E}[|X|^p]\big)^{1/p} < \infty.$$

The quantity $\|X\|_p$ is called the L^p-norm. HW 55 verifies the norm properties, with equality of random variables understood almost surely.

HW 55 — The L^p Norm

Statement. For $1 \leq p < \infty$, prove that $\|\cdot\|_p$ is nonnegative, that $\|X\|_p = 0$ if and only if $X = 0$, that $\|\alpha X\|_p = |\alpha|\|X\|_p$ for $\alpha \in \mathbb{R}$, and that

$$\|X + Y\|_p \leq \|X\|_p + \|Y\|_p \qquad \text{for } X, Y \in L^p(\Omega, \mathcal{F}, \mathbb{P}).$$

Here $X = 0$ means $X = 0$ $\mathbb{P}$-almost surely: on individual measurable functions this is only a seminorm, while it is a norm on L^p classes modulo almost-sure equality.

Expanded Proof.

Let $X, Y \in L^p(\Omega, \mathcal{F}, \mathbb{P})$. Since $|X|^p \geq 0$, its expectation is nonnegative and thus $\|X\|_p \geq 0$. Moreover,

$$\|X\|_p = 0 \iff \mathbb{E}[|X|^p] = 0 \iff |X|^p = 0 \quad \mathbb{P}\text{-a.s.} \iff X = 0 \quad \mathbb{P}\text{-a.s.}$$

The middle equivalence follows from Lemma 5.1 for the nonnegative random variable $|X|^p$; the last follows because $p \geq 1$.

For any $\alpha \in \mathbb{R}$,

$$\|\alpha X\|_p = \big(\mathbb{E}[|\alpha X|^p]\big)^{1/p} = \big(|\alpha|^p \mathbb{E}[|X|^p]\big)^{1/p} = |\alpha| \|X\|_p.$$

The second equality uses the pointwise identity $|\alpha X|^p = |\alpha|^p |X|^p$ and homogeneity of the positive integral (Lemma 5.6).

For the triangle inequality, set $a := \|X\|_p$ and $b := \|Y\|_p$. If $a = 0$ or $b = 0$, the corresponding random variable vanishes almost surely, and Lemma 5.3 gives the required equality of norms. Assume therefore $a, b > 0$ and put $\lambda := a/(a+b) \in (0,1)$. The function $t \mapsto t^p$ is convex on $[0, \infty)$ for $p \geq 1$ (it is linear for $p = 1$, and its second derivative is nonnegative for $p > 1$). Hence, pointwise,

$$\left(\frac{|X| + |Y|}{a+b}\right)^p = \left(\lambda \frac{|X|}{a} + (1-\lambda)\frac{|Y|}{b}\right)^p \leq \lambda \frac{|X|^p}{a^p} + (1-\lambda)\frac{|Y|^p}{b^p}.$$

The equality follows from $\lambda = a/(a+b)$; the inequality is the convexity inequality applied to the nonnegative numbers $|X|/a$ and $|Y|/b$. Taking expectations gives

$$\mathbb{E}\left[\left(\frac{|X| + |Y|}{a+b}\right)^p\right] \leq \lambda \frac{\mathbb{E}[|X|^p]}{a^p} + (1-\lambda)\frac{\mathbb{E}[|Y|^p]}{b^p} = \lambda + (1-\lambda) = 1.$$

The inequality follows from monotonicity and positive linearity of expectation (Lemma 5.6); the equality uses $\mathbb{E}[|X|^p] = a^p$ and $\mathbb{E}[|Y|^p] = b^p$. Since $|X + Y| \leq |X| + |Y|$ pointwise, it follows that

$$\|X + Y\|_p = \big(\mathbb{E}[|X+Y|^p]\big)^{1/p} \leq \big(\mathbb{E}[(|X| + |Y|)^p]\big)^{1/p} \leq a + b = \|X\|_p + \|Y\|_p.$$

The first inequality follows from the pointwise triangle inequality and monotonicity; the second is the expectation bound just established. In particular, this argument also shows that $X + Y \in L^p$, so all terms are well-defined. Therefore, the norm properties hold on almost-sure equivalence classes. The statement is proved.

Lemma 6.3 — Monotonicity of L^p-Norms

Statement. Fix $1 \leq p < r < \infty$. If $Y \in L^p(\Omega, \mathcal{F}, \mathbb{P})$ and $Y \in L^r(\Omega, \mathcal{F}, \mathbb{P})$, then

$$\|Y\|_p \leq \|Y\|_r.$$

Expanded Proof.

For $n \in \mathbb{N}$, set

$$X_n(\omega) := \big(|Y(\omega)| \wedge n\big)^p, \qquad \text{where } a \wedge b := \min\{a, b\}.$$

Since Y is measurable and $0 \leq X_n \leq n^p$, we have $X_n \in L^1$. Moreover, $X_n^{r/p} = (|Y| \wedge n)^r \leq n^r$, so $X_n^{r/p} \in L^1$ as well. As $r/p > 1$, the function $c(x) := (x \vee 0)^{r/p}$, where $x \vee 0 := \max\{x, 0\}$, is convex on $\mathbb{R}$ and satisfies $c(X_n) = X_n^{r/p}$. Thus Jensen's inequality (Theorem 6.2) gives

$$\mathbb{E}[X_n^{r/p}] \geq \big(\mathbb{E}[X_n]\big)^{r/p}.$$

Letting $n \to \infty$, we obtain

$$\begin{aligned} \mathbb{E}[|Y|^r] &\geq \big(\mathbb{E}[|Y|^p]\big)^{r/p} \\ \implies \quad \big(\mathbb{E}[|Y|^r]\big)^{1/r} &\geq \big(\mathbb{E}[|Y|^p]\big)^{1/p}. \end{aligned}$$

Indeed, $X_n \uparrow |Y|^p$ and $X_n^{r/p} \uparrow |Y|^r$ pointwise, so the first inequality follows from MON (Theorem 5.2) and continuity of $t \mapsto t^{r/p}$ on $[0, \infty)$. The implication follows by taking the increasing $1/r$-power on both sides. By the definition of the L^p-norm, this is precisely $\|Y\|_p \leq \|Y\|_r$. Therefore, the statement is proved.

HW 56 — Closure of L^p under Linear Combinations

Statement. Let $1 \leq p < \infty$, $X, Y \in L^p(\Omega, \mathcal{F}, \mathbb{P})$ and $\alpha, \beta \in \mathbb{R}$. Prove that

$$\alpha X + \beta Y \in L^p(\Omega, \mathcal{F}, \mathbb{P}).$$

Expanded Proof.

The random variable $\alpha X + \beta Y$ is measurable by Lemma 3.4. We have

$$\begin{aligned} \|\alpha X + \beta Y\|_p &\leq \|\alpha X\|_p + \|\beta Y\|_p \\ &= |\alpha| \, \|X\|_p + |\beta| \, \|Y\|_p \\ &< \infty. \end{aligned}$$

The first inequality is the triangle inequality from HW 55, and the equality is its homogeneity property. Both scaled random variables belong to L^p, and the final expression is finite because $X, Y \in L^p$. Therefore $\alpha X + \beta Y \in L^p$ by the definition of L^p. The statement is proved.

Remark — Vector Space Structure of L^p

HW 56 shows that L^p is closed under linear combinations, and hence is a vector space (with equality understood almost surely when using the L^p-norm). The next result provides another important inequality.

Theorem 6.4 — Cauchy–Schwarz Inequality

Statement. If $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then $XY \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and

$$\mathbb{E}[|XY|] \leq \|X\|_2 \, \|Y\|_2.$$

Expanded Proof.

For $n \in \mathbb{N}$, set

$$X_n := |X| \wedge n, \qquad Y_n := |Y| \wedge n, \qquad \text{where } a \wedge b := \min\{a, b\}.$$

These random variables are measurable, nonnegative, and bounded by n. For any $\gamma \in \mathbb{R}$, we obtain

$$0 \leq \mathbb{E}[(\gamma X_n + Y_n)^2] = \gamma^2 \mathbb{E}[X_n^2] + 2\gamma \mathbb{E}[X_n Y_n] + \mathbb{E}[Y_n^2] =: F_n(\gamma).$$

The inequality holds because the square is nonnegative, while the identity follows by expanding the square and using linearity of expectation on L^1. All terms are integrable since X_n, Y_n are bounded random variables on a probability space.

If $\mathbb{E}[X_n^2] = 0$, then $X_n = 0$ $\mathbb{P}$-almost surely by Lemma 5.1. Hence $\mathbb{E}[X_n Y_n] = 0$, so the required inequality holds for this n. Otherwise $\mathbb{E}[X_n^2] > 0$, and the parabola F_n has its unique minimum at

$$\gamma_* = -\frac{\mathbb{E}[X_n Y_n]}{\mathbb{E}[X_n^2]}.$$

Substituting γ_* gives

$$\begin{aligned} 0 \leq F_n(\gamma_*) = \mathbb{E}[Y_n^2] - \frac{\big(\mathbb{E}[X_n Y_n]\big)^2}{\mathbb{E}[X_n^2]} \\ \implies \quad \mathbb{E}[X_n Y_n] \leq \big(\mathbb{E}[X_n^2] \, \mathbb{E}[Y_n^2]\big)^{1/2}. \end{aligned}$$

The implication follows by multiplying by the positive number $\mathbb{E}[X_n^2]$ and taking square roots; both sides are nonnegative because $X_n, Y_n \geq 0$.

Since $X_nY_n \uparrow |XY|$, $X_n^2 \uparrow |X|^2$, and $Y_n^2 \uparrow |Y|^2$ pointwise, MON (Theorem 5.2) yields

$$\mathbb{E}[|XY|] = \lim_{n\to\infty} \mathbb{E}[X_nY_n] \le \lim_{n\to\infty} \big(\mathbb{E}[X_n^2]\,\mathbb{E}[Y_n^2]\big)^{1/2} = \big(\mathbb{E}[|X|^2]\,\mathbb{E}[|Y|^2]\big)^{1/2} = \|X\|_2\,\|Y\|_2 < \infty.$$

The first equality and the penultimate equality follow from MON; the latter also uses continuity of the square root and the finiteness of the two second moments. Since XY is measurable by Lemma 3.4 and $\mathbb{E}[|XY|] < \infty$, we have $XY \in L^1$. Therefore, the statement is proved.

HW 57 — The Triangle Inequality in L^2 via Cauchy–Schwarz

Statement. Let $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Use Theorem 6.4 to prove that

$$\|X + Y\|_2 \le \|X\|_2 + \|Y\|_2.$$

Expanded Proof.

By HW 56, $X + Y \in L^2$, and by Theorem 6.4, $XY \in L^1$. We obtain

$$\begin{aligned}\|X+Y\|_2^2 &= \mathbb{E}[(X+Y)^2] = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] \\ &\le \mathbb{E}[X^2] + 2\mathbb{E}[|XY|] + \mathbb{E}[Y^2] \le \|X\|_2^2 + 2\|X\|_2\,\|Y\|_2 + \|Y\|_2^2 = \big(\|X\|_2 + \|Y\|_2\big)^2.\end{aligned}$$

The second equality follows by expanding the square and using linearity of the integral on L^1. The first inequality follows from $XY \le |XY|$ and monotonicity of expectation; the second is Cauchy–Schwarz (Theorem 6.4). All expectations are finite under the stated hypotheses. Taking nonnegative square roots gives

$$\|X + Y\|_2 \le \|X\|_2 + \|Y\|_2.$$

Therefore, the statement is proved.

Variance and Covariance

Let $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. By Cauchy–Schwarz (Theorem 6.4), $XY \in L^1$. Moreover, since $\mathbf{1}_\Omega \in L^2$ and $\|\mathbf{1}_\Omega\|_2 = 1$, the same theorem gives

$$\mathbb{E}[|X|] \le \|X\|_2 < \infty, \qquad \mathbb{E}[|Y|] \le \|Y\|_2 < \infty.$$

Thus $X, Y \in L^1$, consistently with the monotonicity of norms, and their means

$$\mu_X := \mathbb{E}[X], \qquad \mu_Y := \mathbb{E}[Y]$$

are finite. Define the centered random variables

$$\widetilde{X}(\omega) := X(\omega) - \mu_X, \qquad \widetilde{Y}(\omega) := Y(\omega) - \mu_Y.$$

They belong to L^2 by HW 56, since the constants μ_X and μ_Y belong to L^2 on a probability space. By linearity of expectation on L^1,

$$\mathbb{E}[\widetilde{X}] = \mathbb{E}[X] - \mu_X = 0, \qquad \mathbb{E}[\widetilde{Y}] = \mathbb{E}[Y] - \mu_Y = 0.$$

We define the variance and covariance by

$$\mathrm{Var}(X) := \mathbb{E}[\widetilde{X}^2], \qquad \mathrm{Var}(Y) := \mathbb{E}[\widetilde{Y}^2], \qquad \mathrm{Cov}(X,Y) := \mathbb{E}[\widetilde{X}\widetilde{Y}].$$

These quantities are finite: the centered variables are in L^2, and their product is in L^1 by Theorem 6.4. Expanding the products and using linearity gives

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2] - \mu_X^2, \qquad \mathrm{Var}(Y) = \mathbb{E}[Y^2] - \mu_Y^2, \\ \mathrm{Cov}(X,Y) &= \mathbb{E}[XY] - \mu_X\mu_Y.\end{aligned}$$

Applying Cauchy–Schwarz to $\widetilde{X}, \widetilde{Y}$ yields

$$|\,\mathrm{Cov}(X,Y)| = \big|\mathbb{E}[\widetilde{X}\widetilde{Y}]\big| \le \mathbb{E}[|\widetilde{X}\widetilde{Y}|] \le \big(\mathbb{E}[\widetilde{X}^2]\big)^{1/2}\big(\mathbb{E}[\widetilde{Y}^2]\big)^{1/2} = \mathrm{Var}(X)^{1/2}\,\mathrm{Var}(Y)^{1/2}.$$

The first inequality follows from $|\mathbb{E}[W]| \le \mathbb{E}[|W|]$ (HW 50); the second is Theorem 6.4, and the last equality follows from the definitions of variance. If both variances are positive, we may divide to obtain the lecturer's nonnegative correlation coefficient

$$cc := \frac{|\operatorname{Cov}(X,Y)|}{\operatorname{Var}(X)^{1/2}\operatorname{Var}(Y)^{1/2}} \le 1.$$

This is the absolute value of the usual signed correlation coefficient. If either variance is zero, the quotient is undefined, although the covariance inequality remains valid and forces $\operatorname{Cov}(X,Y) = 0$.

Scalar Product in L^2

By HW 56, L^2 is a vector space. Define

$$\langle X, Y\rangle := \mathbb{E}[XY], \qquad X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P}).$$

The scalar product is well-defined since $XY \in L^1$ by Theorem 6.4. It is symmetric and bilinear, because multiplication is commutative and expectation is linear on L^1:

$$\langle X, Y\rangle = \langle Y, X\rangle, \qquad \langle \alpha X + \beta Y, Z\rangle = \alpha\langle X, Z\rangle + \beta\langle Y, Z\rangle.$$

All the products appearing here are integrable by HW 56 and Theorem 6.4. Moreover,

$$\langle X, X\rangle = \mathbb{E}[X^2] = \|X\|_2^2 \ge 0,$$

with equality precisely when $X = 0$ almost surely, by HW 55. Thus it is an inner product on L^2 modulo almost-sure equality (and only a positive semidefinite form on individual representatives). We say that $X, Y \in L^2$ are orthogonal if $\langle X, Y\rangle = 0$.

Pythagoras' theorem. For $X, Y \in L^2$,

$$\|X + Y\|_2^2 = \|X\|_2^2 + \|Y\|_2^2 \quad \Longleftrightarrow \quad \langle X, Y\rangle = 0.$$

HW 58 — Proof of Pythagoras' Theorem

Statement. Prove the preceding equivalence for $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$.

Expanded Proof.

By HW 56, $X + Y \in L^2$, and by Theorem 6.4, $XY \in L^1$. We obtain

$$\|X + Y\|_2^2 = \mathbb{E}[(X + Y)^2] = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] = \|X\|_2^2 + 2\langle X, Y\rangle + \|Y\|_2^2.$$

The second equality follows by expanding the square and applying linearity of expectation on L^1; the last follows from the definitions of the L^2-norm and scalar product. All quantities are finite. Therefore,

$$\|X + Y\|_2^2 = \|X\|_2^2 + \|Y\|_2^2 \iff 2\langle X, Y\rangle = 0 \iff \langle X, Y\rangle = 0.$$

Both implications follow by subtracting the finite squared norms from the preceding identity. Therefore, the statement is proved.

Completeness — Cauchy Sequences in L^p

Fix $1 \le p < \infty$. A sequence $(X_n)_{n\in\mathbb{N}} \subseteq L^p(\Omega, \mathcal{F}, \mathbb{P})$ is called a Cauchy sequence if for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that

$$\|X_j - X_k\|_p \le \varepsilon, \qquad \forall j, k \ge n_\varepsilon.$$

The space L^p is called complete if every Cauchy sequence converges in L^p-norm to some $X \in L^p$, that is,

$$\lim_{n\to\infty} \|X_n - X\|_p = 0.$$

As observed in HW 55, L^p is a normed space when random variables equal $\mathbb{P}$-almost surely are identified.

Lemma 6.5 — Completeness of L^p

Statement. For every $1 \le p < \infty$, the space $L^p(\Omega, \mathcal{F}, \mathbb{P})$ is complete.

Expanded Proof.

Let $(X_n)_{n\in\mathbb{N}} \subseteq L^p$ be a Cauchy sequence. For each $n \ge 0$, choose an index $k_n \in \mathbb{N}$, with $k_{n+1} > k_n$, such that

$$\|X_r - X_s\|_p \le \frac{1}{2^n}, \qquad \forall r, s \ge k_n.$$

Such indices exist by the Cauchy property; each k_n may be chosen larger than the preceding one. In particular, since $k_{n+1} \ge k_n$, we obtain

$$\mathbb{E}[|X_{k_{n+1}} - X_{k_n}|] = \|X_{k_{n+1}} - X_{k_n}\|_1 \le \|X_{k_{n+1}} - X_{k_n}\|_p \le \frac{1}{2^n}.$$

The middle inequality is the monotonicity of norms (Lemma 6.3, with equality when $p = 1$). The L^1-norm exists: for $p > 1$, $|z| \le 1 + |z|^p$ and $\mathbb{P}(\Omega) = 1$ show that every L^p random variable is also in L^1.

Set $D_n := |X_{k_{n+1}} - X_{k_n}|$. Then

$$\mathbb{E}\left[\sum_{n=0}^{\infty} D_n\right] = \sum_{n=0}^{\infty} \mathbb{E}[D_n] \le \sum_{n=0}^{\infty} \frac{1}{2^n} < \infty.$$

The equality follows from HW 54(b), and the inequality from the preceding estimate; the geometric sum is finite, as we will see in HW 59. By HW 54(c), $\sum_{n=0}^{\infty} D_n < \infty$ $\mathbb{P}$-almost surely. Hence the series of signed increments converges absolutely and, by telescoping,

$$X_{k_m} = X_{k_0} + \sum_{n=0}^{m-1} (X_{k_{n+1}} - X_{k_n}) \longrightarrow X \qquad \mathbb{P}\text{-a.s.}$$

Define X to be this limit on the event of convergence and zero elsewhere. This event is measurable, and X is measurable because it is the almost-sure limit of measurable random variables, with the indicated definition on the exceptional measurable null set.

Fix $n \ge 0$, take $m \ge n$ and $r \ge k_n$. Since $k_m \ge k_n$, the choice of k_n gives

$$\mathbb{E}[|X_r - X_{k_m}|^p] = \|X_r - X_{k_m}\|_p^p \le 2^{-np}.$$

Letting $m \to \infty$ and applying Fatou's lemma (Lemma 5.4) to the nonnegative functions $|X_r - X_{k_m}|^p$, we obtain

$$\mathbb{E}[|X_r - X|^p] = \mathbb{E}\left[\liminf_{m\to\infty} |X_r - X_{k_m}|^p\right] \le \liminf_{m\to\infty} \mathbb{E}[|X_r - X_{k_m}|^p] \le 2^{-np}, \qquad r \ge k_n.$$

The first equality uses $X_{k_m} \to X$ almost surely and invariance of nonnegative integrals under almost-sure equality (Lemma 5.3); the inequality is Fatou, followed by the bound valid for every $m \ge n$.

In particular, taking $n = 0$ and $r = k_0$ shows that $X - X_{k_0} \in L^p$. Since $X_{k_0} \in L^p$, HW 56 implies $X \in L^p$. Finally, taking p-th roots in the preceding estimate yields

$$\|X_r - X\|_p \le 2^{-n}, \qquad \forall r \ge k_n.$$

As $n \to \infty$, the right-hand side tends to zero; hence $\|X_r - X\|_p \to 0$ as $r \to \infty$. Thus the original Cauchy sequence converges to X in L^p, and L^p is complete. Therefore, the statement is proved.

HW 59 — Geometric Series

Statement. Calculate

$$\sum_{n=0}^{\infty} \frac{1}{2^n}.$$

Expanded Proof.

Set $S_N := \sum_{n=0}^{N} 2^{-n}$. We obtain

$$S_N - \frac{1}{2}S_N = 1 - \frac{1}{2^{N+1}},$$
$$S_N = 2\left(1 - \frac{1}{2^{N+1}}\right) \xrightarrow[N\to\infty]{} 2.$$

The first equality follows by subtracting the shifted finite sum from S_N; the limit follows since $2^{-(N+1)} \to 0$. Therefore,

$$\sum_{n=0}^{\infty} \frac{1}{2^n} = 2.$$

The statement is proved.

Lemma 6.6 — Expectation with Respect to the Law of a Random Variable

How does the abstract integration theory recover the classical computation of expectations? Recall that the law of X is the probability measure $\mathcal{L}_X = \mathbb{P} \circ X^{-1}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Statement. Let $h : \mathbb{R} \to \mathbb{R}$ be Borel-measurable and let $X \in m\mathcal{F}$. Then $h(X) \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ if and only if $h \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$. In that case,

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)\, \mathcal{L}_X(dx).$$

Expanded Proof.

We follow the standard machinery, beginning with indicators. Since h is Borel-measurable and X is a random variable, the composition $h(X)$ is measurable.

If $h = \mathbf{1}_A$ for $A \in \mathcal{B}(\mathbb{R})$, then

$$\mathbb{E}[h(X)] = \mathbb{E}[\mathbf{1}_A(X)] = \mathbb{P}(X \in A) = \mathcal{L}_X(A) = \int_{\mathbb{R}} \mathbf{1}_A(x)\, \mathcal{L}_X(dx).$$

The first equality substitutes $h = \mathbf{1}_A$; the second is the expectation of an indicator, the third is the definition of $\mathcal{L}_X$, and the last is the integral of an indicator.

If $h \in SF^+$, write $h = \sum_{j=1}^{m} a_j \mathbf{1}_{A_j}$ with $a_j \geq 0$ and $A_j \in \mathcal{B}(\mathbb{R})$. Then

$$\mathbb{E}[h(X)] = \sum_{j=1}^{m} a_j\, \mathbb{E}[\mathbf{1}_{A_j}(X)] = \sum_{j=1}^{m} a_j\, \mathcal{L}_X(A_j) = \int_{\mathbb{R}} h(x)\, \mathcal{L}_X(dx).$$

The first equality uses linearity for nonnegative simple functions (HW 44(ii)); the second is the indicator case, and the last is the definition of the simple-function integral.

Now let $h : \mathbb{R} \to [0, \infty]$ be Borel-measurable. Take simple approximations $h^{(r)} \uparrow h$ from formula (5.1). Then $h^{(r)}(X) \uparrow h(X)$ pointwise, and MON (Theorem 5.2) gives

$$\mathbb{E}[h(X)] = \lim_{r\to\infty} \mathbb{E}[h^{(r)}(X)] = \lim_{r\to\infty} \int_{\mathbb{R}} h^{(r)}(x)\, \mathcal{L}_X(dx) = \int_{\mathbb{R}} h(x)\, \mathcal{L}_X(dx).$$

The first and last equalities follow from MON, respectively under $\mathbb{P}$ and $\mathcal{L}_X$; the middle equality is the simple-function case. These equalities are valid even when the integrals are infinite.

Finally, apply the nonnegative case to $|h|$. We obtain

$$\mathbb{E}[|h(X)|] = \int_{\mathbb{R}} |h(x)|\, \mathcal{L}_X(dx).$$

Since $h(X)$ and h are measurable, the two sides are finite simultaneously. This proves both directions of the stated L^1 equivalence. When they are finite, apply the nonnegative result to h^+ and h^-. Thus

$$\mathbb{E}[h(X)] = \mathbb{E}[h^+(X)] - \mathbb{E}[h^-(X)] = \int_{\mathbb{R}} h^+(x)\, \mathcal{L}_X(dx) - \int_{\mathbb{R}} h^-(x)\, \mathcal{L}_X(dx) = \int_{\mathbb{R}} h(x)\, \mathcal{L}_X(dx).$$

The first and last equalities follow from the definition of the signed integral; the middle equality follows from the nonnegative case. Both subtractions involve finite quantities. Therefore, the statement is proved.

Special Case — Expectation Using a Probability Density

Suppose that X admits a probability density function f_X with respect to Lebesgue measure, so that for every $A \in \mathcal{B}(\mathbb{R})$,

$$\mathcal{L}_X(A) = \int_A f_X(x)\,dx.$$

For an indicator $h = \mathbf{1}_A$, this is precisely the equality

$$\int_{\mathbb{R}} \mathbf{1}_A(x)\,\mathcal{L}_X(dx) = \int_{\mathbb{R}} \mathbf{1}_A(x) f_X(x)\,dx.$$

The same standard machinery—linearity for simple functions, MON for nonnegative measurable functions, and positive/negative parts for integrable signed functions—extends it to any $h \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathcal{L}_X)$. Consequently,

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)\,\mathcal{L}_X(dx) = \int_{\mathbb{R}} h(x) f_X(x)\,dx.$$

In the terminology of the Radon–Nikodym theorem (Theorem 5.8), f_X is a Radon–Nikodym derivative of $\mathcal{L}_X$ with respect to Lebesgue measure:

$$f_X = \frac{d\mathcal{L}_X}{d\mathrm{Leb}} = \frac{d\mathcal{L}_X}{dx} \qquad \text{Leb-a.e.}$$

The derivative is determined only up to Lebesgue-null sets; the formula is the usual density-based computation of expectation.

Lemma 6.7 — Hölder's and Minkowski's Inequalities

Statement. Let $1 < p < \infty$ and $q > 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Suppose $f, g \in L^p(S, \Sigma, \mu)$ and $h \in L^q(S, \Sigma, \mu)$. Then the following inequalities hold.

Hölder's inequality:

$$|\mu(fh)| \le \mu(|fh|) \le \|f\|_p \, \|h\|_q.$$

In particular, $fh \in L^1(S, \Sigma, \mu)$.

Minkowski's inequality:

$$\|f + g\|_p \le \|f\|_p + \|g\|_p.$$

Expanded Proof of Hölder's Inequality.

We first establish the elementary inequality

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}, \qquad a, b \ge 0.$$

Let $t := b^{q-1}$. The supporting-line inequality for the convex function $x \mapsto x^p/p$ on $[0, \infty)$ gives

$$\frac{a^p}{p} \ge \frac{t^p}{p} + t^{p-1}(a - t) = ab - \frac{b^q}{q}.$$

The last equality uses $t^{p-1} = b$, $t^p = b^q$, and $1 - 1/p = 1/q$. Rearranging proves the displayed inequality.

Put $A := \|f\|_p$ and $B := \|h\|_q$, which are finite. If $A = 0$ or $B = 0$, Lemma 5.1 implies that $f = 0$ or $h = 0$ μ-almost everywhere, respectively. Thus $fh = 0$ almost everywhere and $\mu(|fh|) = 0$ by

Lemma 5.3. Otherwise $A, B > 0$. Applying the preceding inequality pointwise with $a = |f|/A$ and $b = |h|/B$, we obtain

$$\frac{|fh|}{AB} \le \frac{|f|^p}{pA^p} + \frac{|h|^q}{qB^q}.$$

Taking nonnegative integrals yields

$$\frac{\mu(|fh|)}{AB} \le \frac{\mu(|f|^p)}{pA^p} + \frac{\mu(|h|^q)}{qB^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

The inequality follows from HW 45 and the equality from positive linearity (Lemma 5.6) and the definitions $A^p = \mu(|f|^p)$ and $B^q = \mu(|h|^q)$. Consequently $\mu(|fh|) \le AB < \infty$, so $fh \in L^1$. By HW 50,

$$|\mu(fh)| \le \mu(|fh|) \le \|f\|_p \, \|h\|_q.$$

This proves Hölder's inequality.

Expanded Proof of Minkowski's Inequality.

By the pointwise triangle inequality and convexity of $x \mapsto x^p$ on $[0, \infty)$,

$$|f+g|^p \le (|f| + |g|)^p = 2^p \left(\frac{|f| + |g|}{2} \right)^p \le 2^{p-1}(|f|^p + |g|^p).$$

The first inequality is the ordinary triangle inequality, and the last follows from convexity with weights $1/2$. Its integral is finite by HW 45 and Lemma 5.6, since $f, g \in L^p$; hence $f + g \in L^p$. Set $C := \|f+g\|_p < \infty$. If $C = 0$, the desired inequality is immediate. Otherwise, pointwise,

$$|f+g|^p \le (|f| + |g|)\, |f+g|^{p-1}.$$

Moreover, $|f+g|^{p-1} \in L^q$, because $(p-1)q = p$, and

$$\big\| |f+g|^{p-1} \big\|_q = \big(\mu(|f+g|^p)\big)^{1/q} = C^{p-1}.$$

Applying nonnegative linearity and the Hölder inequality already proved, we obtain

$$C^p = \mu(|f+g|^p) \le \mu\big(|f|\,|f+g|^{p-1}\big) + \mu\big(|g|\,|f+g|^{p-1}\big) \le \big(\|f\|_p + \|g\|_p\big) C^{p-1}.$$

The first inequality follows from the preceding pointwise bound and Lemma 5.6, while the second applies Hölder to each product. Dividing by $C^{p-1} > 0$ gives

$$\|f+g\|_p \le \|f\|_p + \|g\|_p.$$

Therefore, both inequalities are proved.

Lemma 6.8 — Chebyshev's Inequality

Statement. If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $c > 0$, then

$$\mathbb{P}\big(|X - \mathbb{E}[X]| \ge c\big) \le \frac{\mathrm{Var}(X)}{c^2}.$$

The proof uses Markov's inequality for the squared deviation, as we will see in HW 60.

HW 60 — Proof of Chebyshev's Inequality

Statement. Prove Lemma 6.8.

Expanded Proof.

Let $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $c > 0$. By Lemma 6.3, $X \in L^1$, so $m := \mathbb{E}[X]$ is finite. The centered variable $X - m$ belongs to L^2 by HW 56; therefore $Z := (X-m)^2$ is nonnegative, measurable, and integrable, with $\mathbb{E}[Z] = \mathrm{Var}(X)$. We obtain

$$\mathbb{P}\big(|X - \mathbb{E}[X]| \ge c\big) = \mathbb{P}\big((X-m)^2 \ge c^2\big) \le \frac{\mathbb{E}[(X-m)^2]}{c^2} = \frac{\mathrm{Var}(X)}{c^2}.$$

The first equality uses $c > 0$ and $m = \mathbb{E}[X]$; the inequality is Markov's inequality (Example 4(1)) applied to Z with the positive threshold c^2, and the last equality is the definition of variance.

Therefore, the statement is proved. The next theorem shows how independence impacts the covariance of random variables.

Theorem 6.9 — Independence, Expectation, and Variance

Statement. Let $X, Y \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ be independent random variables. Then $XY \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

If, in addition, $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then

$$\mathrm{Cov}(X, Y) = 0, \qquad \mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Expanded Proof.

We first prove the expectation identity for $X, Y \geq 0$, following the simple-function approximation in formula (5.1). Take $X^{(r)} \uparrow X$ and $Y^{(r)} \uparrow Y$, with

$$X^{(r)}(\omega) = \sum_{i=1}^{k_r} x_i \mathbf{1}_{A_i}(\omega), \qquad Y^{(r)}(\omega) = \sum_{j=1}^{\ell_r} y_j \mathbf{1}_{B_j}(\omega),$$

where $x_i, y_j \geq 0$, $A_i \in \sigma(X)$ and $B_j \in \sigma(Y)$. The level sets in formula (5.1) belong to $\sigma(X)$ and $\sigma(Y)$, respectively, because they are pre-images of Borel sets under X and Y. We obtain

$$\mathbb{E}[X^{(r)}Y^{(r)}] = \sum_{i=1}^{k_r}\sum_{j=1}^{\ell_r} x_i y_j\, \mathbb{P}(A_i \cap B_j) = \sum_{i=1}^{k_r}\sum_{j=1}^{\ell_r} x_i y_j\, \mathbb{P}(A_i)\mathbb{P}(B_j) = \mathbb{E}[X^{(r)}]\,\mathbb{E}[Y^{(r)}].$$

The first equality follows by expanding the product and integrating the resulting simple function. The second follows because independence of X and Y means independence of $\sigma(X)$ and $\sigma(Y)$; the last follows from the simple-function integral and finite factorisation of the double sum.

Since both approximations are nonnegative and increasing, $X^{(r)}Y^{(r)} \uparrow XY$ pointwise. Consequently,

$$\mathbb{E}[XY] = \lim_{r\to\infty} \mathbb{E}[X^{(r)}Y^{(r)}] = \lim_{r\to\infty} \big(\mathbb{E}[X^{(r)}]\mathbb{E}[Y^{(r)}]\big) = \mathbb{E}[X]\,\mathbb{E}[Y].$$

The first equality follows from MON (Theorem 5.2); the second is the simple-function identity above. The last uses MON for each factor and continuity of multiplication, since both expectations are finite.

For general signed $X, Y \in L^1$, apply the nonnegative case first to $|X|$ and $|Y|$. These random variables are independent because they are measurable functions of X and Y, respectively. Thus

$$\mathbb{E}[|XY|] = \mathbb{E}[|X|]\,\mathbb{E}[|Y|] < \infty,$$

so $XY \in L^1$. Next write

$$XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+.$$

Each product is integrable because it is nonnegative and bounded by $|XY|$. The positive and negative parts of X and Y remain independent across the two variables, so the nonnegative identity applies to all four products. Using linearity of expectation on L^1, we conclude that

$$\begin{aligned}\mathbb{E}[XY] &= \mathbb{E}[X^+]\mathbb{E}[Y^+] + \mathbb{E}[X^-]\mathbb{E}[Y^-] \\ &\quad - \mathbb{E}[X^+]\mathbb{E}[Y^-] - \mathbb{E}[X^-]\mathbb{E}[Y^+] \\ &= \big(\mathbb{E}[X^+] - \mathbb{E}[X^-]\big)\big(\mathbb{E}[Y^+] - \mathbb{E}[Y^-]\big) \\ &= \mathbb{E}[X]\,\mathbb{E}[Y].\end{aligned}$$

The first equality follows from the preceding product decomposition and the nonnegative case, the second by factoring finite quantities, and the last by the definition of signed expectation.

Finally, suppose $X, Y \in L^2$. Their means and second moments are finite, and the covariance identity gives

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Expanding the variance, we obtain

$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathbb{E}[(X+Y)^2] - \big(\mathbb{E}[X] + \mathbb{E}[Y]\big)^2 \\ &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\big(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]\big) = \mathrm{Var}(X) + \mathrm{Var}(Y).\end{aligned}$$

The first equality is the variance definition and linearity of expectation. The second follows by expanding the finite square terms and collecting them; the last uses the covariance identity just proved. Therefore, the statement is proved.

Remark — Dependence Does Not Ensure Integrability of a Product

If $X, Y \in L^1$ are not independent, their product need not belong to L^1. For example, on $((0,1), \mathcal{B}((0,1)), \mathrm{Leb})$, let $X(\omega) = Y(\omega) = \omega^{-1/2}$. Then

$$\mathbb{E}[|X|] = \mathbb{E}[|Y|] = \int_0^1 \omega^{-1/2}\, d\omega = 2 < \infty, \qquad \mathbb{E}[|XY|] = \int_0^1 \omega^{-1}\, d\omega = \infty.$$

Thus $X, Y \in L^1$ but $XY \notin L^1$.

Introduction — A Strong Law of Large Numbers

We next prove a version of the Strong Law of Large Numbers under stronger assumptions than necessary, namely a uniform bound on the fourth moments. These stronger assumptions allow a simpler proof.

Theorem 6.10 — A Strong Law of Large Numbers

Statement. Let $(X_n)_{n\in\mathbb{N}} \subseteq L^4(\Omega, \mathcal{F}, \mathbb{P})$ be independent random variables such that

$$\mathbb{E}[X_n] = 0, \qquad \sup_{n\in\mathbb{N}} \mathbb{E}[X_n^4] \le K$$

for some $K > 0$. Setting $S_n := \sum_{i=1}^n X_i$, we have

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{S_n}{n} = 0\right) = 1.$$

Expanded Proof.

We first expand the fourth moment of S_n. We obtain

$$\mathbb{E}[S_n^4] = \mathbb{E}\left[\left(\sum_{i=1}^n X_i\right)^4\right] = \sum_{i=1}^n \mathbb{E}[X_i^4] + 6 \sum_{1\le i<j\le n} \mathbb{E}[X_i^2 X_j^2] = \sum_{i=1}^n \mathbb{E}[X_i^4] + 6 \sum_{1\le i<j\le n} \mathbb{E}[X_i^2]\mathbb{E}[X_j^2].$$

In the expansion, terms containing an index appearing exactly once have zero expectation by independence and $\mathbb{E}[X_i] = 0$. For instance, when $i \neq j$,

$$\mathbb{E}[X_i^3 X_j] = \mathbb{E}[X_i^3]\mathbb{E}[X_j] = 0,$$

and the same reasoning applies to terms with three or four distinct indices. Thus only fourth powers and products $X_i^2 X_j^2$ remain; the coefficient 6 counts the ways to choose the two positions of X_i. The last equality uses independence of the squares. All products are integrable because each product of four absolute factors is bounded by one quarter of the sum of their fourth powers. Applying Jensen's inequality (Theorem 6.2) to X_i^2 with the convex function $c(x) = x^2$, we obtain

$$\mathbb{E}[X_i^2] \le \big(\mathbb{E}[X_i^4]\big)^{1/2} \le K^{1/2}.$$

The first inequality follows by taking square roots in $(\mathbb{E}[X_i^2])^2 \le \mathbb{E}[X_i^4]$; the second uses the uniform fourth-moment bound. Consequently,

$$\mathbb{E}[S_n^4] \le nK + 6\binom{n}{2}K = nK + 3n(n-1)K \le 3Kn^2.$$

The first inequality uses the preceding moment bounds; the equality uses $\binom{n}{2} = n(n-1)/2$, and the last inequality holds for $n \ge 1$.

Consider the nonnegative series $\sum_{n=1}^{\infty}(S_n/n)^4$. We obtain

$$\mathbb{E}\left[\sum_{n=1}^{\infty}\left(\frac{S_n}{n}\right)^4\right] = \lim_{m\to\infty}\mathbb{E}\left[\sum_{n=1}^{m}\left(\frac{S_n}{n}\right)^4\right] = \lim_{m\to\infty}\sum_{n=1}^{m}\frac{\mathbb{E}[S_n^4]}{n^4} \le 3K\sum_{n=1}^{\infty}\frac{1}{n^2} < \infty.$$

The first equality follows from MON (Theorem 5.2), since the partial sums increase pointwise. The second uses finite linearity of expectation; the inequality uses $\mathbb{E}[S_n^4] \le 3Kn^2$. The last series converges: for $n \ge 2$, $1/n^2 \le 1/[n(n-1)]$, and the latter series telescopes.

By HW 54(a), a nonnegative random variable with finite expectation is finite almost surely. Therefore,

$$\sum_{n=1}^{\infty}\left(\frac{S_n}{n}\right)^4 < \infty \qquad \mathbb{P}\text{-a.s.}$$

For every outcome on this probability-one event, the terms of the convergent nonnegative series tend to zero. Hence

$$\left(\frac{S_n}{n}\right)^4 \longrightarrow 0 \quad \Longrightarrow \quad \frac{S_n}{n} \longrightarrow 0 \qquad \mathbb{P}\text{-a.s.}$$

Thus

$$\mathbb{P}\left(\lim_{n\to\infty}\frac{S_n}{n} = 0\right) = 1.$$

Therefore, the statement is proved.

HW 61 — Strong Law with a Nonzero Common Mean

Statement. Under the hypotheses of Theorem 6.10, replace $\mathbb{E}[X_n] = 0$ with $\mathbb{E}[X_n] = \mu$ for every $n \in \mathbb{N}$, where $\mu \in \mathbb{R}$. Prove that

$$\frac{S_n}{n} \longrightarrow \mu \qquad \mathbb{P}\text{-a.s.}$$

Expanded Proof.

Define $Y_n := X_n - \mu$. By HW 56, $Y_n \in L^4$, and

$$\mathbb{E}[Y_n] = \mathbb{E}[X_n] - \mu = 0.$$

The equality follows from linearity of expectation. The variables Y_n remain independent because they are measurable functions of the independent variables X_n. We also obtain

$$\mathbb{E}[Y_n^4] = \mathbb{E}[|X_n - \mu|^4] \le 8\big(\mathbb{E}[|X_n|^4] + |\mu|^4\big) \le 8(K + |\mu|^4) < \infty.$$

The first inequality follows from $|a-b|^4 \le 8(|a|^4 + |b|^4)$, a consequence of the triangle inequality and the convexity of $t \mapsto t^4$ on $[0,\infty)$. The second uses the uniform fourth-moment bound.

Thus (Y_n) satisfies Theorem 6.10, which gives

$$\frac{1}{n}\sum_{i=1}^{n} Y_i = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu) = \frac{S_n}{n} - \mu \longrightarrow 0 \qquad \mathbb{P}\text{-a.s.}$$

The second equality follows from the definition of S_n and the presence of n copies of μ in the sum. Consequently,

$$\frac{S_n}{n} \longrightarrow \mu \qquad \mathbb{P}\text{-a.s.}$$

Therefore, the statement is proved.